

Silicon Nitride Arrayed Waveguide Gratings

Thèse

Qi Han

Doctorat en génie électrique
Philosophiæ doctor (Ph. D.)

Québec, Canada

© Qi Han, 2023

Silicon Nitride Arrayed Waveguide Gratings

Thèse

Qi Han

Sous la direction de:

Wei Shi, directeur de recherche
Michaël Ménard, codirecteur de recherche

Résumé

Le développement des télécommunications optiques à haute capacité fait des multiplexeurs en longueur d'onde un sujet brûlant des récentes recherches. Dans cette thèse, nous proposons et démontrons des réseaux sélectifs planaires ou (Arrayed Waveguide Grating, AWG) basés sur une plateforme de Nitrure de silicium (SiN) comme multiplexeur ou démultiplexeur.

Dans le premier chapitre, nous comparons les guides d'onde en silicium et en nitrure de silicium et confirmons que le SiN sera considéré comme la plateforme principale de ce travail. Nous présentons des simulations des guides d'onde de SiN qui forme les AWGs, incluant les guides d'onde planaires, les guides d'onde à bande, les guides d'onde courbés et les guides d'onde fuselée utilisant *FDTD solution* et *Mode solution* d'*Ansys Lumerical*. L'influence des paramètres de conceptions des AWGs en SiN tels que la longueur focale, la distance séparant les guides d'onde, l'espacement entre les ouvertures adjacentes et les pertes de propagation liées à la fabrication sur les performances est aussi étudié en utilisant sur un modèle semi-analytique.

Les AWGs communs sont typiquement conçus pour les modes électriques transverses (TE). Pour améliorer la capacité de transmission des réseaux WDM, dans le second chapitre, nous présentons un AWG insensible à la polarisation conçus avec des guides d'onde en SiN. L'insensibilité à la polarisation des AWGs est obtenue lorsque l'espace intercanal et la longueur d'onde centrale des deux modes sont alignés pour un même AWG. L'alignement de l'espace intercanal entre les deux états de polarisations est obtenu en optimisant la géométrie du réseau de guide d'onde, alors que l'insensibilité de la longueur d'onde centrale est obtenue en séparant les deux états de polarisations et en ajustant leur angle d'incidence à l'entrée du coupleur en étoile pour compenser la dispersion entre les modes dans l'AWG. Un multiplexeur de longueur d'onde 1×8 avec un espacement entre les canaux de 100 GHz et une diaphonie de -16 dB est démontré expérimentalement.

Dans la conception d'un AWGs en SiN, un espacement d'une largeur supérieur à 10 μm entre des guides d'ondes identiques minimise le couplage parasite augmentant ainsi leur empreinte. Dans le troisième chapitre, nous présentons un AWG ultra-compact 1×8 ayant une séparation de 100 GHz entre les canax rendu possible grâce à des guides d'onde en super-réseaux supprimant le couplage entre les guides d'onde. Bénéficiant de la haute densité du super-réseau,

cet AWG possède une empreinte compacte de $4.3mm \times 0.6mm$, ce qui est plus que 2 fois plus petit qu'un AWG conventionnel ayant des performances similaires à celui présenté dans le second chapitre. Le SL-AWG montre aussi une faible perte d'insertion de 3.4 dB et une faible diaphonie de -18 dB.

À part le couplage entre les guides d'onde discuté dans le troisième chapitre, l'erreur de phase produite par les variations de fabrication a l'impact le plus important sur les performances de l'AWG. Il a été étudié que leurs performances sont liées à la longueur du réseau de guide d'onde déterminant l'erreur de phase. Toutefois, il existe encore un écart de quantification de l'impact de la longueur du réseau et les variations de fabrication sur les performances de l'AWG. Dans le quatrième chapitre, nous présentons une analyse statistique de l'AWG en présence d'erreurs de phase dans les guides d'onde. Des figures de mérites importantes pour la performance incluant les pertes d'insertion, la diaphonie et la non-uniformité sont paramétrées en fonction de la longueur de cohérence, un paramètre physique qui caractérise l'accumulation d'erreur de phase dans les guides d'ondes optique. Une longueur de cohérence de 23.7 mm au niveau de la matrice pour les guides d'onde de SiN peut être extraite en mesurant les variations dans la longueur d'onde de résonnance d'un interféromètre de Mach-Zhender. Au travers de simulations Monte-Carlo, nous examinons l'impact de l'erreur de phase sur les performances de l'AWG avec une espace entre les canaux de 100 GHz et 200 GHz.

Abstract

The development of optical communications with high transmission capacity makes wavelength division multiplexing (WDM) systems a hot topic of recent research. In this thesis, we propose and demonstrate arrayed waveguide gratings (AWGs) based on a SiN platform as the multiplexers or demultiplexers.

In the first chapter, we compare the material and waveguides between silicon and silicon nitride. We present numerical simulations of the SiN waveguides, including slab waveguides, strip waveguides, bent waveguides and tapered waveguides, using FDTD solutions and MODE solutions from Ansys Lumerical. These waveguides are used to form an AWG in this thesis. The influences of SiN AWGs designed parameters including focal length, separation of arrayed waveguides, gaps between adjacent apertures and propagation loss on the performances are studied based on a semi-analytical model.

Common AWGs are typically designed in TE mode. In order to improve the transmission capacity in WDM system, in the second chapter, we present a polarization insensitive AWG built with SiN waveguides. The polarization insensitive AWGs are obtained when both the channel spacing and the center wavelength are aligned for TE and TM modes in a single AWG. The channel spacing polarization insensitivity is obtained by optimizing the geometry of the arrayed waveguides whereas the central wavelength polarization insensitivity is obtained by splitting the two polarization states and adjusting their angle of incidence at the input star coupler. A 100 GHz 1×8 AWG with crosstalk below -16 dB is demonstrated experimentally.

In the design of SiN AWGs, the gaps of wider than 10 μm between adjacent identical waveguides are designed to minimize parasitic coupling. However, these gaps suppress further shrinking the footprint of AWGs. In the third chapter, we present an ultra-compact 100 GHz 1×8 SiN AWG enabled by a novel concept of the waveguide superlattice suppressing coupling between waveguides. Benefiting from the densely arrayed waveguides patterning with waveguide superlattice, this superlattice AWG has a compact footprint of 4.3 mm $\times$ 0.6 mm, which is more than two times smaller than a conventional AWG with similar performance. The SL-AWG also shows a low insertion loss of 3.4 dB and a low crosstalk level of -18 dB.

Beside the coupling between waveguides discussed in the third chapter, the phase errors due

to fabrication variations have a considerable impact on the performance of AWGs. It is shown that their performances are related to the length of arrayed waveguides determining the phase errors. However, there lacked a practical way to quantify the impact of arrayed waveguide length and fabrication variations on the performances of AWGs. In the fourth chapter, we present a statistical analysis of AWGs in presence of phase errors of arrayed waveguides. The important figures of merits including insertion loss, crosstalk and non-uniformity, are parameterized by the coherence length, a physical parameter that characterizes the accumulated phase errors in an optical waveguide. A die-level coherence length of 23.7 mm for the SiN waveguides is extracted by measuring variation of resonant wavelength of Mach-Zehnder interferometers. Through Monte Carlo simulations, we present the impacts of phase errors on performance of 1×4 AWGs with 200 GHz and 100 GHz channel spacings.

Contents

Résumé ii

Abstract iv

Contents vi

List of Tables viii

List of Figures ix

List of Abbreviations and Symbols xii

Acknowledgements xvii

Foreword xviii

Introduction 1
- I.1 Wavelength division multiplexing systems . . . 1
- I.2 Silicon photonics . . . 2
- I.3 Optical multiplexers and demultiplexers . . . 3
- I.4 Objective and thesis outline . . . 7

1 Fundamentals of silicon photonic waveguides and AWGs 8
- 1.1 Silicon-on-insulator and silicon nitride . . . 8
- 1.2 SiN waveguides . . . 10
- 1.3 Operating principle of arrayed waveguide gratings . . . 15

2 Polarization-insensitive silicon nitride arrayed waveguide grating 30
- 2.1 Résumé . . . 30
- 2.2 Abstract . . . 30
- 2.3 Introduction . . . 31
- 2.4 Operating principle . . . 32
- 2.5 Experimental results . . . 35
- 2.6 Conclusion . . . 38

3 Superlattice Arrayed Waveguide Grating in Silicon Nitride 39
- 3.1 Résumé . . . 39
- 3.2 Abstract . . . 39
- 3.3 Introduction . . . 40

3.4 Operating principle of superlattice AWG 40
3.5 Experimental results 44
3.6 Conclusion 48

4 Phase errors and statistical analysis of silicon-nitride arrayed waveguide gratings 49
4.1 Résumé 49
4.2 Abstract 50
4.3 Introduction 50
4.4 Phase error and coherence length 52
4.5 Statistical analysis of AWGs 56
4.6 Conclusion 65
4.7 Appendix 65

Conclusion and outlook 67

Bibliography 70

List of Tables

I.1 Summary of state for AWGs on various platforms 6

List of Figures

I.1 WDM operating principle. 1
I.2 Schematic cross-section of a SOI waveguide. 2
I.3 SEM figure of (a) cascaded micro-ring resonators published in [17], (b) an AWG, (c) fabricated cascaded MZIs and (d) an echelle grating published in [12]. . . . 4
I.4 Schematic of (a) 1 × N AWG, (b) star coupler by Dragone in [19], (c) phased array by Smit [20] and (d) AWG by Dragone [21]. 5

1.1 Cross-sectional view of a silicon nitride waveguide. 10
1.2 Simulation of the effective index of the waveguide modes versus thickness for a silicon nitride slab waveguide with a thickness of 450 nm at a wavelength of (a) 1550 nm and (b) 1600 nm. 11
1.3 Simulation of the effective index of the waveguide modes versus width for a silicon nitride strip waveguide with a thickness of 450 nm at a wavelength of (a) 1550 nm and (b) 1600 nm. 11
1.4 Simulation of the effective index of the waveguide modes versus thickness for a silicon nitride strip waveguide with a width of 850 nm at a wavelength of (a) 1550 nm and (b) 1600 nm. 12
1.5 Waveguide bending loss for the fundamental mode versus radius for a silicon nitride bent waveguide with a width of 450 nm and 850 nm. 13
1.6 Transmission ratio versus length for a silicon nitride linear waveguide taper with $w_i = 5$ μm and $w_o = 850$ nm. 14
1.7 Effective index of the waveguide and its dependence at 1550 nm on the waveguide width for the (a) 220 nm thick SOI waveguide and 450 nm thick SiN waveguide; on waveguide thickness for the (b) 500 nm wide SOI waveguide and 850 nm wide SiN waveguide. 15
1.8 Schematic of the star coupler. S, P are the launching point and focusing point on the Rowland circle, respectively. The point O and G are two points of reflection on the grating circle. 16
1.9 Schematic of the (a) 1 × N AWG, (b) input star coupler, and (c) output star coupler. 18
1.10 Input and output star coupler geometry and coordinate system. 20
1.11 Transmission spectrum of 1 × 4 AWG with performance metrics labeled. 22
1.12 Schematic of input star coupler of (a) appropriate design, with (b) an insufficient resolution and (c) insufficient numbers of arrayed waveguides with the field distribution of light. 23
1.13 (a) Schematic of the star coupler with focal length labeled and (b) separation d and fill factor. 26

1.14 Insertion loss for separation d of 3.4 μm, 4.4 μm and 5.4 μm with various (a) gaps between adjacent apertures of arrayed waveguides and (b) fill factor. . . . 27
1.15 Insertion loss for AWGs with different (a) focal length of star couplers and (b) separations. . . . 28
1.16 Insertion loss as a function of propagation loss in the arrayed waveguides for 1×4 AWGs with channel spacing of 400 GHz, 200 GHz and 100 GHz. . . . 29

2.1 (a) Microscope image of the polarization insensitive AWG; (b) microscope image of the polarization beam splitter; (c) schematics of the input star couplers with different injection angles for TE and TM modes; (d) Schematic of the output star coupler (not in scale). . . . 33
2.2 Ratio (Γ2) as a function of the width of the waveguide. . . . 34
2.3 (a) Measured power transmission spectra of the polarization beam splitter for the TE and TM modes; (b) layout schematic of the polarization beam splitter. 36
2.4 Measured power transmission spectra of the AWG for the TE and TM polarizations. . . . 37
2.5 Simulated center wavelength deviations due to fabrication errors in waveguide width. . . . 38

3.1 Schematic of a waveguide superlattice. . . . 41
3.2 Simulated results of (a) the coupling ratio in a supercell, (b) the coupling ratio in the directional coupler, (c) the field distribution in the supercell and (d) the field distribution in the directional coupler. . . . 42
3.3 Schematic of (a) the AWG with an arrayed waveguide superlattice, of (b) the output star coupler, of (c) an arrayed waveguide superlattice and of (d) conventional arrayed waveguides with identical widths. . . . 43
3.4 Microscrope image of (a) the SL-AWG and of (b) the conventional AWG. (c) Scanning electron microscope (SEM) image of a cross-section of the superlattice. (d) SEM image of the cross-section of a waveguide with a 800 nm width and (e) of a waveguide with a 900 nm width. . . . 45
3.5 Measured spectrum of (a) the conventional MZI and of (b) the MZI with waveguide supercell arms. . . . 46
3.6 Measured spectrum of (a) the SL-AWG and of (b) the reference conventional AWG. . . . 47

4.1 Procedure for the proposed statistical analysis. . . . 51
4.2 (a) Schematic of the SiN waveguide; effective index of the waveguide and its dependence at 1550 nm on (b) the waveguide width, (c) waveguide thickness and (d) waveguide sidewall angle. . . . 54
4.3 Simulated spectra of MZIs in 100 Monte Carlo samples with (a) various coherence lengths and a total waveguide length $L_1 + L_2 = 2$ mm, (b) various waveguide lengths and a coherence length $L_{coh} = 20$ mm; (c) measured spectra of the MZI with waveguide length of 2.3 nm within dies; (d) the phase error variances extracted based on MZIs. . . . 56
4.4 (a) Schematic of an AWG; (b) measured spectrum for the 1 × 4 200 GHz AWG and (c) 1000 Monte Carlo calculated responses in light color, one specific response in dark color, and average crosstalk floor shown with a black dash line. 57
4.5 Insertion loss for 200 GHz and 100 GHz AWGs with different propagation loss. 59

4.6 (a) and (b) show the simulated insertion loss as a function of the number of waveguides in the array (the equivalent length of the middle waveguide is also shown) at various coherence lengths for the AWG with a channel spacing of (a) 200 GHz and (b) 100 GHz. Each scatter plot contains 1000 Monte Carlo simulations. The standard deviations at 25 arrayed waveguides are labeled. . . 60
4.7 Crosstalk distributions of AWGs with 17 waveguides based on a coherence length of 20 mm in 1000 Monte Carlo samples for (a) the 200 GHz AWG and (b) the 100 GHz AWG; probability of 0.5, 0.68 and 0.97 to obtain the maximum crosstalk for (c) the 200 GHz AWG and (d) the 100 GHz AWG. 62
4.8 (a) and (b) show the simulated average crosstalk as a function of the number of waveguides in the array (the equivalent length of the middle waveguide is also shown) at various coherence lengths for the AWG with a channel spacing of (a) 200 GHz and (b) 100 GHz. (c) and (d) show the worst-case crosstalk for (c) the 200 GHz AWG and (d) the 100 GHz AWG. Each scatter plot contains 1000 Monte Carlo simulations. The standard deviations at 25 arrayed waveguides are labeled. 63
4.9 (a) and (b) show the simulated non-uniformity as a function of the number of waveguides in the array (the equivalent length of the middle waveguide is also shown) at various coherence lengths for AWG with a channel spacing of (a) 200 GHz and (b) 100 GHz. Each scatter plot contains 1000 Monte Carlo simulations. The standard deviations at 25 arrayed waveguides are labeled. . . 64

List of Abbreviations and Symbols

Abbreviations and acronyms:

AWG Arrayed waveguide grating

BOX Buried oxide

CMOS Complementary metal–oxide semiconductor

CWDM Course wavelength division multiplexing

DC Directional coupler

DWDM Dense wavelength division multiplexing

EME Eigenmode expansion

FDTD Full-wave finite difference time domain

FIR Finite impulse response

FPR Free propagation region

FSR Free spectral range

IIR Infinite impulse response

ITU International Telecommunication Union

LPCVD Low pressure chemical vapour deposition

MMI Multimode interference coupler

MZI Mach-Zehnder interferometer

NIR Near-infrared

PBS Polarization beam splitter

PECVD Plasma enhanced chemical vapour deposition

PIC Photonic integrated circuit

PSR Polarization splitter and rotator

SEM Scanning electron microscope

SGC Surface grating coupler

SiN Silicon nitride

SiP Silicon photonic

SL-AWG Superlattice arrayed waveguide grating

SOI Silicon-on-insulator

WDM Wavelength division multiplexing

WDM-PON Wavelength division multiplexing passive optical network

WG Waveguide

Symbols:

β Propagation constant

ΔL Length difference between adjacent waveguides

Δl Path length difference

δn_{eff} Random error of effective index

$\Delta\lambda_{FSR}$ FSR

$\delta\phi$ Phase error

λ Wavelength

ϕ Phase

θ Angle

D	Separation between adjacent output channels
d	Separation between adjacent arrayed waveguides
f	Focal length
I	Power intensity
K	Wave number
L	Waveguide length
L_{coh}	Coherence length
L_c	Correlation length
m	Grating order
N	Arrayed waveguides number
N_o	Output channels number
n_{eff}	Effective index in arrayed waveguides
n_g	Group index
n_s	Effective index in slab waveguide
t	Waveguide thickness
U	Field of supported mode
w	Waveguide width

When optimizations cannot provide good results, something new is being discovered.

Whether you can observe a thing or not depends on the theory which you use. It is the theory which decides what can be observed.

Albert Einstein

Acknowledgements

I am so lucky that I can get the help from many people during my Ph.D. project. I clearly see that I cannot finish my Ph.D. degree without their help. Now, it is a good time to thank all of them. Firstly, I would like to thank my family for encouraging and supporting me to focus on my researches.

Secondly, I would like to thank my supervisor professor Wei Shi for guiding me with great patience. He has given lots of suggestions to improve my English and professional skills. His insightful discussions make me face all challenges with confidence.

Thirdly, I would like to thank my co-supervisor professor Michaël Ménard for his selfless assistance and discussion in most of my work. He is always glad to help me and gives me suggestions on researches and paper writing.

I would like to thank the members of the jury for spending the time reading my thesis and for valuable comments and suggestions.

Moreover, I am grateful to my colleagues in COPL for their friendship and support. I would like to thank Jonathan St-Yves, Simon Levasseur, Nathalie Bacon for helping me make the setup. Some of my experiments cannot be finished without their help.

I am grateful to my friend, David Turgeon who helped me to translate some sections of this thesis in French. I also would like to thank my good friends Jonathan St-Yves, Daniel Robin, Xun Guang, Ye Tian, Zhongjin Lin, Huijie Wang, Yuxuan Chen, Hui Wang, Xingguo Xiao, Zibo Zheng, Rizan Homayoun Nejad, Omid Jafari, Mohammad Ahmadi, Dominic Hould, Simon Bélanger-de Villers, and Alessandro Corsi for their friendship and moral support.

Foreword

Chapters 2-4 of this thesis are based on materials published in journal papers. Most of the contents in these three chapters are the same as the journal papers while the chapters include minor modifications to provide better coherence to the thesis. I was the first author of these papers. In the following, the responsibility and contribution of each co-author of the papers used for this thesis will be explained.

Chapter 2: Qi Han, Jonathan St-Yves, Yuxuan Chen, Michaël Ménard and Wei Shi. "Polarization-insensitive silicon nitride arrayed waveguide grating." *Optics letters*, Vol. 44, No. 16 (2019): 3976-3979. This paper is submitted on June 21 2019 and published on August 7 2019. This paper, we design and experimentally demonstrate a polarization-insensitive arrayed waveguide grating based on SiN platform. I performed all the simulations and experiments. I analyzed the captured data. Yuxuan Chen helped me design the polarization splitters. Jonathan St-Yves helped me build the experimental setup. Wei Shi, who was the supervisor of the project, provided overall guidance. The paper was written by me and revised by Michaël Ménard and Wei Shi.

Chapter 3: Qi Han, Michaël Ménard and Wei Shi. "Superlattice Arrayed Waveguide Grating in Silicon Nitride." *IEEE Photonics Technology Letters*, Vol. 32, No. 22 (2020): 1411-1414. This paper is submitted on August 3 2020 and published on November 15 2020. The device was designed by me. All the experiments were performed by me as well as most analysis of the captured data. Wei Shi, who was the supervisor of the project, provided overall guidance. The paper was written by me and revised by Michaël Ménard and Wei Shi.

Chapter 4: Qi Han, Daniel Robin, Antoine Gervais, Michaël Ménard, and Wei Shi. "Phase errors and statistical analysis of silicon-nitride arrayed waveguide gratings." *Optics express*, Vol. 30, no. 24 (2022): 42784-42800. This paper is submitted on June 14 2022 and published on November 7 2022. I performed all the simulations and experiments. I analyzed the captured data and built the theoretical model. Daniel Robin helped me build the AWGs model. Antoine Gervais helped me the data analysis. Wei Shi, who was the supervisor of the project, provided overall guidance. The paper was written by me and revised by Michaël Ménard and Wei Shi.

Introduction

I.1 Wavelength division multiplexing systems

Optical communications systems are widely utilized, thanks to their low power loss and ultra-high bandwidth. In order to increase the data transmission capacity, wavelength division multiplexing systems are widely used in optical telecommunication [1]. WDM is a technology which transmits a number of optical signals through a fiber or a waveguide (WG) carried by different wavelengths (or frequencies) [2].

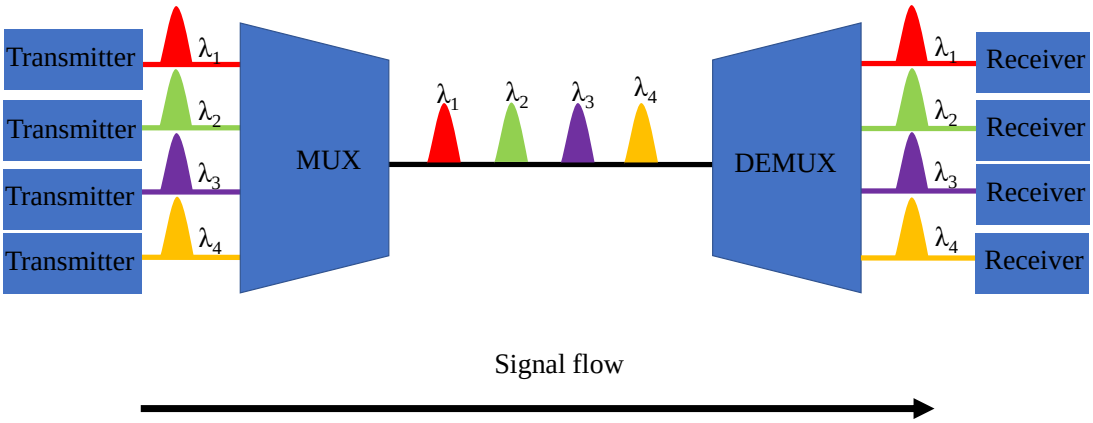


Figure I.1: WDM operating principle.

A WDM system requires a multiplexer at the transmitter to combine the optical signals carried by different wavelengths together and a demultiplexer at the receiver to separate them apart, as shown in Fig. I.1. WDM systems are based on the concept of multiplexing multiple wavelengths onto one fiber but differ in transmission capacity because they have different channel spacing and number of channels. Two different versions of WDM including coarse WDM (CWDM) and dense WDM (DWDM), are defined by standards of the International

Telecommunication Union (ITU). CWDM (ITU standard G.694.2) are WDM systems with small number of channels (e.g. four or eight) and a large channel spacing of 20 nm on each fiber. Fewer channels and wider channel spacing make CWDMs less sophisticated and more compact, therefore, they are widely used for systems in which transmission capacity is not an important requirement.

DWDM (ITU standard G.694.1) uses a small channel spacing of 12.5 GHz, 25 GHz, 50 GHz or 100 GHz. The smaller wavelength spacing makes them support more channels onto a single fiber. Therefore, a DWDM system can implement 40 channels with 100 GHz spacing [3] or 80 channels with 50 GHz spacing [4] in the C-band (1530 nm - 1565 nm) wavelength range. 12.5 GHz spacing or ultra-dense WDM is also being developed with some technologies [5]. New technologies used extend the applicable wavelengths into the L-band (1565 nm - 1625 nm) to improve the transmission capacity [6]. In this thesis, the main research field focuses on the DWDM system with 100 GHz in the C-band wavelength range.

I.2 Silicon photonics

Optical devices with high performances can improve optical communications. Semi-conductor lasers, optical fibers and fiber amplifiers are essential in long-range communication [7]. Photonics integrated circuits (PICs) provide complex functionalities on a single chip. They are also explored for inter-chips and intra-chips communications.

Integrated photonics has been developed on several material platforms such as silica-on-silicon and silicon-on-insulator. Devices based on silica-on-silicon are widely utilized due to their high performance. However, their large footprints limit further integration. Silicon-on-insulator

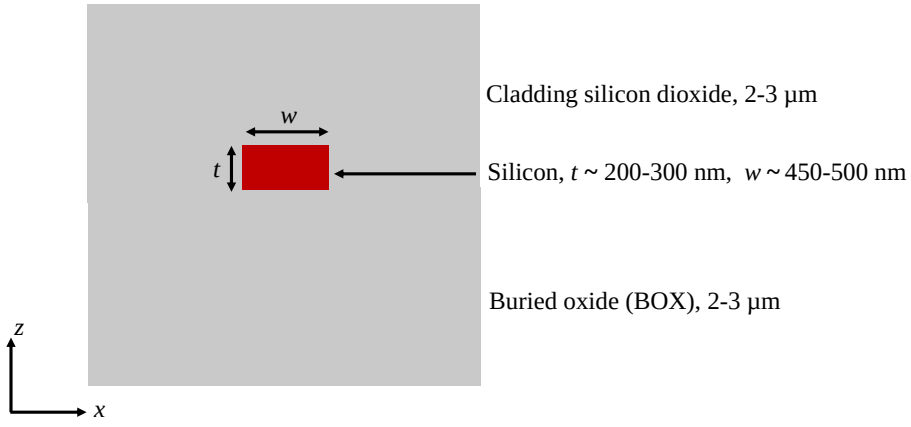


Figure I.2: Schematic cross-section of a SOI waveguide.

(SOI) platforms are attractive due to unprecedented compactness, integration density and low-cost fabrication. A typical SOI platform consists of a silicon layer and a silicon dioxide cladding, as shown in Fig. I.2. SOI photonic integrated circuits have much smaller footprints thanks to the high refractive index contrast between the waveguide core and cladding.

Silicon photonic circuits can be fabricated with existing semiconductor fabrication techniques and thus they can be affordable and highly integrated. The silicon is typically patterned with sub-micrometre precision to implement microphotonic components. Silicon waveguides can be used to design photonic devices, such as wavelength filters, modulators, and detectors. These components are commonly operated at a wavelength of 1.55 micrometer in optical systems.

However, SOI waveguides are sensitive to fabrication variations which limits their developments. SiN waveguides have a moderate refractive index contrast, which results in better tolerances to fabrication variations. Therefore, they can be used as an alternative platform. A detailed discussion is presented in the Chapter 1. In this thesis, we use a SiN platform as the main platform.

I.3 Optical multiplexers and demultiplexers

I.3.1 Optical filters

Multiplexers and demultiplexers can be implemented using various filters, as shown in Fig. I.3. There are several types of wavelength filters, but all of them are based on the interference of light. Finite impulse response (FIR) filters such as Mach-Zehnder interferometers (MZI) filters [8; 9], arrayed waveguide gratings filters and echelle gratings filters [10; 11] rely on interferences obtained through a finite number of delay waveguides. We can also implement infinite impulse response (IIR) filters based on feed-back mechanisms which provide multiple self-interference of light through the delay waveguide [12]. IIR filters include ring resonators and Fabry-Perot cavities [13]. Compared to FIR wavelength filters, IIR filters usually have compact footprints since they use the same optical delay waveguide multiple times. However, the feed-back mechanisms create a high peak power which makes the IIR filters suffer from nonlinear effects [12]. Moreover, the impacts of fabrication variations on the delay waveguides used are amplified. Thus, IIR filters suffer from issues related to their sensitivity to fabrication variations [14].

FIR filters are based on two or more interfering optical delay lines. The input light is separated into dispersive lines of various length, and the light waves from the dispersive lines are recombined at different outputs depending on their wavelength. A Mach-Zehnder interferometer filter separates the light into two waveguides, in which one delay stage is introduced. In WDM systems, MZI filters are used in multiple cascaded stages [15; 16]. An AWG uses a star coupler to separate the light into dispersive optical delay waveguides (array waveguides), and

an echelle grating has reflecting facets along a grating circle at various distances as the delay lines [11]. For both AWGs and echelle gratings, the optical delay length determines the free spectral range (FSR). The slab waveguide has a lower group index than the strip waveguides. Thus, the delay length needs to be longer in an echelle grating than in an AWG to obtain the same FSR. The footprint of an echelle grating increases faster than a comparable AWG with smaller FSR. For a large FSR and few channels, an echelle grating can be extremely compact making it suitable to realize CWDM (de)multiplexers with a few wide channels. Compared to echelle gratings, AWGs can usually achieve narrow channel spacing with good performances.

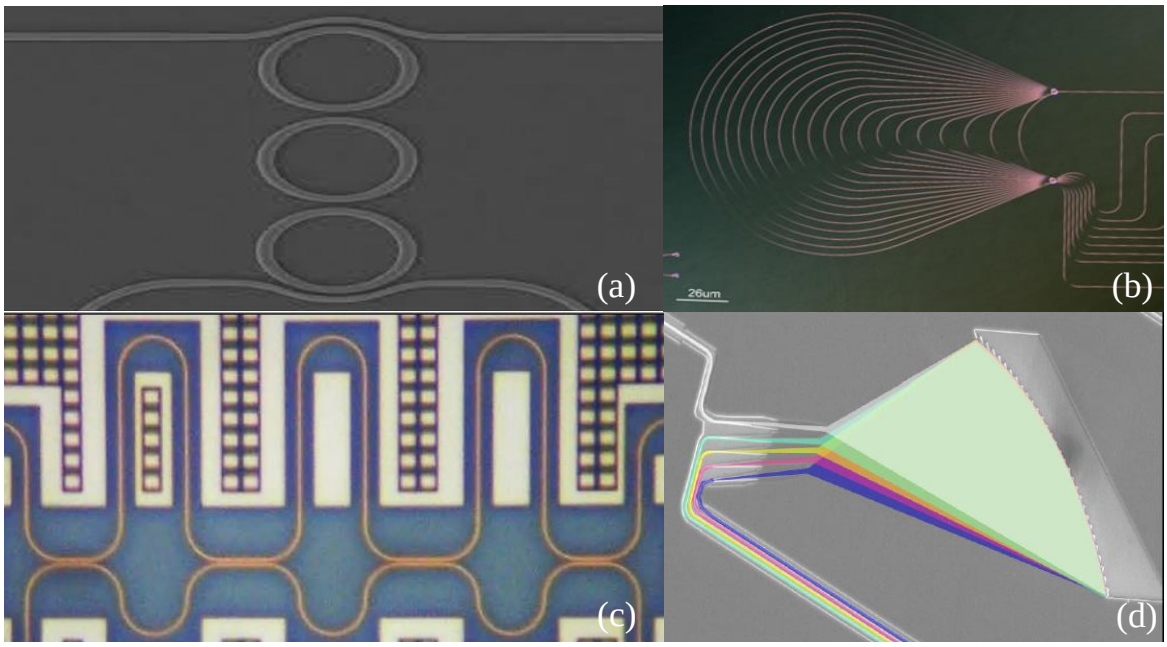


Figure I.3: SEM figure of (a) cascaded micro-ring resonators published in [17], (b) an AWG, (c) fabricated cascaded MZIs and (d) an echelle grating published in [12].

One important performance metric for optical filters is crosstalk. For AWGs and echelle gratings, the crosstalk is determined by how well the image from the input port is reproduced and captured at the aperture of the output ports. The quality of the image is based on the number of optical delay lines since more optical delay lines increase resolution. However, more delay lines also mean longer delay lines, and higher phase errors causing more crosstalk. Moreover, the footprint is increased with a large amount of delay lines. Therefore, in an AWG or echelle grating with a small number of optical delay lines, a reduction in crosstalk is obtained by increasing the path number until an optimum number is reached. Above that crosstalk will increase with the number of delay lines [18].

Insertion loss is another important metric for a filter. The insertion loss of MZIs is very sensitive to phase errors which introduce shifts in the resonant wavelengths resulting in the power not being coupled into the expected ports. The insertion loss of AWGs is caused by propagation losses in the arrayed waveguides and the transition loss between the star coupler and the arrayed waveguides. The insertion loss of echelle gratings depends on the quality of the reflecting facets, which are difficult to define accurately.

In [18], a 1 × 4 echelle grating with a channel spacing of 6.4 nm has a footprint of 700 μm × 420 μm, more than 4 times larger than that of a 1 × 4 AWG (275 μm × 245 μm) with a channel spacing of 6.4 nm in the same SOI platform. However, the crosstalk of the echelle grating is −18.7 dB. As a comparison, the AWG has a crosstalk of −27.1 dB. The echelle grating with small channel spacing needs a very large slab region, which introduces large phase errors. When the FSR or channel spacing increases, the gaps between adjacent arrayed waveguides will become small in an AWG, which results in excessive coupling. Using S-shaped waveguides as an alternative layout makes it possible to design an AWG with very low channel spacing. However, the waveguides of the S-shaped AWG are very long, since they contain two additional bends. High phase errors are introduced with the very long arrayed waveguides. A 1 × 4 echelle grating with a channel spacing of 20 nm has a footprint of 250 μm × 155 μm which is half of that of a 1 × 4 AWG (305 μm × 260 μm) with a channel spacing of 24 nm. The crosstalk of the echelle grating is −23.3 dB which is better than that of −20.6 dB for AWG. Therefore, echelle gratings have a good performance in low resolution (CWDM) de/multiplexer applications. For high resolution (DWMD) de/multiplexer applications, AWGs are normally a better choice.

I.3.2 Arrayed waveguide gratings

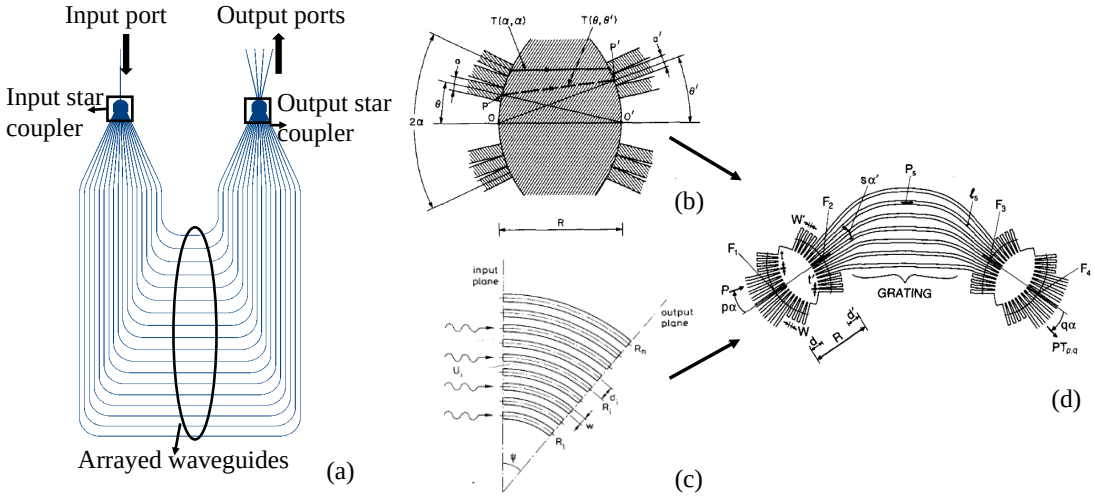


Figure I.4: Schematic of (a) 1 × N AWG, (b) star coupler by Dragone in [19], (c) phased array by Smit [20] and (d) AWG by Dragone [21].

The schematic of a 1 × N AWG is shown in Fig. I.4(a). The AWG consists of the input port, the input star coupler, the arrayed waveguides, the output star coupler, and the output ports. The light carrying the signals with multiple wavelengths propagates into the input star coupler through the input channel port. The light will no longer be confined and expands in the input star coupler. The light is split and coupled into the arrayed waveguides. The arrayed waveguides have a constant length difference between adjacent waveguides, thus a constant phase difference is introduced between the light waves. The multiple light waves with a constant phase difference interfere in the output star coupler. Different wavelengths will have different focused points. The output ports collect the light in the focused points along the edge of the output star coupler. With this method, the demultiplexing process of signals with multiple wavelengths is completed.

Research on AWGs started in 1988. Dragone presented an N × N star coupler [19]. The star coupler is realized in free space confined by two array connecting N single mode fibers each array. In the same year, Smit demonstrated a focusing and dispersive component based on a phased array of bent optical waveguides [20]. In 1991, combing the star coupler and phased array, Dragone experimentally demonstrated a N × N AWG with an insertion loss of 7 dB and a crosstalk of −25 dB [21]. The schematic of the N × N star coupler, phased array and AWGs is shown in Fig. I.4(b), (c) and (d), respectively.

More than two decades later, AWGs have been developed on various platforms, including SiO_2, SOI, InP, polymer, SiON and SiN. Table I.1 shows a comparison of the state-of-the-art for AWGs on these platforms. The InP AWG has the most compact footprint but the worst performances. It is evident that the silica AWG has good performances including a low insertion loss and crosstalk, but a large footprint. SOI AWGs have a compact footprint which motivates people to keep improving their performances. In [22], the SiN waveguides have a thickness of 50 nm and a width of 5.5 μm, which minimizes the impacts of sidewall roughness and fabrication variations, and thus, improve the performances. As a tradeoff, its footprint is comparable to a conventional silica AWG. The devices in our work are implemented on a SiN platform with a waveguide core thickness of 440 nm based on the process available at the commercial foundry. The performances of our AWGs are more sensitive to fabrication quality than that of the SiN AWG in [22], but less sensitive than that of SOI AWGs. Detailed discussions are presented in Chapter 1.

Table I.1: Summary of state for AWGs on various platforms

Platform	Insertion loss	Crosstalk	Non-uniformity	channel number × channel spacing	WG thickness / BOX thickness	Footprint
Silica [23]	1.3 dB	−32 dB	0.6 dB	40 × 100 GHz	4.5 μm / ∼25 μm	31 mm × 23 mm
Polymer [24]	5.1 dB	−24 dB	0.8 dB	8 × 100 GHz	2.0 μm / 3.0 μm	9 mm × 8 mm
InP [25]	5 dB	−12 dB	2 dB	4 × 400 GHz	0.5 μm / ∼1.2 μm	0.23 mm × 0.33 mm
SOI [26]	2.7 dB	−21 dB	2.5 dB	8 × 400 GHz	0.22 μm / 2 μm	0.73 mm × 0.58 mm
SiON [27]	1.3 dB	−21 dB	2.9 dB	16 × 200 GHz	1.0 μm / 4.0 μm	3 mm × 3.5 mm
SiN [22]	0.5 dB	−30 dB	∼4 dB	16 × 200 GHz	0.05 μm / 15 μm	20 mm × 15 mm
SiN (our work) [28]	3.4 dB	−18 dB	2.8 dB	8 × 100 GHz	0.44 μm / 3.2 μm	4.3 mm × 0.6 mm

I.4 Objective and thesis outline

The primary goal is to study and explore novel design methodologies for arrayed waveguide gratings in the SiN platform. This work can improve the performances of AWGs with a small channel spacing and a large number of channels, and allow ones using these devices to create a better and cheaper network. Another goal is to develop a system of yield prediction, not only for AWGs based on SiN platform but also on other material systems, and allow to extend the predictions to other optical devices, at other wavelength ranges.

Due to the constraints in design and fabrication process of silicon nitride, realizing an AWG working for DWDM systems with high performance is not trivial. We propose and demonstrate novel AWGs to improve the performances without increasing cost.

Our work begins with comparing SOI and SiN waveguides. SiN waveguides show better performances in the wavelength range of interest. The SiN waveguides including slab waveguides, strip waveguides, bent waveguides, and tapered waveguides are studied with Lumerical FDTD and MODE solutions. Metrics, including insertion loss, crosstalk and non-uniformity, are introduced to characterize the performances of an AWG. The design strategy and parameters affecting the performances of AWGs are studied in Chapter 1.

Conventional AWGs are only designed to work with the TE mode. Thus, polarization insensitive AWGs with various configurations were studied to improve the transmission capacity of WDM systems. A new polarization insensitivity AWG based on SiN waveguides is proposed and experimentally validated in Chapter 2.

Because of the low effective index contrast of SiN waveguides, gaps wider than 10 μm between adjacent waveguides were normally required to suppress the parasitic coupling between waveguides. As a result, SiN AWGs usually have a large footprint. In order to shrink the footprint while maintaining high performances, a waveguide superlattice made of a periodic structure consisting of alternate waveguide geometries is introduced to form the arrayed waveguides and minimize the couplings between waveguides in Chapter 3. A dense superlattice AWG can suppress the footprint by a factor of two, while keeping the same performances.

The phase errors caused by fabrication imperfections and their impacts of the performances of AWGs are discussed in Chapter 4. Besides parasitic coupling, the phase errors accumulated in arrayed waveguides have an important impact on the performances of AWGs. It is known that longer arrayed waveguides with higher phase errors decrease performances including increased insertion loss and crosstalk. However, this impact was never quantified. Coherence length, a physical parameter that characterizes the accumulated phase errors, previously measured in silica waveguides and SOI waveguides, is adopted. We measure the coherence length of SiN waveguides and the impacts of the phase errors on the performances of AWGs through Monte Carlo simulations.

Chapter 1

Fundamentals of silicon photonic waveguides and AWGs

1.1 Silicon-on-insulator and silicon nitride

The implementation of passive optical components based on SOI waveguides includes splitters [29], filters [30], and resonators [31]. Micro-heaters on the SOI platform can be used to change the refractive index of silicon and implement tunable components [32]. III-V light sources are integrated through wafer-scale processes on this platform [33]. Although silicon waveguides are the mostly used in silicon photonics, alternative CMOS-compatible materials are being explored for the spectral bands for which silicon or silicon dioxide are not transparent and absorb light. Silicon nitride waveguides, where the silicon core is replaced by a silicon nitride core, are being accepted as alternative waveguides. Nevertheless, SiN waveguides still remain CMOS-compatible [34; 35; 36].

Silicon nitride is deposited by low pressure chemical vapour deposition (LPCVD) at high temperature ($>$ 700 °C) or by plasma enhanced chemical vapour deposition (PECVD) at low temperature ($<$ 400 °C) [37]. SiN deposited by LPCVD can be close to stoichiometric Si_3N_4, and shows a good stability on the homogeneity of the material index. It has a refractive index of about 2.0 at a wavelength of 1550 nm. The layer thickness is also well controlled for this type of SiN [38]. However due to the increasing tensile stress during deposition, the thickness deposited is limited to around 400 nm, beyond which cracks would happen [39]. A crack-free SiN film with thickness up to 1.0 μm can be grown by PECVD [40], however the intrinsic absorption loss is high for the PECVD-based SiN. The comparisons of the key properties between the SiN and SOI platform are discussed below.

Applicable wavelength range: SiN waveguides have a wider transparency range than SOI waveguides. SOI waveguides are transparent or have low absorption losses in a wavelength range between 1.1 μm and about 3.7 μm. Therefore, SOI waveguides are not usable for

application at shorter wavelengths. In comparison, SiN waveguides are transparent down to a wavelength of at least 500 nm. Thus, they can be used to implement photonics components at wavelengths below 1.1 μm [41] including spectroscopic sensor [42; 43] and filters at near-infrared wavelengths [44].

Refractive index contrast: SiN waveguides have a lower index contrast than SOI waveguides. Comparing the high index contrast between the index of the Si core (3.5) and that of the silica cladding (1.5) in SOI waveguides, that in SiN waveguides is low since SiN cores have a index of 2. Because of their high index contrast, SOI devices can have a compact footprint. However, their performance is very sensitive to the quality of the fabrication. The high contrast leads to waveguides having high scattering losses resulting from roughness of the sidewalls. Even only a few nm of sidewall roughness can introduce almost 1 dB/cm of loss in SOI waveguides [37]. The high contrast also results in a high sensitivity of the effective index to variations in the geometry of the waveguide, including waveguide thickness and width variations [45]. The effective index variations cause unwanted resonant wavelength shifts for devices such as Mach-Zehnders or ring resonators [46]. It also results in crosstalk in arrayed waveguide gratings [47]. The lower index contrast of SiN waveguides means that these devices usually have a larger footprint than that of SOI devices. Nevertheless, their effective index is less sensitive to fabrication variations and it is easier to obtain better performances with these devices.

Low loss: SiN waveguides have lower losses than SOI waveguides. SOI strip waveguides have typical propagation losses of 1 to 5 dB/cm [48; 49]. In SiN photonic waveguides, losses can be down to about 1 dB/m [22; 50].

In this section, the SiN platform uses a waveguide core thickness of 450 nm. The thickness was set by the process available at the commercial foundry contracted by our industrial collaborator. The fabrication begins with low pressure chemical vapour deposition of 3.2 μm of SiO_2 obtained from tetraethyl orthosilicate on a Si wafer. The SiN layer is deposited using LPCVD, patterned and dry etched. During the final step, the wafer is covered with another 3.2 μm of SiO_2 deposited by plasma enhanced chemical vapour deposition to form the top cladding, as shown in Fig. 1.1.

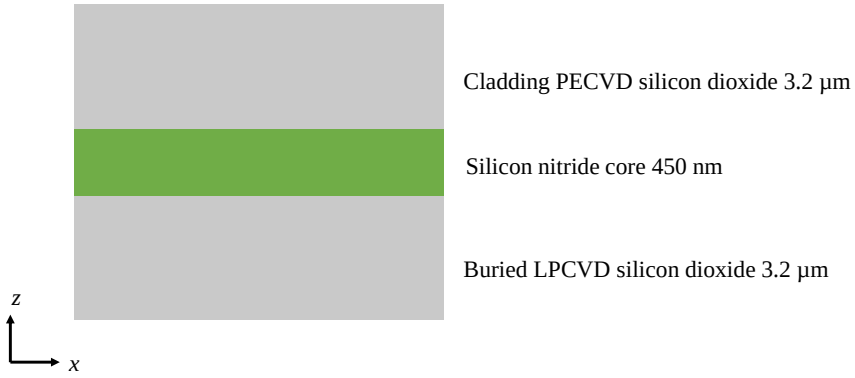


Figure 1.1: Cross-sectional view of a silicon nitride waveguide.

1.2 SiN waveguides

There are various types of waveguides implemented in photonics. In this section, we will study the SiN waveguides used to form the AWGs in this thesis. Slab waveguides form the free propagation region (FPR) of star couplers; strip waveguides only supporting a single mode are used as arrayed waveguides; bent waveguides with a sufficiently large radius minimize the bending loss of strip waveguides; and tapered waveguides minimize the transmission loss between star couplers and arrayed waveguides.

1.2.1 SiN slab waveguides

The effective index of slabs with various thickness is studied using MODE Solution from Ansys Lumerical in this subsection. The effective index of the TE modes and TM modes is discussed. Pure TE or pure TM modes are supported in a 1D slab waveguide. This is different from 2D strip or rib waveguides, where TE like or TM like rather than pure TE or TM modes can be guided [51]. The refractive index of the silicon dioxide bottom and top cladding is plotted with a dotted line, as shown in Fig. 1.2. The modes with effective index below the refractive index of silicon dioxide are not guided. As shown in the Fig. 1.2, slab waveguides support only a single TE and TM mode when the thicknesses are below 576 nm for 1550 nm and 599 nm for 1600 nm, indicated by the black vertical line. Comparing Fig. 1.2(a) and (b), the thickness of slab waveguides that support a single mode is higher at longer wavelength than that at shorter wavelength.

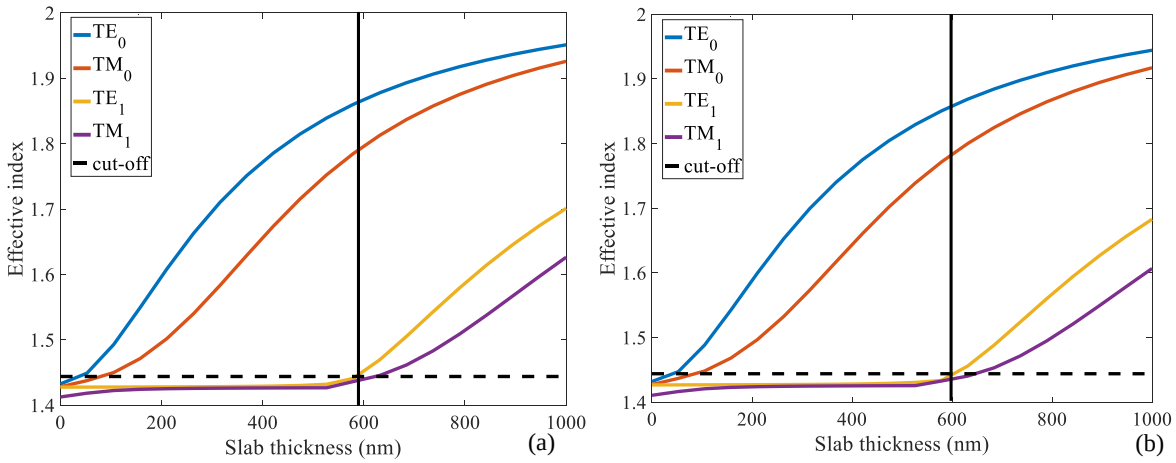


Figure 1.2: Simulation of the effective index of the waveguide modes versus thickness for a silicon nitride slab waveguide with a thickness of 450 nm at a wavelength of (a) 1550 nm and (b) 1600 nm.

1.2.2 SiN strip waveguides

The effective index with different width w of strip waveguides from 500 nm to 1500 nm, using MODE Solution, is studied in this subsection. The effective index determines the condition when a single mode is supported by the waveguides. The effective index for different modes is calculated and plotted in Fig. 1.3(a) for 1550 nm, and in Fig. 1.3(b) for 1600 nm, respectively.

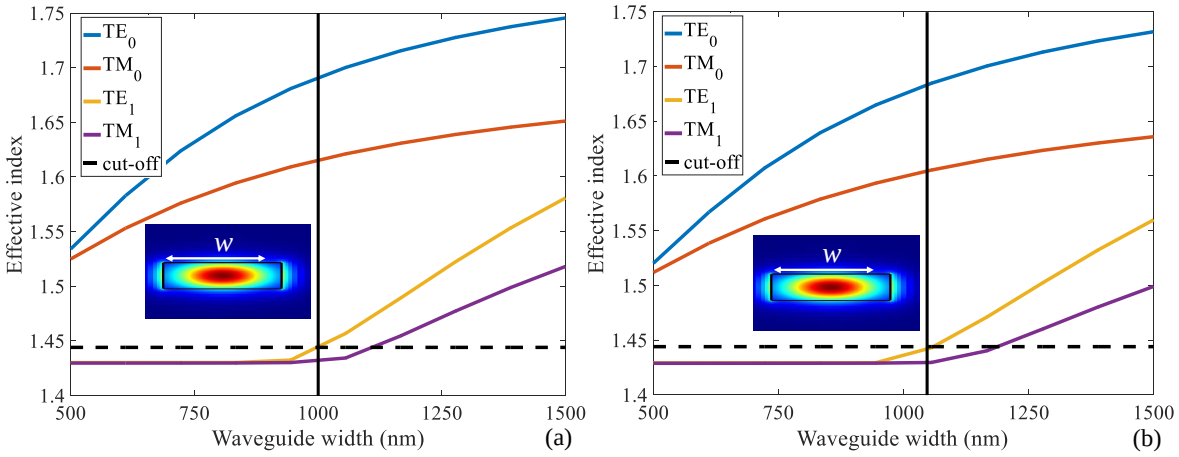


Figure 1.3: Simulation of the effective index of the waveguide modes versus width for a silicon nitride strip waveguide with a thickness of 450 nm at a wavelength of (a) 1550 nm and (b) 1600 nm.

In order to support a single TE-like mode light at 1550 nm, a strip waveguide with a thickness of 450 nm requires a width of less than 1005 nm. For this kind of waveguides, only one single TE-like and one TM-like mode can be supported. The second TE-like mode and second TM-like mode are guided in strip waveguides wider than 1115 nm. For a wavelength of 1600 nm, the single-mode condition is reached when the width is lower than 1060 nm. The width limit is about 60 nm wider at 1600 nm than that at 1550 nm for strip waveguides.

We also study the impact of the thickness t from 100 nm to 800 nm for strip waveguides with a width of 850 nm. The indexes are shown in Fig. 1.4(a) for 1550 nm, and in Fig. 1.4(b) for 1600 nm, respectively. To support a single TE-like mode light at 1550 nm, a strip waveguide requires a thickness thinner than 646 nm. For a wavelength of 1600 nm, a waveguide with a width of 850 nm must have a thickness of less than 714 nm to support a single mode.

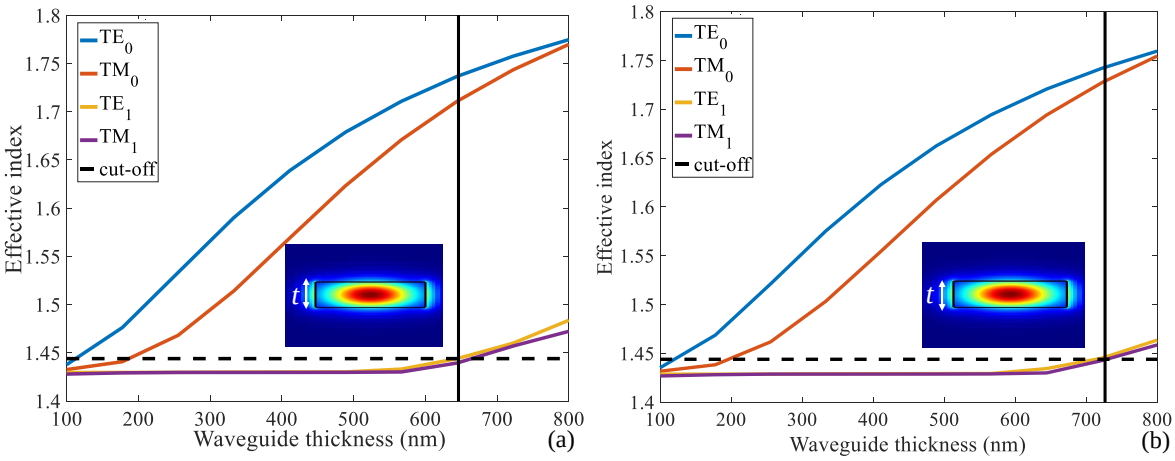


Figure 1.4: Simulation of the effective index of the waveguide modes versus thickness for a silicon nitride strip waveguide with a width of 850 nm at a wavelength of (a) 1550 nm and (b) 1600 nm.

1.2.3 SiN bent waveguides

Bent waveguides are required in optical devices, e.g., for signal routing, and ring/racetrack resonators. Thus, bent waveguides with low losses are discussed in this subsection. We calculated the bending losses using 3D FDTD Solution from Ansys Lumerical. The main contributions to the bending losses include: scattering losses and substrate leakage for bent waveguides with short bending radius, and mode mismatch loss due to imperfect mode overlapped between the straight and the bent waveguides [51]. This leads to scattering in the abrupt radius transition regions e.g., start and end of bent waveguides.

The bending loss for the fundamental mode guided in strip waveguides with a waveguide width of 450 nm and 850 nm is calculated and plotted in Fig. 1.5. For the 850 nm width, the

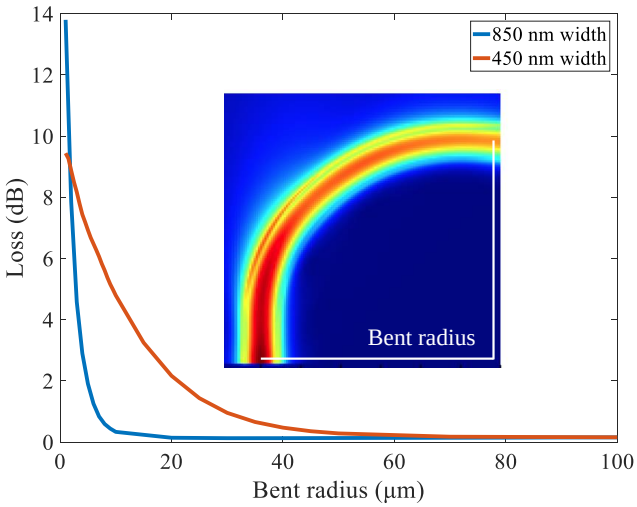


Figure 1.5: Waveguide bending loss for the fundamental mode versus radius for a silicon nitride bent waveguide with a width of 450 nm and 850 nm.

bending loss decreases rapidly as the radius increases. The loss is down to a low level when the bending radius is larger than 40 μm. However, the bending loss decrease slowly until the bend radius is larger than 80 μm for waveguides with a 450 nm width. In this work, we use a bending radius of 100 μm to minimize the impact of bending losses.

1.2.4 SiN tapered waveguides

Tapered waveguides are used to achieve a high coupling efficiency between waveguides with various width or thickness. The shapes of the guided modes change while light propagate in a tapered waveguide. In order to suppress mode conversion from the first order mode of the waveguide to high order modes or radiation modes, an adiabatic taper is needed [52]. The typical tapered waveguide design is linear. An adiabatic linearly tapered waveguide can be designed by increasing/decreasing the taper width very slowly. Therefore, linear adiabatic tapers must be extended long enough to achieve a low propagation loss.

As shown in Fig. 1.6(a), a linearly tapered waveguide connecting a 5 μm wide (i.e. w_i) input waveguide and a 850 nm wide (i.e. w_o) output waveguide was designed. We used the EME solver in Lumerical MODE Solution to calculate the transmission. The simulated transmission of the fundamental TE mode as a function of the tapered waveguide length for a 450 nm thick SiN core is shown in Fig. 1.6(b). The transmission ratio is very low for a short tapered waveguide. The high loss results from a large amount of power coupling into higher-order modes and radiation modes for the tapered waveguides with a length shorter than 40 μm. Tapered waveguides longer than 400 μm have a high transmission ratio of 0.98. We used 400 μm long tapered waveguides in our design to minimize the losses.

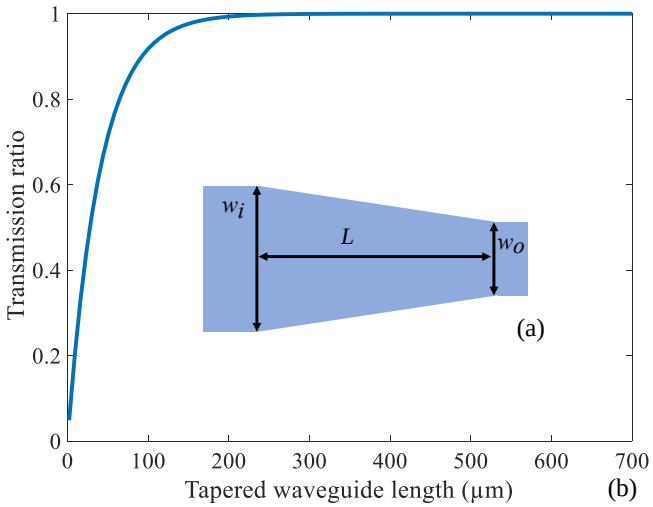


Figure 1.6: Transmission ratio versus length for a silicon nitride linear waveguide taper with $w_i = 5$ μm and $w_o = 850$ nm.

1.2.5 Effective index dependence on geometric variations

The effective index of the waveguide changes with variations in its geometry, including its thickness t and width w. The effective index of the TE mode for both SOI and SiN waveguides is calculated using MODE Solution. The reference waveguide has a nominal width of 500 nm and a thickness of 220 nm for the SOI waveguide, and a width of 850 nm and a thickness of 450 nm for the SiN waveguide. The effective index of the waveguide is in red and its dependence on variations in geometry is in blue, as shown in Fig. 1.7. The detailed effective index and the dependence for SiN waveguides are plotted in Fig. 4.2 in Chapter 4. The sensitivity to width variations is lower in wider waveguides for both the SOI waveguide and SiN waveguide, as shown in Fig. 1.7(a). Wider waveguides reduce the impact of fabrication variations since their effective index is less sensitive to changes in dimensions [47]. The effective index dependence on the width has a peak at 298 nm for the SOI waveguide and at 470 nm for the SiN waveguide. In the simulations, the effective index dependence on the waveguide width dn_{eff}/dw has a peak value of 4.9×10^{-3} nm^{-1} for the SOI waveguide, whereas the one for the SiN is 4.7×10^{-4} nm^{-1}. For the reference waveguides used in this work, the effective index shows a dependence of 3.7×10^{-3} nm^{-1} on the thickness and 1.6×10^{-3} nm^{-1} on the width for the SOI waveguide; and 5.0×10^{-4} nm^{-1} on the thickness, of 2.4×10^{-4} nm^{-1} on the width for the SiN waveguide. Therefore, the effective index dependence is stronger in SOI waveguides than in SiN waveguides, which implies that the effective index of SiN waveguides is less sensitive on fabrication variations than that of SOI waveguides, as discussed in the previous section. Comparing the Fig. 1.7(a) and (b), the effective index shows a stronger dependence on the waveguide thickness than width, indicating that the effective index variation is more sensitive to the thickness for both reference SOI and SiN waveguides.

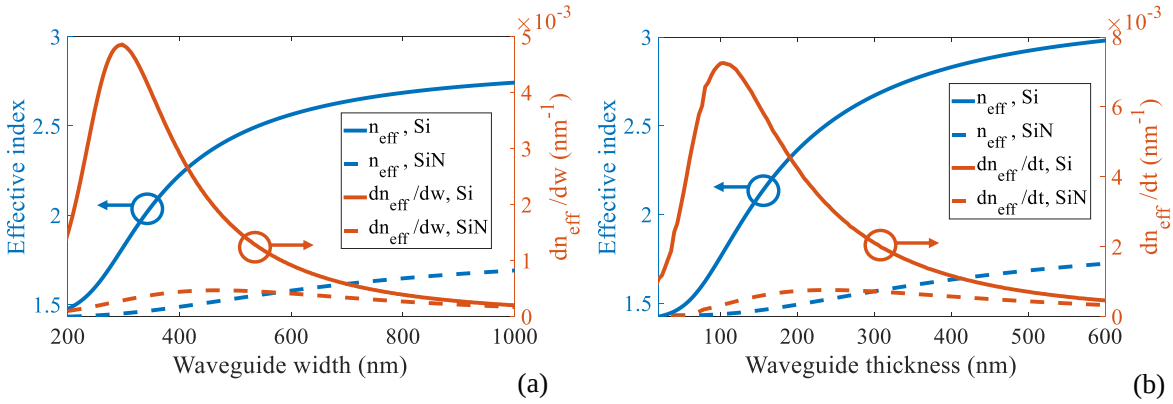


Figure 1.7: Effective index of the waveguide and its dependence at 1550 nm on the waveguide width for the (a) 220 nm thick SOI waveguide and 450 nm thick SiN waveguide; on waveguide thickness for the (b) 500 nm wide SOI waveguide and 850 nm wide SiN waveguide.

1.3 Operating principle of arrayed waveguide gratings

In this section, the operating principle of the star coupler with a Rowland circle configuration is presented. We discuss the strategy for designing an AWG with higher performances. Based on a semi-analytical AWG model, the impact of the design parameters is studied with an example of SiN AWG.

1.3.1 Star couplers

Star couplers are typically used as free propagation regions in an AWG to provide both diffraction and focusing functions. The free propagation region of a star coupler is defined by two spherical surfaces with various radii, as shown in Fig. 1.8. The incident light waves from the spherical grating with a radius of f are focused on points located on another circle with a radius of $f/2$ [53]. The circle in which the focusing points are located is named the Rowland circle, whereas the circle of the spherical grating with a radius of f is called the grating circle [54; 55]. In the design of an AWG, both the incident launching points and focusing points of the light are located along the Rowland circle.

Figure 1.8 shows a schematic of a star coupler. The point C represents the center of curvature of the Rowland circle. The radius of the grating circle and the Rowland circle is f and $f/2$, respectively. The point O is the center of grating circle and also the axis pole.

The incident light starts from point S with a coordinates of $(x_S\ ,\ y_S)$ on the Rowland circle, is reflected by the spherical grating, and then focuses on the point P with coordinates $(x_P,\ \ y_P)$ back to the Rowland circle. The point G with coordinates $(x_G\ ,\ y_G)$ is one point in the grating circle where light reflected, as shown in Fig. 1.8. The light paths of SG and GP are given as:

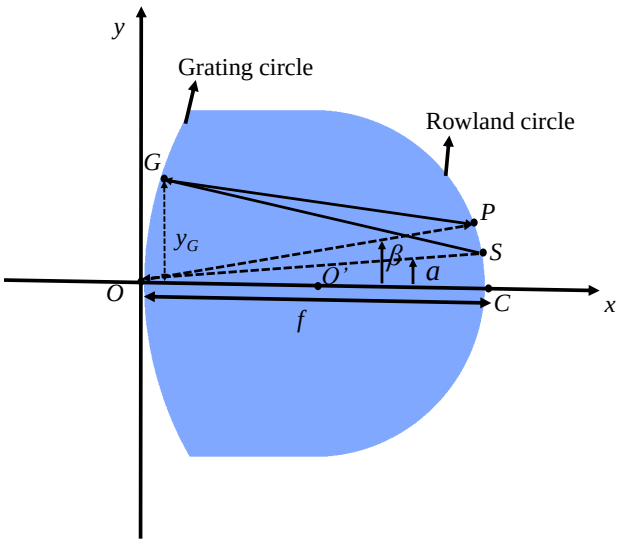


Figure 1.8: Schematic of the star coupler. S, P are the launching point and focusing point on the Rowland circle, respectively. The point O and G are two points of reflection on the grating circle.

$$SG = \sqrt{(x_G - x_S)^2 + (y_G - y_S)^2} \tag{1.1}$$

$$GP = \sqrt{(x_P - x_G)^2 + (y_P - y_G)^2} \tag{1.2}$$

The coordinates for the point G (x_G , y_G) are expressed as:

$$(x_G - f)^2 + {y_G}^2 = f^2 \tag{1.3}$$

We assume that the light from the launching point S reflected back to the focusing point P has another point of reflection O with coordinates (0, 0) in the grating circle. The light paths SO and OP are given by:

$$OS = f \cos\alpha \tag{1.4}$$

$$PO = f \cos\beta \tag{1.5}$$

where α i.e., $\angle$SOC and β i.e., $\angle$COP are the incident and reflected angles for SO and OP. The optical path difference Δl between light paths SGP and SOP is given by:

$$\Delta l = SG + GP - (SO + OP) \tag{1.6}$$

The launching point S and the focusing point P have the coordinates:

$$\begin{cases} x_S = f\cos^2\alpha \\ y_S = f\cos\alpha\sin\alpha \end{cases} \tag{1.7}$$

and

$$\begin{cases} x_P = f\cos^2\beta \\ y_P = f\cos\beta\sin\beta \end{cases} \tag{1.8}$$

Therefore, combing Equation 1.3, 1.6, 1.7 and 1.8, the path difference Δl is calculated by [54]:

$$\Delta l = -(\sin\alpha + \sin\beta)y_G + \frac{1}{8}\frac{1}{f^3}(\frac{\sin^2\alpha}{\cos\beta} + \frac{\sin^2\beta}{\cos\beta})y_G^4 + \text{high order} \tag{1.9}$$

Using the paraxial approximation of $y_P \ll f$, the high order terms including y_G^4 can be neglected. Therefore, Equation 1.9 can be simplified as:

$$\Delta l = -(\sin\alpha + \sin\beta)y_G \tag{1.10}$$

For light at a specific wavelength λ, the interference points along the Rowland circle appear where the light path difference Δl is equal to integer multiple of wavelength. The condition is given by:

$$n_s(\sin\alpha + \sin\beta)d = m\lambda \tag{1.11}$$

where $d = y_G$, in which the arrayed waveguides apertures are located to match the interference on the Rowland circle.

1.3.2 AWG design strategy

The design parameters includes the focal length f, the separations between adjacent output channels D, the separations between adjacent arrayed waveguides d at the interface with the FPR of star couplers, the length difference ΔL between adjacent waveguides and the number of arrayed waveguides N, as shown in Fig. 1.9.

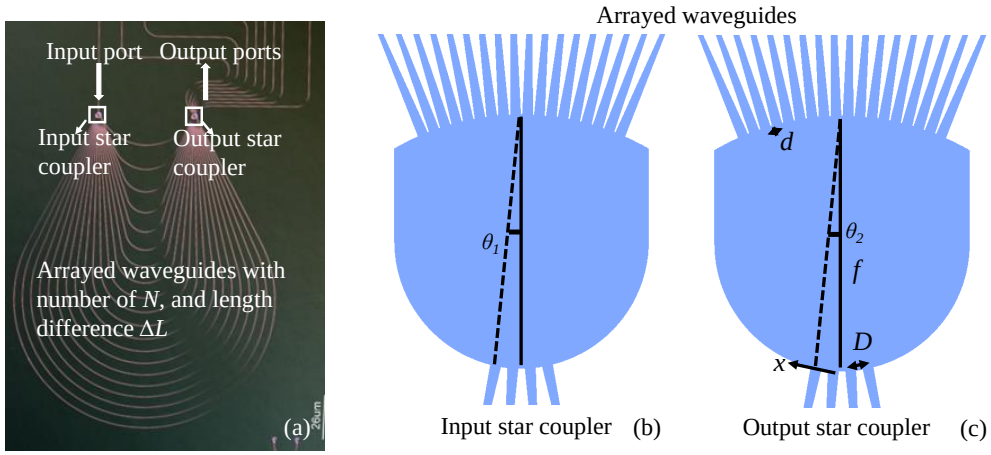


Figure 1.9: Schematic of the (a) 1 × N AWG, (b) input star coupler, and (c) output star coupler.

For the light waves propagating through an AWG, the path difference is equal to an integer multiple of 2π to create an constructive interference at the output star coupler. The grating equation is given by:

$$\beta_{slab} d \sin\theta_1 + \beta_{wg} \Delta L + \beta_{slab} d \sin\theta_2 = 2\pi m \tag{1.12}$$

where β_{slab} and β_{wg} are the propagation constants of the slab waveguides of star couplers and the arrayed waveguides, respectively, θ_1 and θ_2 are the angles of the input port and output port with respect to the center axis of the star couplers and m is the grating order. Eq. 1.12 can also be written by replacing the propagation constants with the effective indexes:

$$n_s d \sin\theta_1 + n_{eff} \Delta L + n_s d \sin\theta_2 = m\lambda \tag{1.13}$$

where n_s and n_{eff} are the effective indexes of the slab waveguide and the arrayed waveguides, respectively. The light launching from the center of the input plane at center wavelength λc typically has an interference point at the center of the output plane. Therefore, θ_1 and θ_2 equal zero for the center wavelength. Eq. 1.13 can be further simplified to:

$$n_c \Delta L = m\lambda_c \tag{1.14}$$

where n_c is the effective index of the waveguide at the center wavelength.

The FSR is defined as the wavelength difference between two adjacent orders in the same output port. The adjacent wavelengths of constructive-interference have a phase difference of

2π, therefore, they arrive at the same focused position at the output star coupler. Given that the difference between two adjacent orders is 1, based on Eq. 1.13, we find:

$$\frac{n_{eff,m+1}}{\lambda_{m+1}}\Delta L - \frac{n_{eff,m}}{\lambda_m}\Delta L = 1 \tag{1.15}$$

Defining the FSR $\Delta\lambda_{FSR} = \lambda_m - \lambda_{m+1}$, from Eq. 1.15 we can derive:

$$n_{eff,m+1} = n_{eff,m} - \frac{dn}{d\lambda}\Delta\lambda_{FSR} \tag{1.16}$$

We know the group index n_g is given by:

$$n_g = n_{eff}(\lambda) - \lambda\frac{dn_{eff}}{d\lambda} \tag{1.17}$$

Combing Eq. 1.15, 1.16 and 1.17, the FSR is given by:

$$\Delta\lambda_{FSR} = \frac{\lambda^2}{n_g\Delta L} \tag{1.18}$$

From Eq. 1.18, we can see that the FSR is inversely proportional to the length difference ΔL and the group index of the waveguide.

After propagating through the arrayed waveguides, the light is focused on the focal line of the output star coupler, where the output channel ports are located, as shown in Fig. 1.9(c). The light with wavelength λ will interfere at point x on Rowland circle if it satisfies [56]:

$$n_{eff}\Delta L + n_s d\frac{x}{f} = m\lambda \tag{1.19}$$

Therefore, the dispersion dx/df on the focal plane is derived by [57]:

$$\frac{dx}{d\lambda} = \frac{fm}{dn_s} \tag{1.20}$$

The separation of the output ports can be obtained with $dx/d\lambda = D/\Delta\lambda$ [58]:

$$D = \frac{fm\Delta\lambda}{dn_s} \tag{1.21}$$

where the channel spacing $\Delta\lambda$ is defined as the peak wavelength difference between the adjacent output ports. Therefore, the channel spacing is determined by the design parameters

including the focal length f, the separations between adjacent output channels D, the separations between adjacent arrayed waveguides d and the grating order m. In an AWG design, the channel spacing is usually controlled by the separation D. However, a small distance between adjacent output ports for AWGs with tight channel spacing may cause extra crosstalk through the exponential tails of their field distributions [59]. Expanding the focal length or decreasing the separation d will induce an extra insertion loss. Due to this trade-off, AWGs with a dense channel spacing usually have lower performance than that with wide channel spacings.

The focal length f of the star couplers controls the field distribution of the light propagating in the input star coupler and the focusing locations in the output star coupler. The expansion of light in the input star coupler is a function of the propagating distance. Therefore, these parameters such as focal length, separation of the arrayed waveguides and the number of arrayed waveguides need to be carefully engineered. A detailed discussion on SiN AWGs will be presented later in this chapter.

1.3.3 Modeling the AWG

We use a semi-analytical approach following the general procedure outlined in [60]. The three-dimensional device is modeled by a series of transfer functions representing the transmission of light, assuming the complex field profile is matched across each interface. Only the fundamental mode is considered. The field distributions are collapsed to a single dimension by considering the effective waveguide index.

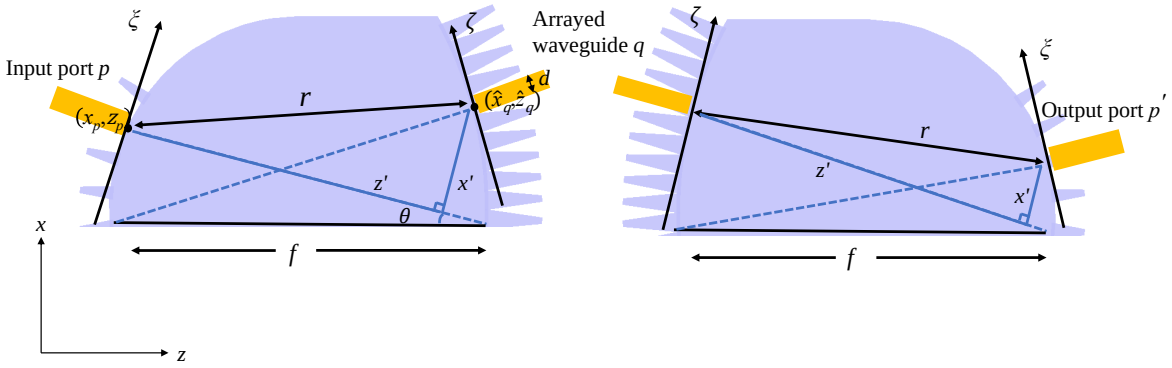


Figure 1.10: Input and output star coupler geometry and coordinate system.

First, we calculate the coupling from an input waveguide p to each arrayed waveguide q by considering the overlap of the diffracted field $\hat{U}_p$, originating from the input, with the modal field U_a at the waveguide array aperture. This is achieved by numerically solving the following overlap integral over the aperture coordinate ζ:

$$C_{p,q} = \int_{\zeta_q - d/2}^{\zeta_q + d/2} \hat{U}_p(\zeta) U_a(\zeta) d\zeta, \tag{1.22}$$

where the integral is taken over the range $\pm d/2$ representing the aperture center-to-center spacing. Only the fundamental mode is kept in our analysis. This process is repeated for every waveguide q, where $\hat{U}_p$ is recalculated at each aperture. The diffracted field from the input is described by the following Huygens-Fresnel integral:

$$\hat{U}_p(\hat{x}, \hat{z}) = \sqrt{\frac{k_0 n_s}{2j\pi}} \int_{-\infty}^{\infty} U_p(\xi) \frac{z'}{r} \frac{\exp(jk_0 n_s r)}{\sqrt{r}} d\xi, \tag{1.23}$$

where $r = \sqrt{[\hat{x}(\zeta) - x(\xi)]^2 + [\hat{z}(\zeta) - z(\xi)]^2}$ is the propagation distance, $k_0 = 2\pi/\lambda$ is the free space wave number and n_s is the slab effective index, shown in Fig. 1.10. The wave vector originating from the input waveguide is oriented along the angle θ, such that the aperture coordinates $(\hat{x}_q, \hat{z}_q)$ are transformed into the local input coordinate system at (x_p, z_p) through:

$$\begin{bmatrix} x' \\ z' \end{bmatrix} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{pmatrix} \begin{bmatrix} \hat{x}_q - x_p \\ \hat{z}_q - z_p \end{bmatrix} \tag{1.24}$$

The light in the waveguide array accumulates phase proportional to the waveguide length $L_q = L_0 + q\Delta L$. The mode field exiting each waveguide q is the product of the excited mode with the phase term:

$$U_q = C_{p,q} U_a \exp\left(-jk_0 n_{eff} L_q\right) \tag{1.25}$$

where n_{eff} is the mode effective index. Here, cross-coupling to adjacent waveguides is assumed to be negligible and waveguide losses are ignored but may be included in $C_{p,q}$ for completeness.

By reciprocity, coupling to an output waveguide p' can be calculated in the reverse direction by propagating the output waveguide mode field and computing $C_{q,p'}$ from the overlap of the diffracted field with the mode exiting the array. The procedure is the same as the input star coupler.

The total transmission from an input waveguide p to output waveguide p' is then the sum of the transmissions for every array waveguide q:

$$t_{p,p'} = \sum_{q=1}^{N} C_{p,q} C_{q,p'} \exp\left(-jk_0 n_{eff} L_q\right). \tag{1.26}$$

1.3.4 Characterization

The insertion loss, crosstalk, and non-uniformity are typically used to measure the performances of AWGs, as shown in Fig. 1.11. In this subsection, the sources contributing to insertion loss, crosstalk, and non-uniformity are studied.

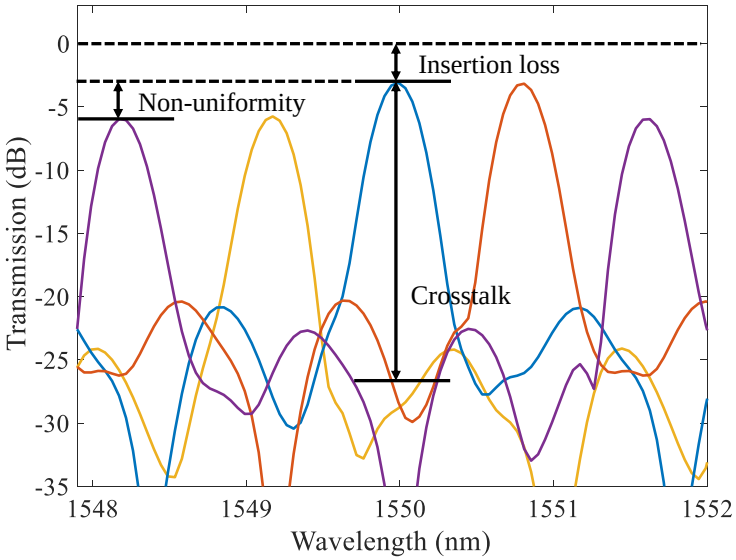


Figure 1.11: Transmission spectrum of 1 × 4 AWG with performance metrics labeled.

Insertion loss

The main contributions to the insertion loss of an AWG are:

- Propagation losses in the arrayed waveguides.
- The bending losses in bent waveguides.
- Transition losses between star couplers and arrayed waveguides.
- Insufficient spatial resolutions or arrayed waveguides numbers.
- Mode field fluctuations at the focusing points due to phase errors in the output star coupler.

Insertion loss of an AWG in this thesis is defined as the insertion loss of the channel with the lowest loss, which is normally given in decibels. The sidewall roughness of the waveguide results in propagation losses. Using thinner waveguides and improved fabrication process can reduce the insertion loss of an AWG. The bending loss in bent waveguides can be reduced significantly by enlarging the bending radius [61]. Another contribution is the transition losses

between the star couplers and arrayed waveguides. When the light propagates from the star coupler to the arrayed waveguides, a part of it is not coupled but reflected from the gaps between the adjacent waveguide apertures, creating transition losses. An adiabatic tapers between the star couplers and the arrayed waveguides can be used to reduce this loss [62]. However, the adiabatic tapers need to be sufficiently long to minimize the loss when light propagates between the arrayed waveguides and the free space region of the star couplers. Because of the limitation of the fabrication process, a minimum gap of a few hundred nm still exists between adjacent waveguides. A parameter called the fill factor is defined as the ratio of the aperture width of the tapers to the separation d to discuss this loss in the later section. Increasing the fill factor and shrinking the gaps between adjacent apertures can minimize the transition loss, as discussed below.

Another source of loss comes from an insufficient spatial resolution or number of arms. The light entering the free propagation region is no longer confined and becomes divergent, as shown in Fig. 1.12(a). The far field distribution of the light has a near Gaussian shape along the edge between the free propagation region and waveguide apertures. The input waveguide aperture width controls the diffraction angle, and the star coupler focal length f affect the extending width of field distribution. An input star coupler with an insufficiently long focal length will couple light to only a couple of arms, as shown in Fig. 1.12(b). This results in a low resolution that suppresses the power in the main lobe and increase the power in the side lobe in the output channels. If the focal length of input star coupler is too long, the field distribution will be wider than all the apertures of the arrayed waveguides. Thus, a part of light cannot be coupled into the arrayed waveguides, as shown in Fig. 1.12(c). An extra loss is created due to an insufficient number of array waveguides.

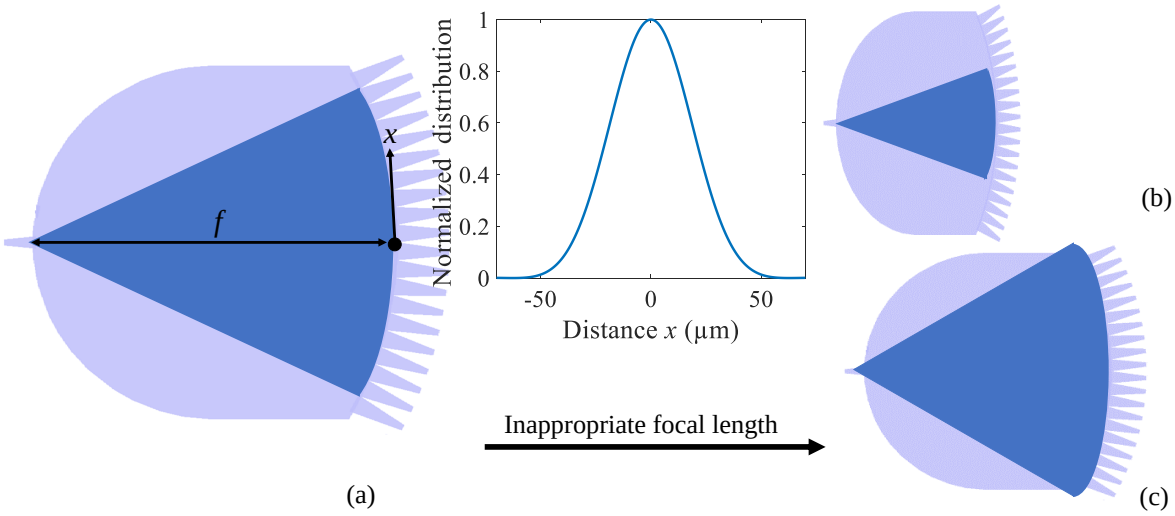


Figure 1.12: Schematic of input star coupler of (a) appropriate design, with (b) an insufficient resolution and (c) insufficient numbers of arrayed waveguides with the field distribution of light.

Another contribution to insertion loss occurs when light couples into different grating orders. A part of light with grating order m will couple into the adjacent $m \pm 1$ orders [59]. The phase errors due to fabrication variations also cause losses. The phase errors accumulated in the arrayed waveguides shift the focal points at the output star coupler. A part of the power may be coupled into other channels, which raise the noise floor and introduce the extra loss in the AWG.

Crosstalk

The crosstalk in an AWG is the result of various contributions:

- Insufficient spatial resolutions.
- Coupling of evanescent fields between the channel ports.
- Phase errors due to fabrication variations.
- Coupling between adjacent arrayed waveguides.

Crosstalk is defined as the power difference between the desired and undesired light at the center wavelength for a given channel. Insufficient resolution due to a low number of arms increase the power in the side lobes, which can be decreased by increased number of arrayed waveguides [18]. Another source of crosstalk is the coupling between the output ports through the exponential tails of the field distributions [59]. Increasing the separations between adjacent output ports can suppress this contribution.

Phase errors accumulated in waveguides due to fabrication variations is another source of crosstalk in AWGs. Fabrication variations result in changes in the waveguide geometrical parameters and in variations of the refractive index of the cladding and core material [45]. Fluctuations in the core layer thickness due to non-uniformity in the deposition process and in the waveguide width due to etching imperfections result in variations of the effective index of the waveguide. The effective index variations shift the focal points in the output star coupler. More in-depth studies will be presented in Chapter 5. As a result, part of the power may be coupled into other channels, which increases the noise floor of the AWG spectral response and introduces crosstalk. Wider arrayed waveguides can alleviate the impact of fabrication variations since wider waveguides have a lower effective index dependence on geometry variations, as discussed earlier in this chapter. However, very wide waveguides support higher order modes, which introduce phase errors in bent waveguides. In [63], the bent waveguides only support the fundamental mode, and wide straight waveguides supporting higher order modes are connected to the bent waveguides with the adiabatic tapers to minimize the impact of fabrication variations.

Crosstalk may also be introduced by phase distortions due to parasitic coupling between the arrayed waveguides. Superlattice AWGs, where waveguides with different widths are used to form the waveguides array to suppress parasitic coupling, and thus decreased crosstalk, were demonstrated on a SiN platform in Chapter 3 [28].

Non-uniformity

Non-uniformity of an AWG is defined as the difference in power between the channel with the highest output and the one with the lowest output. Typically, tapered waveguides are used to connect the arrayed waveguides and star couplers to minimize coupling losses. Furthermore, the far field distribution appearing at the input of the arrayed waveguides has a near Gaussian shape. Therefore, a similar field distribution appears at the image plane in the output star coupler. This near Gaussian shaped field distribution results in the outer channel having a higher insertion loss than the center channel [64; 65].

Due to the near Gaussian distribution of the received power in one FSR along the output channels, a method to decrease the non-uniformity is to broaden the FSR. The output channels will capture the center of the Gaussian distribution, and thus, the power difference between channels is decreased. Well-designed waveguides are used to connect arrayed waveguides and star couplers in the work of Yohei *et.al* [65]. The waveguides provide a field distribution with a sinc-function shape at the start of the output star coupler, and the power with flat top distribution among the output channels is obtained. Low non-uniformity is obtained due to this flat top distribution.

Polarization dependence

The polarization dependence of AWGs results from the polarization dependence of the arrayed waveguides and of the slab waveguide forming the star couplers. Birefringent waveguides with different effective indexes for the fundamental TE and TM modes cause a shift in the center wavelength and channel spacing in the spectral responses of the AWGs. To make AWGs polarization independent, several solutions have been developed. One earlier method applied in AWG devices is to use a half-wave plate in the middle of the waveguide array [57]. Light with one polarization will be converted into the other polarization through the half-wave plate in the centers of waveguide array. Therefore, the light that was on the TE mode in the first half of the array propagates as TM polarized light in the second and vice-versa. With this method, the light on the TE and TM modes has the same phase changes negating the polarization dependence of the waveguides array and star couplers [57; 66]. Another earlier solution is to design a square waveguide surrounded by the same cladding. The waveguide becomes polarization independent since the effective indexes are the same along the vertical and horizontal directions [67].

The half-wave plate in the middle of the waveguide array also introduce losses. The method

of using square waveguides cannot achieve perfect polarization independence since it ignores the polarization dispersion in the slab waveguides forming the star couplers. An alternative method to obtain polarization independence is using a polarization splitter and rotator before the AWGs to split light into two polarizations, and then convert one polarization into the other polarization. Another solution is combining two different AWGs, one for each polarization, into one AWG in the form of a bidirectional AWG router. A polarization splitter is placed before the AWG router. The different polarizations are split and go through different AWGs. Another method based on optimizing the geometry of the arrayed waveguides and splitting the two polarization states and adjusting their angle of incidence at the input star coupler to compensate for the polarization mode dispersion of the AWG is discussed in Chapter 2.

1.3.5 Design parameters

The impacts on the performances of AWGs of various parameters, including the focal length f, separation of the arrayed waveguides d, fill factor and propagation loss are studied with an example of a 1 $\times$ 4 SiN AWG based on the semi-analytical model presented above. Other parameters, such as the length of the middle waveguide and phase errors, are discussed in Chapter 4. As shown in Fig. 1.13, the separation d is the center-to-center spacing between the adjacent arrayed waveguides connecting with the FPRs; the gap is the lateral spacing of the edges of adjacent waveguide apertures; the fill factor is the ratio of aperture width w_a to the separation d (i.e. w_a/d), as shown in Fig. 1.13(b). The basic 1 $\times$ 4 SiN AWG used in this study has a focal length f of 150 μm, a separation d of 5.4 μm, a number of arrayed waveguides N equal to 40.

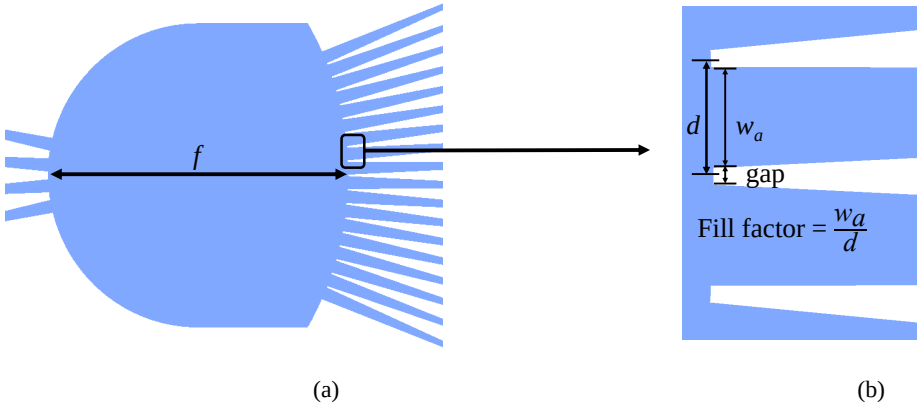


Figure 1.13: (a) Schematic of the star coupler with focal length labeled and (b) separation d and fill factor.

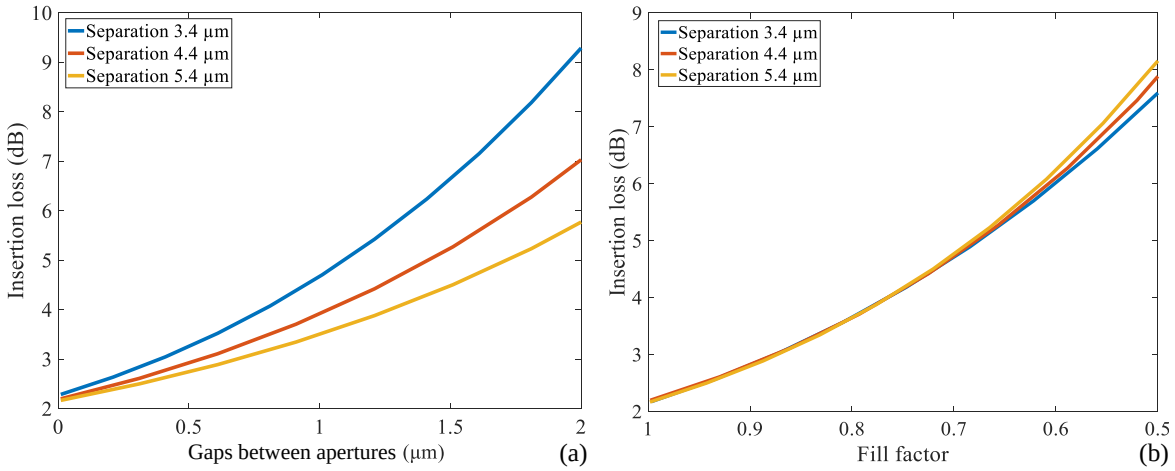


Figure 1.14: Insertion loss for separation d of 3.4 μm, 4.4 μm and 5.4 μm with various (a) gaps between adjacent apertures of arrayed waveguides and (b) fill factor.

Fill factor

Figure 1.14 shows the impact of the gaps between adjacent apertures in arrayed waveguides and of the fill factor on the insertion loss of the basic AWG. The power reflected from the gaps between adjacent taper waveguides introduces losses since it cannot be coupled into the arrayed waveguides. In these simulations, we ignored propagation losses in the arrayed waveguides. Thus, the intrinsic insertion losses are attributed to the coupling loss to other grating orders. As shown in Fig. 1.14(a), the insertion loss is low for narrow gaps in AWGs with different separations of 3.4 μm, 4.4 μm and 5.4 μm. The insertion loss increases with broadening gaps. The insertion loss for these separations when the gaps are 2 μm gaps are 9.3 dB, 7.5 dB and 6.1 dB, respectively. We see that the insertion loss changes for various separations when gaps are constant.

Star couplers with higher fill factor have a high coupling efficiency. The insertion losses increase as the fill factor decreases in all separations, as shown in Fig. 1.14(b). Meanwhile, the increasing trends almost overlap and the insertion loss are 7.6 dB, 7.9 dB and 8.1 dB, for separations of 3.4 μm, 4.4 μm and 5.4 μm, respectively, when the fill factor is 0.5. Therefore, the coupling loss is controlled by the fill factor. The minimum gaps between adjacent apertures is 0.4 μm, which is determined by the fabrication process. Assuming the separation is 5.4 μm, the maximum fill factor is 0.93 in the study.

Focal length and separation

We swept the focal length of the star couplers while keeping the other parameters constant to investigate its impact on insertion loss. The insertion loss of AWGs with different focal length is shown in Fig. 1.15(a). The effect of propagation loss is ignored. The insertion loss is high when the focal length is below 50 μm. A reduction of the insertion loss occurs when

the focal length increases. When the focal length of the star coupler is short, only a part of the apertures of the arrayed waveguides are illuminated and thus the spatial resolution is insufficient, even though the array has 40 waveguides. The insertion losses reach a low level of 2.5 dB when the focal length is between 72 μm and 144 μm. The insertion losses increase again when the star coupler length gets larger than 150 μm. In this case, the increase in losses is attributed to an insufficient number of arrayed waveguides since the light beam has expanded beyond the width of the apertures of the arrayed waveguides, as discussed earlier. We can also add a couple of dummy waveguides at the outer edge of the arrayed waveguides to suppress the reflection of these tails of the field distribution.

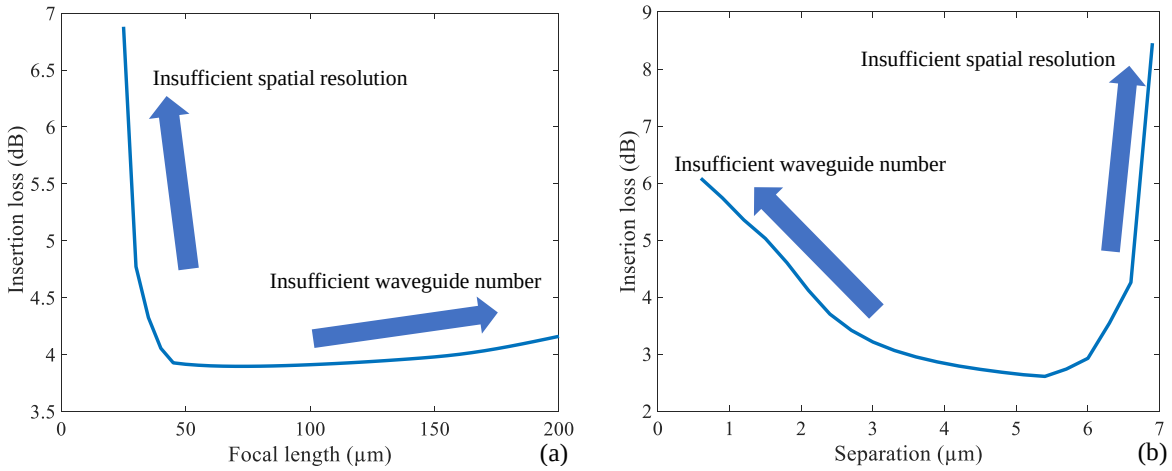


Figure 1.15: Insertion loss for AWGs with different (a) focal length of star couplers and (b) separations.

The impact of the separation between adjacent apertures in the arrayed waveguides on the insertion loss is also studied. Again, for this analysis the propagation loss in the arrayed waveguides was ignored. With a constant 0.4 μm gaps, the insertion loss is high when the separation is narrow, as shown in Fig. 1.15(b). The reason is that a narrower separation and an insufficient number of arrayed waveguides introduce high coupling loss between the star couplers and the arrayed waveguides. Broader separations increase coupling and decrease the insertion loss. This work until the separation reach a value of 5.4 μm, where a minimum insertion loss value of 2.5 dB is obtained. After that the losses increase again. This phenomenon appears due to an insufficient spatial resolution. Although there are 40 arrayed waveguides, only a part of the aperture is used to capture light when the separation becomes large.

Propagation loss

Fig. 1.16 shows the impact of propagation loss on the insertion loss of AWGs with 40 arrayed waveguides and channel spacing of 400 GHz, 200 GHz and 100 GHz for which the length difference ΔL is 87 μm, 173 μm, 346 μm, respectively. The propagation loss calculated

ranges from 0.5 dB/cm to 6 dB/cm to cover the recent fabrication processes reported in the literatures [68]. The insertion loss is about 2.5 dB for all AWGs when the propagation loss is low as 0.5 dB/cm. The insertion loss almost linearly increases with increasing propagation loss for these AWGs. Because it has longer arrayed waveguides for the same number of waveguide in the array, the insertion loss in the AWG with a channel spacing of 100 GHz is higher than that for AWGs with 200 GHz and 400 GHz channel spacing. The insertion loss is 16.6 dB, 8.4 dB and 5.4 dB for the AWGs with channel spacing of 100 GHz, 200 GHz and 400 GHz, respectively. The impact of propagation losses on the insertion loss of AWGs with different number of arrayed waveguides and channel spacings is studied in Chapter 4.

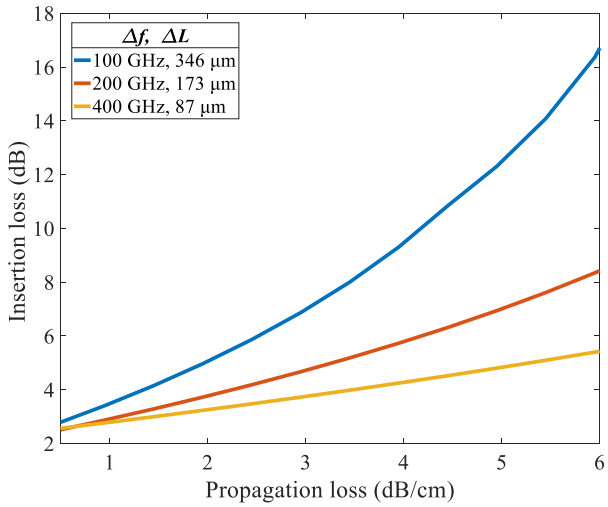


Figure 1.16: Insertion loss as a function of propagation loss in the arrayed waveguides for 1×4 AWGs with channel spacing of 400 GHz, 200 GHz and 100 GHz.

Chapter 2

Polarization-insensitive silicon nitride arrayed waveguide grating

Publié dans Optics Letters, Vol. 44, No. 16 (2019): 3976-3979. [69]

© 2019 Optica Publishing Group. Users may use, reuse, and build upon the article, or use the article for text or data mining, so long as such uses are for non-commercial purposes and appropriate attribution is maintained. All other rights are reserved.

2.1 Résumé

La prochaine génération de réseaux optiques passifs nécessite des solutions intégrées de multiplexage en longueur d'onde, insensible à la polarisation, pour lesquels l'émergence des plateformes nanophotoniques de nitrure de silicium à faible perte possède un grand potentiel. Dans cette lettre, il est présenté un nouveau réseau sélectif planaire (Arrayed Waveguide Grating, AWG) insensible à la polarisation. L'insensibilité de la polarisation est obtenue lorsque l'espace entre les canaux et la longueur d'onde centrale des deux états de polarisation orthogonaux, les modes TE et TM, sont alignés simultanément. Dans notre modèle, l'alignement de l'espace intercanal des deux états de polarisation est obtenu par l'optimisation de la géométrie du réseau de guide d'onde. Alors que pour la longueur d'onde centrale, l'insensibilité de la polarisation est obtenue en séparant les deux états de polarisation et en ajustant leur angle d'incidence à l'entrée du coupleur en étoile pour compenser la dispersion. Une diaphonie inférieure à -16 dB a été démontré pour un multiplexeur en longueur d'onde 1×8 à 100 GHz.

2.2 Abstract

Next-generation passive optical networks require integrated, polarization insensitive wavelength division multiplexing solutions, for which the recently emerging low-loss silicon nitride

nanophotonic platforms hold great potential. A novel polarization insensitive arrayed waveguide grating (AWG) built with silicon nitride waveguides is presented in this chapter. Polarization insensitivity is obtained when both the channel spacing and the center wavelength of the two orthogonal polarization states (i.e., the TE and TM waveguide modes) are simultaneously aligned. In our design, the channel spacing alignment between the polarization states is obtained by optimizing the geometry of the arrayed waveguides, whereas the central wavelength polarization insensitivity is obtained by splitting the two polarization states and adjusting their angle of incidence at the input star coupler to compensate for the polarization mode dispersion of the AWG. A 100 GHz 1×8 wavelength-division multiplexer with crosstalk levels below -16 dB is demonstrated experimentally.

2.3 Introduction

Polarization insensitive multiplexers and demultiplexers are crucial for wavelength division multiplexed (WDM) optical links using standard single mode fibers. Arrayed waveguide gratings (AWGs) are among the most important devices in dense WDM (DWDM) systems. One approach to reduce the polarization sensitivity of AWGs reported previously [67] is based on using square waveguides in the array. However, this approach is limited since the signal still suffers from polarization dispersion in the free-propagation regions (FPR) (i.e., the input and output star couplers). As a result, the channel spacing depends on the polarization state. Thus, complete polarization insensitivity can be difficult to obtain by simply using squared arrayed waveguides. Another solution that has been demonstrated is to insert a half-wave plate in the arrayed waveguides to eliminate the optical path difference between the TE and TM modes [57; 66]. However, the addition of the half wave plate might cause extra loss. Converting one of the input polarizations using a polarization splitter and rotator (PSR) before the AWG is an efficient method to achieve a polarization insensitive demultiplexer [70], but it requires a more complex fabrication process to create an asymmetric waveguide and it increases insertion losses.

Many AWGs have been reported on silicon-on-insulator (SOI) platforms [71; 72; 73; 74; 75]. These AWGs are appealing due to their small footprints. However, the high refractive index difference between the waveguide core and cladding makes them sensitive to phase errors caused by fabrication variations. Thin-film silicon nitride (SiN) AWGs fabricated with a SiO_2 cladding have some advantages over SOI [44; 76; 77; 78; 79]. SiN has a refractive index close to 2.0 at a wavelength of 1550 nm, which is much smaller than that of silicon. Therefore, SiN waveguides and devices are expected to have a better tolerance to fabrication errors. Moreover, SiN can be deposited by low pressure chemical vapour deposition (LPCVD), giving it better optical characteristics at the expense of higher levels of stress, or by plasma enhanced chemical vapour deposition (PECVD), which results in higher propagation losses, but is a process prevalent in complementary metal–oxide–semiconductor (CMOS) foundries [80].

In this letter, we propose and experimentally demonstrate a polarization-insensitive AWG in SiN for DWDM with a 100 GHz channel spacing.

2.4 Operating principle

A microscope image of a fabricated prototype of the proposed AWG is shown in Fig. 2.1(a). Without loss of generality, we discuss its operating principle by considering only the demultiplexing operation. A main difference between this design and a conventional AWG is that the TE and TM modes are launched at different angles in the input star coupler, as shown in Fig. 2.1(c), to compensate for the phase mismatch between the TE and TM polarization states [59]. An on-chip polarization beam splitter is used to separate the two orthogonal polarization states, which can be realized using a directional coupler [81], as shown in Fig. 2.1(b). To obtain polarization-insensitive operation, both the channel spacing and the center wavelength should be polarization independent. This is achieved by (1) choosing an appropriate geometry for the arrayed waveguides (i.e., the waveguide width for a given thickness) and output star coupler, which ensures a polarization-insensitive channel spacing; and (2) calculating the angle required between the TE and TM input ports to make the central wavelength polarization-insensitive. The mathematical principles are detailed below.

We first consider the channel spacing, which is determined by the design of the output star coupler and the arrayed waveguides. A schematic of the output star coupler is shown in Fig. 2.1(d), and the relationship between the channel spacing and the geometry of the coupler is given by [82; 83]:

$$\triangle\lambda = \frac{Ddn_s}{fm} \tag{2.1}$$

where d is the separation between the arrayed waveguides, D is the separation between output waveguides, f is the length of the free propagation region, n_s is the effective index of the slab waveguide used to form the star coupler, m is the grating order, and $\Delta\lambda$ is the channel spacing of the AWG. In our design, we use a thickness for the SiN layer of 436 nm, determined by the wafer used in our fabrication process.

To compensate the channel spacing mismatch, we have to consider the impact of polarization mode dispersion (in both the effective and group indices) in the arrayed waveguides on the free spectral range (FSR) of the AWG. The channel spacing is chosen as $1/N_o$ of the FSR and given by [83]:

$$\Delta\lambda_{FSR} = \frac{\lambda n_{eff}}{mn_g} = N_o \triangle \lambda \tag{2.2}$$

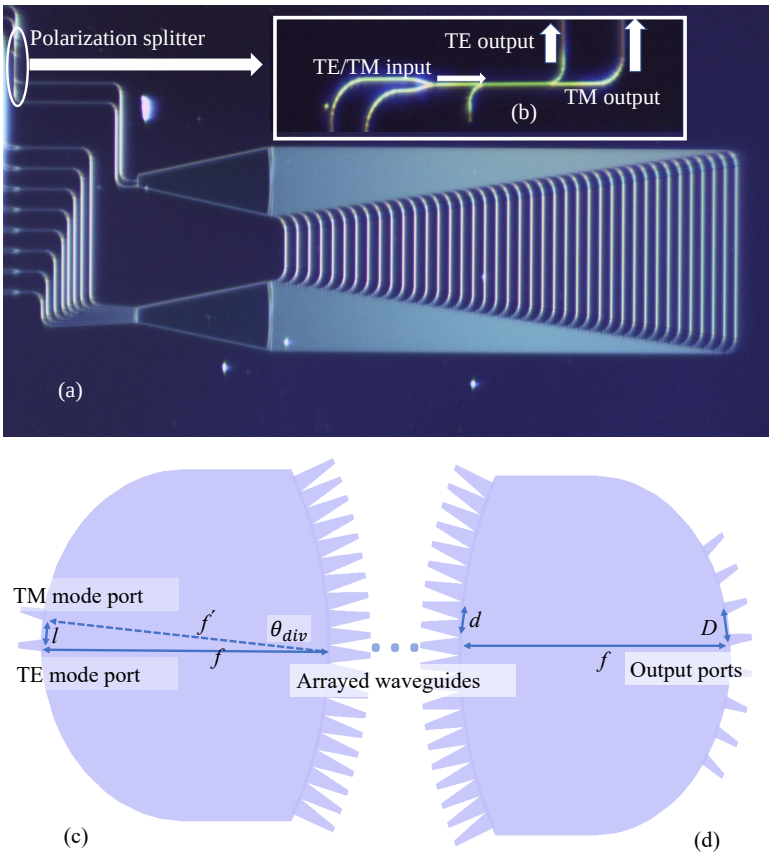


Figure 2.1: (a) Microscope image of the polarization insensitive AWG; (b) microscope image of the polarization beam splitter; (c) schematics of the input star couplers with different injection angles for TE and TM modes; (d) Schematic of the output star coupler (not in scale).

$$\triangle\lambda = \frac{\lambda n_{eff}}{m n_g} \frac{1}{N_o} \tag{2.3}$$

where n_{eff} is the effective index of the arrayed waveguides, N_o is the output channels number of the AWG, and n_g is the group index.

The channel spacing derived from the geometry of the output star coupler (Eq.2.1) needs to match the one of the arrayed waveguides (Eq.2.2 and 2.3). From Eqs.2.1 and 2.3, we can find that the effective index of the slab waveguide must be proportional to the ratio of the effective index to the group index of the arrayed waveguides:

$$\frac{D d n_s}{f m} = \frac{\lambda n_{eff}}{m n_g N_o} \tag{2.4}$$

$$n_s \propto \frac{n_{eff}}{n_g} \tag{2.5}$$

Polarization independent operation requires that $\frac{\triangle\lambda_{TE}}{\triangle\lambda_{TM}} = 1$, and thus, according to Eq. 2.5, the following condition must be met:

$$\frac{n_{eff,TE}}{n_{g,TE}} / \frac{n_{eff,TM}}{n_{g,TM}} = \frac{n_{s,TE}}{n_{s,TM}}. \tag{2.6}$$

To satisfy Eq. 2.6, the width of the arrayed waveguides has to be carefully designed. To simplify the discussion, we define two parameters referring to the left and right sides of Eq. 2.6: $\Gamma 1 = \frac{n_{s,TE}}{n_{s,TM}}$, $\Gamma 2 = \frac{n_{eff,TE}}{n_{group,TE}} / \frac{n_{eff,TM}}{n_{group,TM}}$. The effective index of the slab waveguide is calculated to be 1.75 and 1.67 for the TE and TM modes, respectively, giving $\Gamma 1 = 1.048$. To find a waveguide geometry where $\Gamma 2 = \Gamma 1$, we then calculated the effective and group indices of the SiN waveguide as function of the waveguide width using an Eigenmode solver. The waveguide thickness was kept constant at 436 nm. As shown in Fig. 2.2, $\Gamma 2$ increases with the waveguide width and equals $\Gamma 1$ at 371 nm, which is the value that has been adopted in our fabricated device.

We then consider the polarization response of the center wavelength. In order to achieve a polarization independent response, we need to align the focal points of both the TE and TM modes in the output star coupler. The center wavelength of the center channel is given by [84]:

$$\lambda_{TE/TM} = (n_{eff,TE/TM}\Delta L - n_{s,TE/TM} dl/f)/m_{TE/TM} \tag{2.7}$$

where ΔL is the length difference of adjacent arrayed waveguides, and l is the separation between the input ports, as shown in Fig. 2.1(c). In a conventional AWG design, the input port is set in the center of star coupler for both polarization states (i.e., l=0) and thus $\lambda_{TE/TM} = n_{eff,TE/TM}\Delta L/m_{TE/TM}$. When both the TE and TM modes are launched into the same

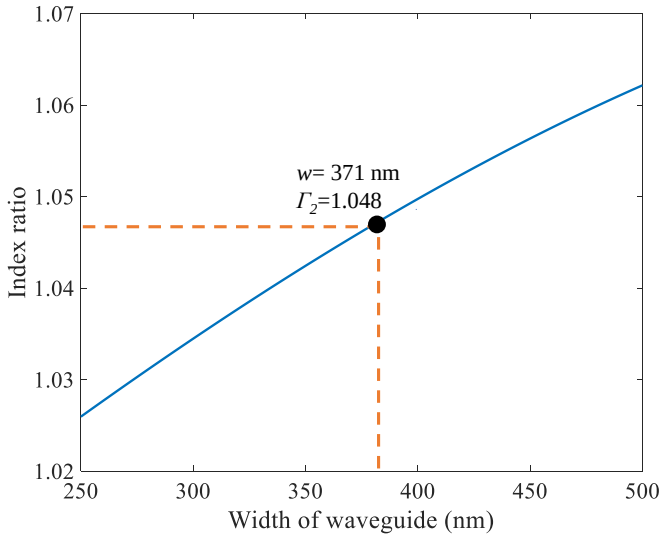


Figure 2.2: Ratio ($\Gamma 2$) as a function of the width of the waveguide.

port, the relationship between the grating orders of the TE and TM modes is given by:

$$\Delta L = \frac{m_{TE}\lambda}{n_{eff,TE}} = \frac{m_{TM}\lambda}{n_{eff,TM}} \tag{2.8}$$

$$m_{TM} = \frac{m_{TE} n_{eff,TM}}{n_{eff,TE}} \tag{2.9}$$

For the waveguide geometry considered here, the ratio $n_{eff,TM}/n_{eff,TE}$=1.006. As a result, it is impossible to obtain an integral grating order for both the TE and TM modes and achieve a polarization insensitive response. For example, for $m_{TE} = 125$, m_{TM} should be 125.75 according to Eq. 2.9 in order to obtain polarization independent operation. Otherwise, the TE and TM modes will be focused at different locations in the output star coupler.

To address this issue, we introduce an extra optical path to obtain an integer value for the TM mode grating order and simultaneously align the focal positions of both polarizations. In this design, we set the position of the TE mode input port at the center of the input star coupler (i.e., normal incidence). The TM mode input port is located at a distance l from the center of the star coupler, as shown in Fig. 2.1(c). We use a polarization splitter to separate the polarization states at the input before the first star coupler as shown in Fig. 2.1(a). This approach was originally proposed in [59] but not demonstrated experimentally. An extra optical path length of $(f' - f)n_{s,TM}$ is introduced for the TM mode, where f is the focal length of the center input port (i.e., the TE mode) and f' is the focal length for the TM mode. As a result, the central wavelengths of operation are the same for both the TE and TM modes. The angle (θ_{div}) required between the inputs is given by [84]:

$$\theta_{div} = \frac{l}{f} = (n_{s,TM} n_{eff,TE} \Delta L - n_{s,TE} n_{eff,TM} \Delta L)/n_{s,TM} n_{s,TE} d \tag{2.10}$$

To demonstrate this approach, we designed an 8-channel polarization-insensitive AWG with a center wavelength of 1550 nm and a channel spacing of 100 GHz. As described above, the arrayed waveguides have a rectangular strip waveguide geometry with a thickness of 436 nm and a width of 371 nm. The angle between the TE and TM inputs is set to 2.79°, and the grating orders for the TE and TM modes are 125 and 131, respectively. The separation between the arrayed waveguides is 6.4 μm, the separation between output waveguides is 10.2 μm and the length of the free propagation region is 1142.5 μm in the output star coupler. Forty arrayed waveguides separated by at least 10 μm are used in the device. In order to minimize bending losses, a large bending radius of 100 μm is used to implement the array. Figure 2.1 shows the entire device.

2.5 Experimental results

The device was fabricated using e-beam lithography and dry etching to pattern a layer of SiN formed on 4 μm of SiO_2; then a top cladding of 4 μm SiO_2 was deposited. Light is coupled to

the chip through surface grating couplers (SGCs). Since the SGCs are efficient for only one polarization, two different designs of SGCs (one for the TE mode and the other for the TM mode) combined by a Y-branch are used for each input/output waveguides of the AWG. Their responses are calibrated using a pair of the same Y-branch-combiners connected by a short waveguide on the same chip. The fabricated polarization-insensitive AWG is characterized using an automated measurement setup equipped with TE and TM sources.

The layout of the polarization splitter, which is an asymmetric directional coupler, is shown in Fig. 2.3 (a microscope image of the fabricated device is shown in Fig. 2.1b). It consists of two strip waveguides with different widths. The TE mode propagates through the input waveguide, while the TM mode is coupled into the cross port. The coupling length is 130 μm, the width of the waveguides is 900 nm, and the gap in the coupler is 0.59 μm. The measured excess loss is 1 dB for the TE mode and approximately 1.5 dB for the TM mode. The best crosstalks are -18 dB at the TE output port and -16 dB at the TM output port.

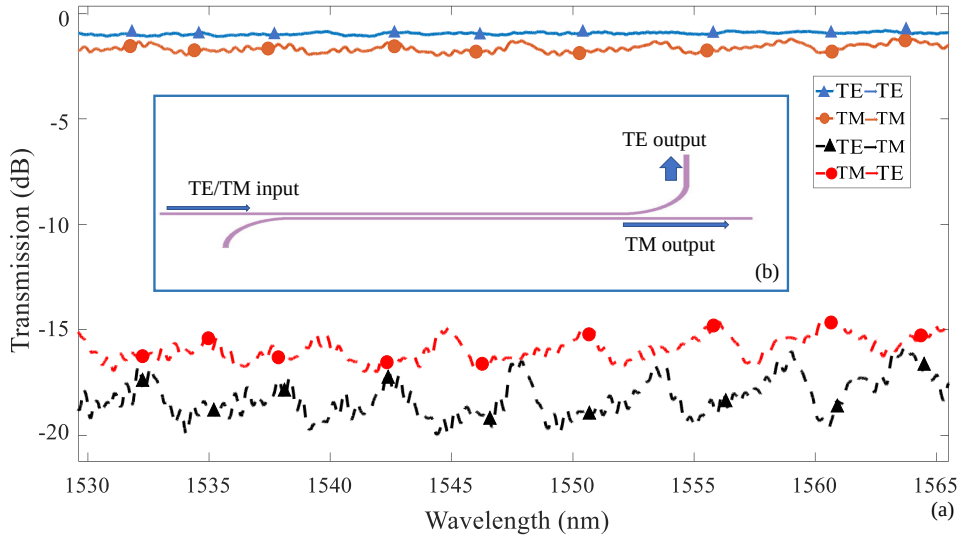


Figure 2.3: (a) Measured power transmission spectra of the polarization beam splitter for the TE and TM modes; (b) layout schematic of the polarization beam splitter.

Fig. 2.4 shows the normalized transmission spectrum of the fabricated polarization-insensitive AWG. The blue curve shows the spectrum of the TE mode, and the red curve shows the spectrum of the TM mode. At the central wavelength, an insertion loss of approximately 6 dB is measured for the TE mode and 6.2 dB for the TM mode. The insertion loss of the AWG includes the waveguide scattering loss, the bending loss in the arrayed waveguides, and the transition coupling loss between the star couplers and the arrayed waveguides. The bending loss of the arrayed waveguides can be decreased by utilizing a larger bend; however, excess insertion loss will be introduced due to a larger overall waveguide length. Furthermore, tapered waveguides are used at the junctions between the arrayed waveguides and the star couplers in order to minimize transition loss. Nevertheless, scattering losses in the fabricated device

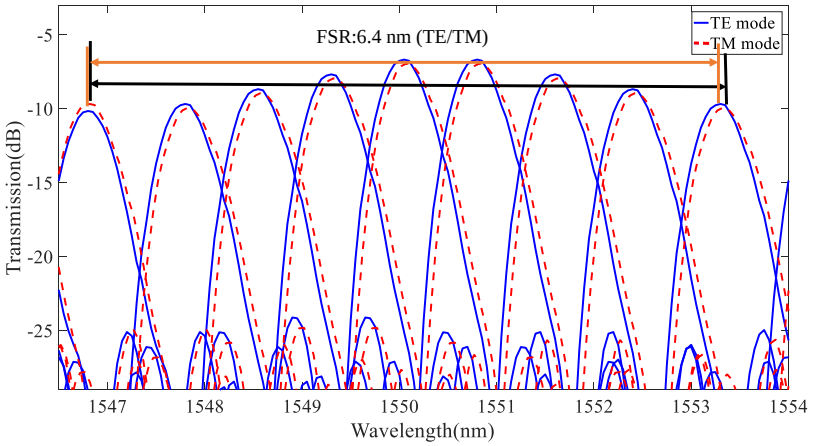


Figure 2.4: Measured power transmission spectra of the AWG for the TE and TM polarizations.

are important due to the roughness of the sidewall. The measured propagation loss is about 1.5 dB/cm for straight waveguides. The total insertion loss could be significantly reduced by improving the reactive ion etching process.

The crosstalk across all 8 channels of the AWG ranges from -16 to -20 dB. It is related to the resolution of the recombined field distribution at the output waveguides. This resolution is a function of the number of optical delay waveguides; a larger number results in a more faithful recreation of the input image. However, it also leads to a larger footprint and increases the probability of phase errors in the fabrication process. Thus, there is an optimum number of waveguides for which the level of crosstalk is minimized, beyond which phase errors overcome the gain in resolution provided by increasing the number of waveguides. As a rule of thumb, the number of arrayed waveguides should be about 3.5 times the number of output channels [18].

As shown in Fig. 2.4, the measured channel spacing is 0.8 nm (100 GHz) with an FSR of 6.4 nm for both the TE and TM modes, as expected. This proves that we achieved polarization insensitivity in the channel spacing through optimizing the geometry of arrayed waveguides. However, the center wavelengths of the TE and TM modes do not match perfectly with each other. The center wavelength is 1550.05 nm for the TE mode and 1550.15 nm for the TM mode. We attribute this deviation to fluctuations in the width of the waveguides due to fabrication errors. Assuming a good uniformity in thickness, we calculated the center wavelength deviation as a function of the variation in width of the waveguides, as shown in Fig. 2.5. The variation of the width of the waveguides affects the center wavelength of the AWG by changing the effective index of the arrayed waveguides. We can see that a width variation of 12 nm causes a drift in the center wavelength by 0.1 nm. This difference in center wavelength between the two polarization states could by reduced by improving the fabrication process or adopting a

flat-top AWG design [85].

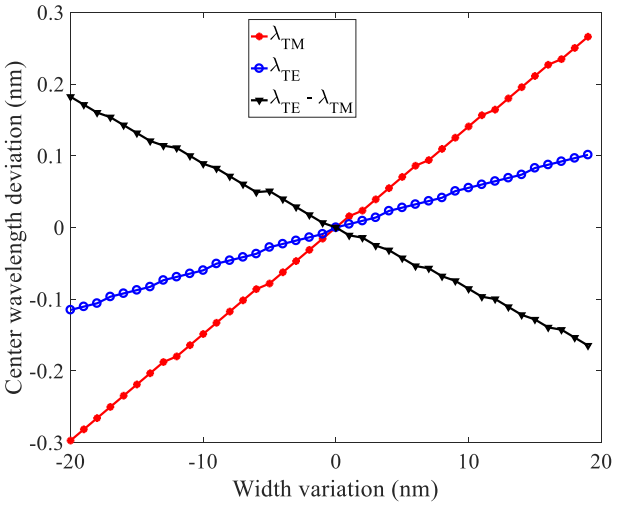


Figure 2.5: Simulated center wavelength deviations due to fabrication errors in waveguide width.

2.6 Conclusion

In summary, we presented a design methodology to obtain a polarization-insensitive AWG that combines two techniques. First, the effective and group indices of the TE and TM modes are engineered by optimizing the geometry of the arrayed waveguides to achieve an identical FSR for both two orthogonal polarization states. Then, to align the central wavelength of each channels for both polarizations, the TE and TM modes are separated into two different waveguides via an on-chip polarization splitter. Each polarization state is launched into the first star coupler of the AWG with a different angle of incidence that allows to compensate for the polarization dispersion of the AWG. Following this methodology, we realized a 1×8 DWDM AWG with SiN waveguides using a CMOS-compatible fabrication process. We achieved a channel spacing of 100 GHz for both polarization states with a crosstalk below -16 dB. The remaining deviation of 0.1 nm in central wavelength between the TE and TM modes could be compensated through further optimization of the fabrication process. Our results show that SiN is promising for the implementation of AWGs targeting DWDM applications, such as WDM passive optical network (WDM-PON), where a low-cost polarization-insensitive WDM solution is required.

Chapter 3

Superlattice Arrayed Waveguide Grating in Silicon Nitride

Publié dans IEEE Photonics Technology Letters, Vol. 32, No. 22 (2020): 1411-1414.

© 2020 IEEE. Reprinted, with permission, from [28].

3.1 Résumé

Les circuits nanophotoniques de nitrure de silicium (SiN) sont une technologie avantageuse pour les systèmes de multiplexage en longueur d'onde, mais possèdent des dimensions supérieures aux composants de silicium sur isolant dû à un plus faible contraste d'indice. Nous présentons un réseau sélectif planaire (Arrayed Waveguide Grating, AWG) 1 × 8 ultra compact pour des canaux séparés par 100 GHz basé sur le nouveau concept de guides d'onde en super-réseau. Une superlattice de guides d'onde est une structure périodique consistant en une alternance de guides d'onde ayant un couplage parasitaire négligeable. Grâce à la densité du réseau de guide d'onde, cet AWG en superlattice (SL-AWG) a des dimensions compactes de 4.3 mm × 0.6 mm, ce qui est plus que deux fois plus petit que les AWG conventionnels pour des performances similaires sur une plateforme de SiN. Le SL-AWG montre aussi de faibles pertes d'insertion de 3.4 dB et une faible diaphonie de −18 dB.

3.2 Abstract

Emerging silicon nitride (SiN) nanophotonic circuits are an advantageous technology for dense wavelength division multiplexing systems but they have larger footprints than silicon-on-insulator devices due to the lower index contrast. We present an ultra-compact 100 GHz 1 × 8 SiN arrayed waveguide grating (AWG) enabled by the novel concept of the waveguide superlattice. A waveguide superlattice is a periodic structure consisting of alternate waveguide

geometries with negligible parasitic couplings. Thanks to the densely arrayed waveguides, this superlattice AWG (SL-AWG) has a compact footprint of 4.3 mm × 0.6 mm, which is more than two times smaller than a conventional AWG with similar performance implemented previously on the same SiN platform. The SL-AWG also shows a low insertion loss of 3.4 dB and a low crosstalk level of −18 dB.

3.3 Introduction

Arrayed waveguide gratings (AWGs) are important components in dense wavelength division multiplexing (DWDM) systems that are used to increase the transmission capacity in optical communications [59]. AWGs can also be used as spectroscopic sensors [75; 86]. Therefore, AWGs were developed using different materials such as silica-on-Si waveguides [87], InP waveguides [25; 88], polymer waveguides [89; 90] and silicon-on-insulator (SOI) waveguides [73; 74]. AWGs built with silica-on-Si waveguides are widely used in commercial systems because of their high performance. Nevertheless, the minimum practical bending radius is large with this type of waveguides, and thus, it results in devices with significant footprints. In silicon-on-insulator devices, the high refractive index contrast allows the use of sharp bends to reduce the footprint of AWGs. However, these SOI AWGs are sensitive to fabrication variations that can result in phase errors in the arrayed waveguides.

Silicon nitride (SiN) platforms are promising for the realization of high performance AWGs [76; 77; 79]. On one hand, SiN waveguides have a lower refractive index contrast, and thus have a better tolerance to fabrication errors compared to SOI. As a result, crosstalk is generally lower in SiN AWGs than in SOI AWG [22]. On the other hand, with conventional SiN AWG designs, the waveguides forming the array must be separated by a distance large enough to suppress parasitic coupling between adjacent waveguides and thus minimize crosstalk. In our previous work [69], a large 10 μm gap was necessary to reduce the crosstalk to an acceptable level, which prevented us from optimizing the AWG footprint.

Here we introduce the concept of the waveguide superlattice based on the asymmetric directional coupler theory [91; 92] into the design of AWGs. A waveguide superlattice comprises waveguides of alternating widths in the waveguide array, which significantly reduces parasitic couplings between adjacent waveguides. Using a superlattice in the arrayed waveguides, we demonstrate a 100-GHz-grid 1 × 8 AWG operating in the transverse electric (TE) mode with a significantly reduced footprint and lower crosstalk.

3.4 Operating principle of superlattice AWG

To understand how a waveguide superlattice suppresses crosstalk, we need to understand how light propagates inside of it. A waveguide superlattice is made of periodic supercells, as shown

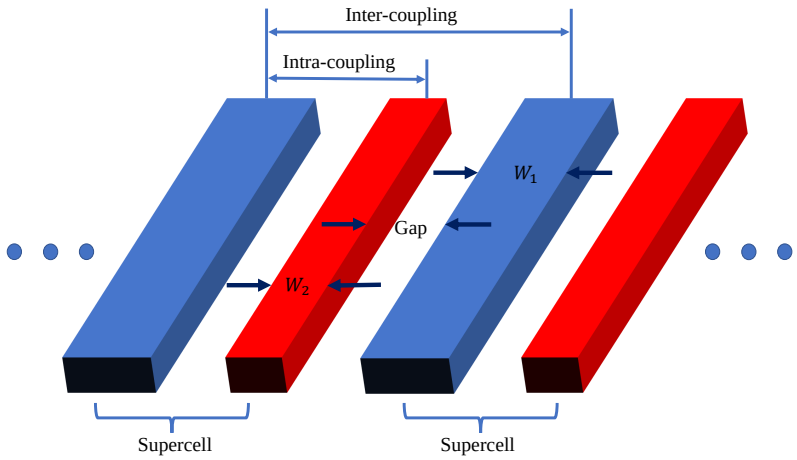


Figure 3.1: Schematic of a waveguide superlattice.

in Fig. 3.1. We use a pair of waveguides with different widths for this supercell. Little to no parasitic coupling occurs between the waveguides as explained by the asymmetric directional coupler theory [92]. In a waveguide superlattice supercell, the phase mismatch caused by the effective index difference due to the difference in width between adjacent waveguides suppresses coupling [91]. Nonetheless, we must also consider coupling between the waveguides with identical widths in adjacent supercells since crosstalk can occur between a waveguide and the next-nearest one in a waveguide array.

In order to calculate the coupling ratio within and between waveguide supercells, we used the MODE Solution EigenMode Expansion (EME) solver from Lumerical. The widths of the waveguides in the superlattice are 800 nm and 900 nm. The gap between adjacent waveguides is 1.5 μm in the waveguide array. The coupling ratio never exceeds -40 dB between adjacent waveguides (i.e., intra-coupling) in the waveguide supercell, as shown in Fig. 3.2(a). Moreover, the coupling ratio between identical waveguides across supercells (i.e., inter-coupling) is similar to that of intra-coupling. By comparison, without alternating waveguide widths, the coupling ratio between two identical 800 nm waveguides with a gap of 1.5 μm exceeds an acceptable level, as demonstrated in Fig. 3.2(b). In order to observe how alternating waveguide widths decrease parasitic coupling, we plot the field distribution, as shown in Fig. 3.2(c) and (d). We observe a reduction in parasitic coupling between adjacent waveguides in the supercell. It is evident that superlattices can effectively suppress coupling and thus improve the crosstalk performance of the AWG.

To demonstrate the performance improvement enabled by the proposed superlattice structure, we fabricated two 1×8 AWGs: a conventional design using arrayed waveguides of identical widths and another using a waveguide superlattice. Fig. 3.3 shows the schematic layout of a 1×8 AWG demultiplexer. An AWG comprises two star couplers and a waveguide array. The

waveguides in the array are designed with a length difference ΔL between adjacent identical waveguides. Thus, the array has a path length difference between each successive waveguide

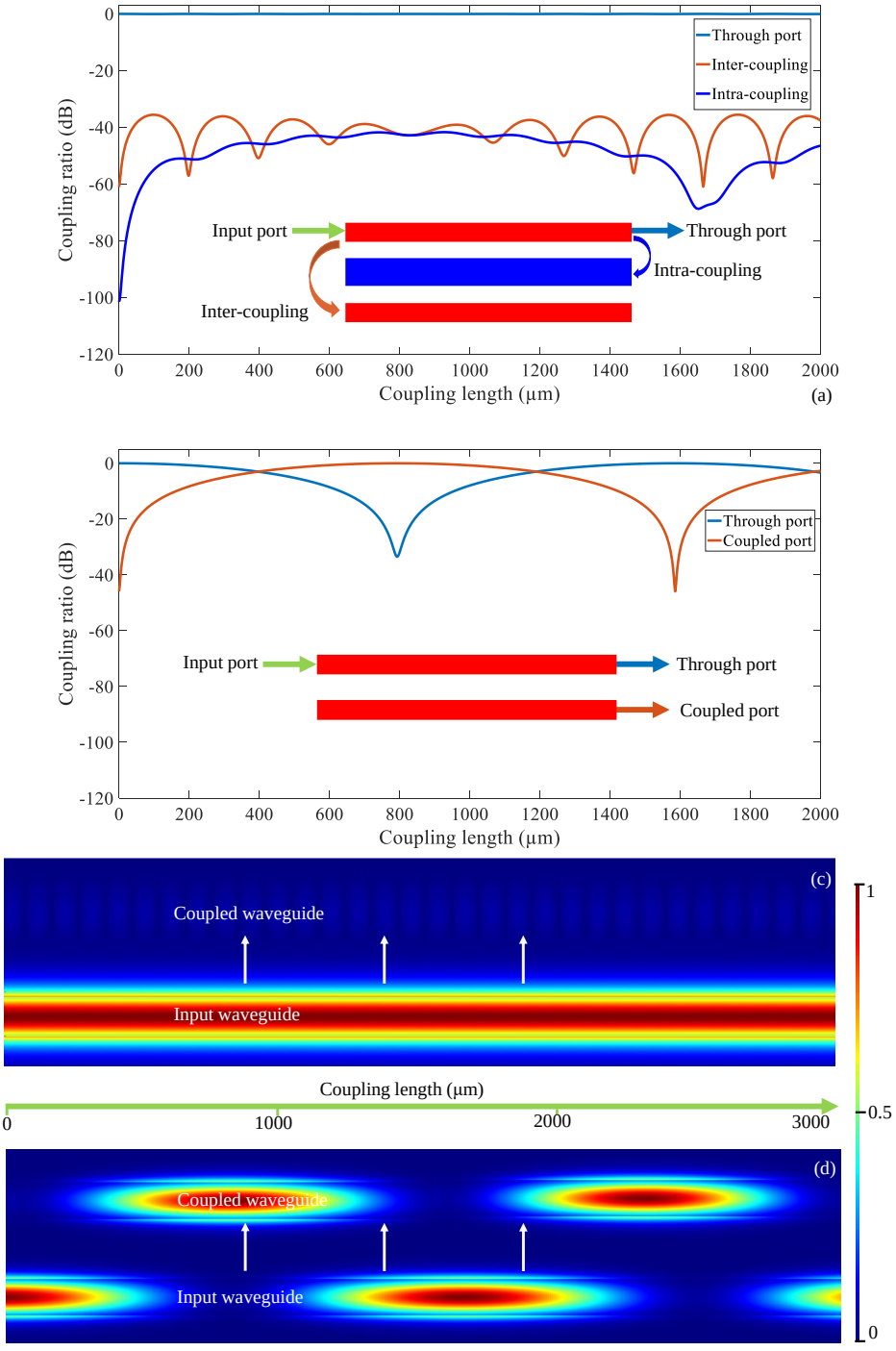


Figure 3.2: Simulated results of (a) the coupling ratio in a supercell, (b) the coupling ratio in the directional coupler, (c) the field distribution in the supercell and (d) the field distribution in the directional coupler.

equal to:

$$m\lambda = n_{eff,w1}\Delta L_{w1} = n_{eff,w2}\Delta L_{w2} \tag{3.1}$$

where m is the grating order, λ is the center wavelength, and $n_{eff,w1}$ and $n_{eff,w2}$ are the effective indexes of waveguides with widths of 800 nm and 900 nm, respectively. The length difference between adjacent 800 nm-wide waveguides is ΔL_{w1}, and ΔL_{w2} is the length difference between adjacent waveguides that are 900 nm in width. The waveguide widths were chosen as a result of a trade-off between tolerance to fabrication errors, parasitic coupling, and single mode operation. Beyond 900 nm in width, the waveguides support higher-order modes.

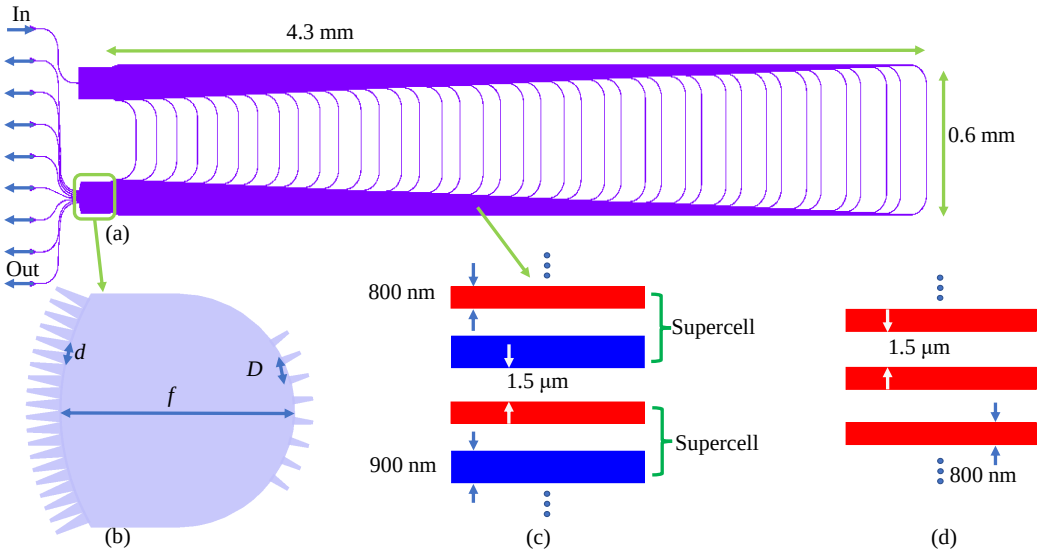


Figure 3.3: Schematic of (a) the AWG with an arrayed waveguide superlattice, of (b) the output star coupler, of (c) an arrayed waveguide superlattice and of (d) conventional arrayed waveguides with identical widths.

We must also consider the grating order m which is determined by the design of the output star coupler and the arrayed waveguides. A schematic of the output star coupler is shown in Fig. 3.3(b) and the grating order is given by [79]:

$$m = \frac{Ddn_s}{f \triangle \lambda} \tag{3.2}$$

where d is the separation between the arrayed waveguides, D is the separation between the output channels, f is the length of the star coupler, n_s is the effective index of the slab waveguide used to form the star coupler, and $\Delta\lambda$ is the channel spacing of the AWG.

Because the effective indexes between the waveguides that are 800 nm and 900 nm wide are different (i.e., $n_{eff,w1} \neq n_{eff,w2}$) but they must have the same grating order, the length

difference between adjacent waveguides must be designed with different values (i.e., $\Delta L_{w1} \neq \Delta L_{w2}$). The optimization of the length difference is a trade-off between footprint and crosstalk. A larger length difference for a higher grating order leads to a larger footprint and higher propagation losses. However, with respect to Eq. (3.2), a smaller length difference for a lower grating order requires a reduction in the separation between the arrayed waveguides or between the output channels. This may cause greater coupling between adjacent apertures and thus higher levels of crosstalk. In our design, the length difference is 472.56 μm between 800 nm-wide waveguides and 461.31 μm between 900 nm-wide waveguides.

For both the conventional AWG and the waveguide superlattice AWG (SL-AWG), an array of 40 waveguides each separated by 1.5 μm is used, as shown in Fig. 3.3. The lengths of parallel arrayed waveguides range from 240 μm to 11 mm. In the superlattice, the width of the waveguides alternates between 800 nm and 900 nm, as shown in Fig. 3.3(c), whereas in the conventional AWG, all the waveguides have a width of 800 nm, as shown in Fig. 3.3(d). Both devices have a center wavelength of 1550 nm and a channel spacing of 100 GHz. Waveguide tapers with a length of 20 μm and that expand to a width of 3 μm are used to connect the arrayed waveguides to the star couplers to reduce coupling losses. A large radius of 100 μm is used for the waveguide bends to minimize bending losses. Both AWGs have a footprint of 4.3 mm $\times$ 0.6 mm.

In order to characterize the anti-coupling effect of the waveguide superlattice, we also designed two Mach-Zehnder interferometers (MZIs): one with a directional coupler made of identical 800 nm-wide waveguides and another using the same configuration as the waveguide supercell, including waveguides with different widths of 800 nm and 900 nm, as shown in Fig. 3.5. Both MZIs have a 1.5 μm gap in the coupler, a coupler length of 464 μm, and a 600 μm imbalance between the two arms.

3.5 Experimental results

The devices were implemented on a SiN platform with a waveguide core thickness of 440 nm. The thickness was set by the process available at the commercial foundry. The fabrication began with low pressure chemical vapour deposition (LPCVD) of 3.2 μm of SiO_2 layer obtained from tetraethyl orthosilicate on a Si wafer. The SiN layer was deposited using LPCVD, and patterned using electron beam lithography and dry etching. During the final step, the wafer was covered with another 3.4 μm of SiO_2 layer deposited by plasma enhanced chemical vapour deposition (PECVD) to form the top cladding. The sidewall angle of the waveguides is approximately 85°, and these non-vertical sidewalls have no to little impact on the performance of the AWGs. Images of the devices are shown in Fig. 3.4. Some cracks in the oxide cladding are present on both sides of the waveguide due to the unoptimized PECVD process. These cracks cause excess losses. This issue will be resolved in future work.

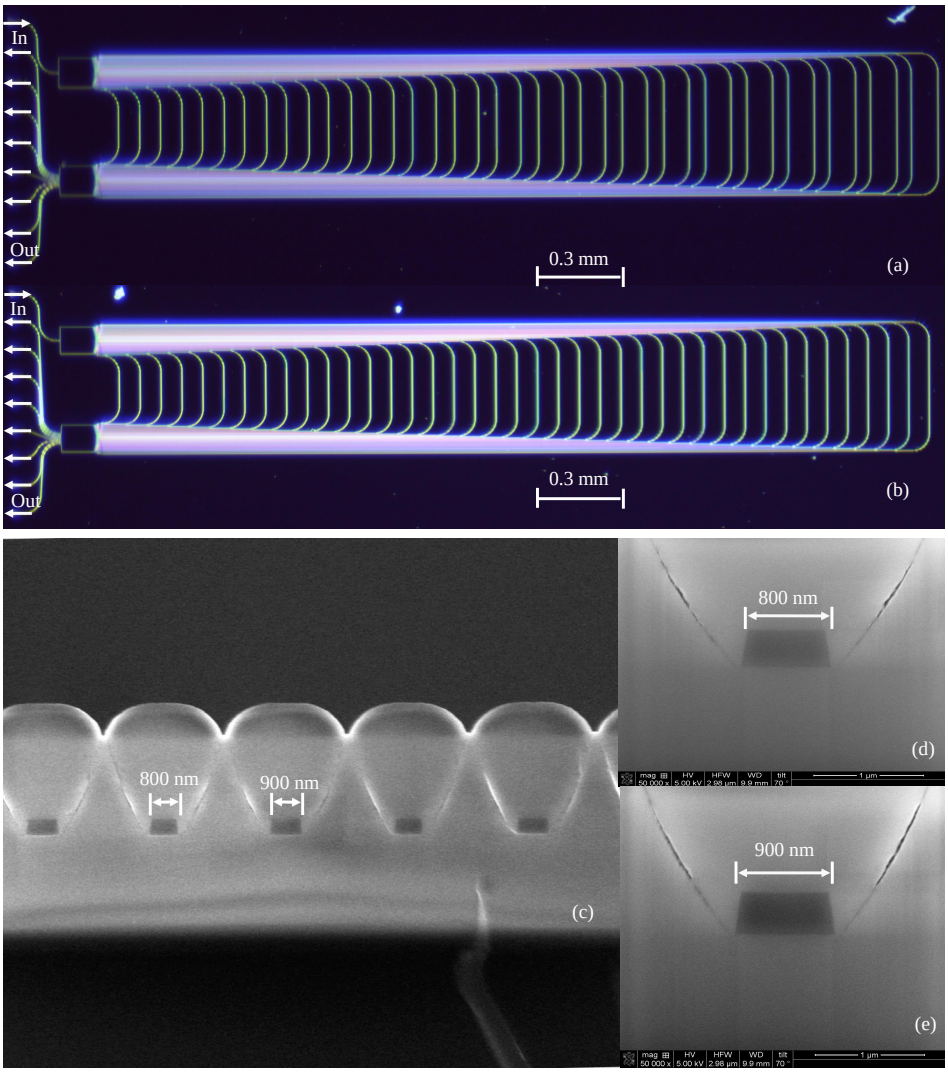


Figure 3.4: Microscrope image of (a) the SL-AWG and of (b) the conventional AWG. (c) Scanning electron microscope (SEM) image of a cross-section of the superlattice. (d) SEM image of the cross-section of a waveguide with a 800 nm width and (e) of a waveguide with a 900 nm width.

Light comes from a laser through an optical polarization maintaining (PM) fiber and is coupled to the chip through a grating coupler. The optical device performs a function on the light and the light is then detected through a second grating coupler. The normalized transmission spectrums of the MZIs are shown in Fig. 3.5. The interference pattern is clearly seen in the conventional MZI, as shown in Fig. 3.5(a), while little interference appears in the MZI with the supercell, as shown in Fig. 3.5(b). This indicates that the waveguide superlattice can effectively prevent parasitic coupling between adjacent waveguides.

The minimum insertion loss is approximately 3.4 dB for the SL-AWG and around 3.6 dB for the conventional AWG, as shown in Fig. 3.6. The non-uniformity in terms of interchannel imbalance is approximately 3 dB. The insertion loss of the AWG includes the waveguide

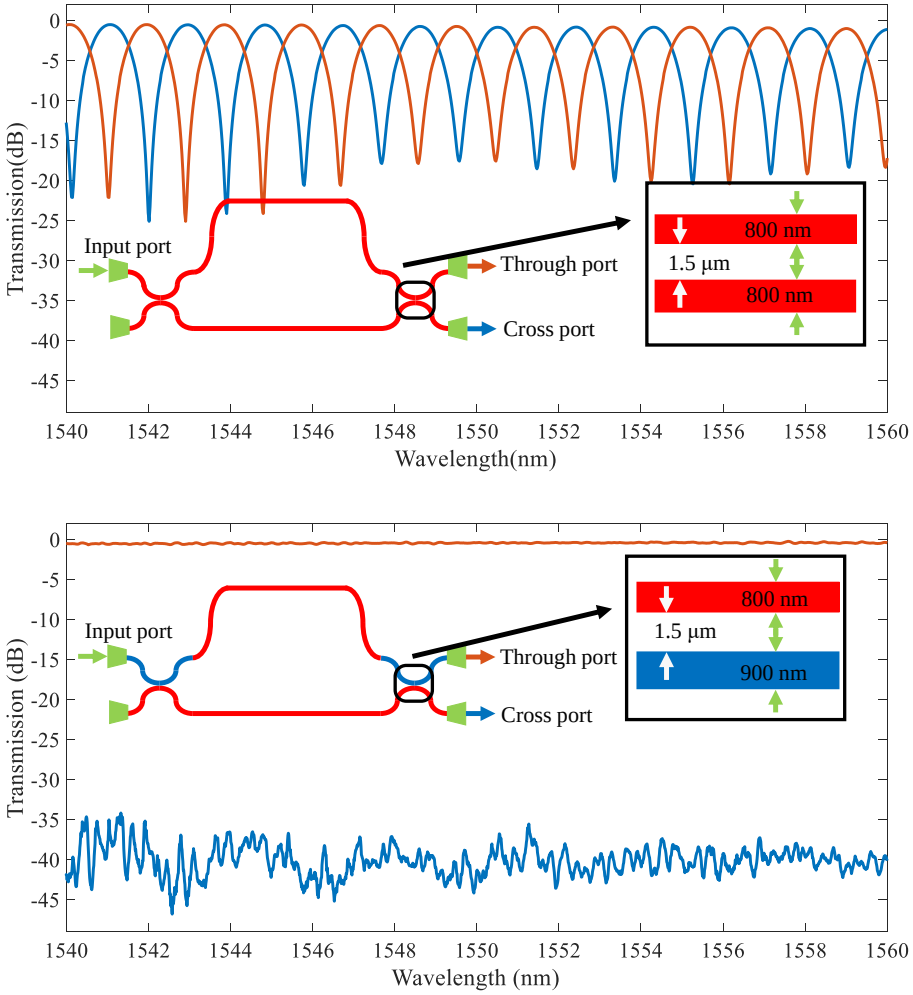


Figure 3.5: Measured spectrum of (a) the conventional MZI and of (b) the MZI with waveguide supercell arms.

scattering loss and the loss due to coupling between adjacent grating orders [59; 77]. The latter is estimated to be 2.24 dB, which could be decreased through further design optimization [59]. The measured waveguide propagation loss is about 1.5 dB/cm. This is mainly due to scattering losses caused by roughness of the sidewalls, which could be reduced by optimizing reactive ion etching process.

The crosstalk level defined by measuring the power difference (in dB) between the crosstalk floor and the channel loss is adopted to examine the anti-coupling effect of superlattice waveguides in SL-AWGs [93]. The crosstalk level across all channels for the SL-AWG is approximately -18 dB, as shown in Fig. 3.6(a). The crosstalk level for the conventional AWG is about -14 dB, as shown in Fig. 3.6(b). In addition to coupling between adjacent waveguides, crosstalk level may be caused by other sources such as phase errors and receiver crosstalk [59]. Phase errors are the results of fabrication variations that can be reduced by improving

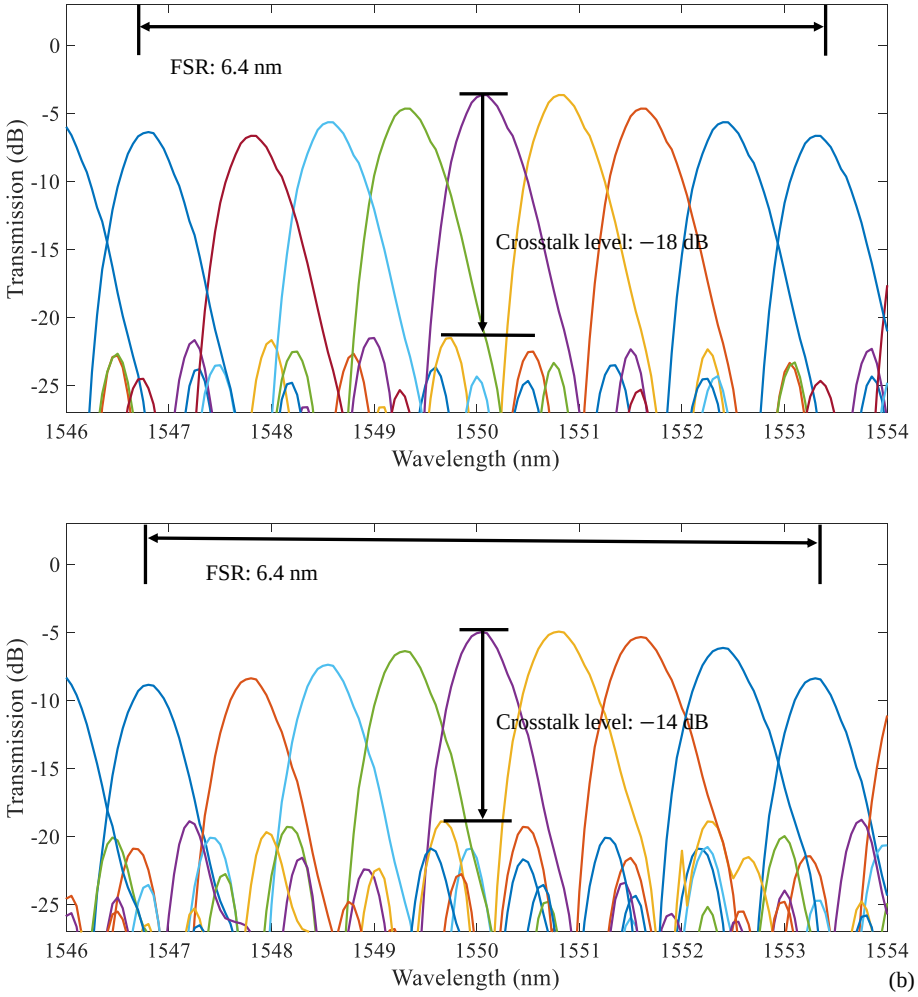


Figure 3.6: Measured spectrum of (a) the SL-AWG and of (b) the reference conventional AWG.

the fabrication process. The crosstalk due to phase errors is estimated to be around −21 dB. It can be improved by optimizing the fabrication process or utilizing a thinner but wider waveguide geometry to reduce sensitivity to fabrication errors [94]. For instance, the crosstalk reported for a 16 × 200 GHz AWG based on 5.5 μm-wide and 50 nm-thick SiN waveguides was approximately −40 dB [22]. Receiver crosstalk is due to coupling between the output channels through the exponential tails of the field distributions inside the star coupler and can be reduced by increasing the separation between the output channels [59]. In the AWGs presented here, the receiver crosstalk is estimated to be around −21 dB. Crosstalk level also depends on the recombined field distribution at the output channels [18]. The suppressed neighborhood crosstalk in the superlattice improves the recombined field distribution at the output channels, which significantly improves the crosstalk performance and also allows for the denser waveguide array of the SL-AWG. The denser waveguide array incidentally also

reduces the SL-AWG device footprint, which helps to minimize the portion of the insertion loss due to the propagation losses in the arrayed waveguides. In our previous work [69], the 1×8 AWG built on the same SiN platform shows similar performance but requires 10 μm gaps in the arrayed waveguides and occupies an area of 4.7 mm $\times$ 1.4 mm. In this work, the SL-AWG measures 4.3 mm $\times$ 0.6 mm and is thus about 2.6 times smaller than our previous AWG .

3.6 Conclusion

In summary, we designed and fabricated a SL-AWG and compared it to a conventional AWG. The minimum insertion loss of 3.4 dB is measured for the SL-AWG and of 3.6 dB for the conventional AWG. The average crosstalk level across 8 channels is approximately -18 dB for the SL-AWG and -14 dB for the conventional AWG. Our results illustrate that the use of a waveguide superlattice significantly improves the performance of SiN AWGs while reducing the device footprint. This shows that these SL-AWGs are promising for DWDM applications, such as WDM passive optical networks.

Chapter 4

Phase errors and statistical analysis of silicon-nitride arrayed waveguide gratings

Publié dans Optics Express, accepted (2022).

© 2022 Optica Publishing Group. Users may use, reuse, and build upon the article, or use the article for text or data mining, so long as such uses are for non-commercial purposes and appropriate attribution is maintained. All other rights are reserved.

4.1 Résumé

Nous présentons une analyse statistique de la performance des réseaux sélectifs planaires (Arrayed Waveguide Grating, AWG) en présence d'erreurs de phase dan les guides d'onde optique provoqués par des variations causées par les processus de fabrication. Les critères de performance importants tels que les pertes d'insertion, la diaphonie et l'uniformité sont analysés en fonction de la longueur de cohérence, un paramètre physique caractérisant l'erreur de phase accumulée dan un guide d'onde optique et qui peut être extrait en mesurant la variation de la longueur d'onde de résonance d'un interféromètre Mach-Zehnder. Au niveau de la matrice de guides d'onde, une longueur de cohérence de 23,7 mm est mesurée pour les guides d'onde en nitrure de silicium (SiN) à épaisseur sousmicrométrique ayant un processus de fabrication utilisant des gaufre de 200 mm. Grâce à des simulations Monte-Carlo, nous avons examiné l'impact de l'erreur de phase sur la performance d'AWG avec des espaces intercanaux de 200 GHz et 100 GHz. Nos résultats montrent que l'erreur de phase des guides d'onde produit des pertes d'insertion et une diaphonie excessive, mais a peu d'impact sur l'uniformité.

4.2 Abstract

We present a statistical analysis of arrayed waveguide gratings (AWGs) in presence of phase errors of optical waveguides due to fabrication process variation. Important figures of merit, such as insertion loss, crosstalk and non-uniformity, are parameterized by the coherence length, a physical parameter that characterizes the accumulated phase error of an optical waveguide and can be extracted by measuring variation in resonant wavelength of Mach-Zehnder interferometers. A die-level coherence length of 23.7 mm is measured for sub-micrometer-thick silicon nitride (SiN) waveguides fabricated using a 200-mm wafer process. Through Monte Carlo simulations, we examine the impact of phase errors on performance of AWGs with 200 GHz and 100 GHz channel spacing. Our results show that the waveguide phase errors cause remarkable excess insertion loss and crosstalk of an AWG, but have less impact on non-uniformity.

4.3 Introduction

In dense wavelength division multiplexing (DWDM) systems, arrayed waveguide gratings (AWGs) are important components used as wavelength multiplexers and demultiplexers [59]. They are also widely used as microspectrometers [86]. Therefore, AWGs have been developed on multiple material platforms including silica-on-silicon [87; 95], silicon-on-insulator (SOI) [71; 72; 73; 74; 75], silicon oxynitride (SiON) [96; 97] and silicon nitride (SiN) [22; 44; 76; 77; 78; 79]. AWGs based on silica-on-silicon are widely utilized because of their high performance. However, their large footprints limit integration with other components. AWGs in SOI photonic integrated circuits (PICs) have much smaller footprints thanks to the high refractive index contrast between the waveguide core and cladding, which enables the implementation of ultra-compact waveguides that can be densely packed. However, the phase errors in SOI arrayed waveguides are usually high due to their high sensitivity to fabrication variations. SiN waveguides on the other hand have a moderate refractive index contrast, which results in better tolerances to fabrication variations compared to SOI waveguides.

Process variations, such as those in wafer thickness, etching depth, waveguide width, and local material refractive index, cause changes in the effective index of waveguides, resulting in phase variations that introduce uncertainty in the responses of photonic devices. Phase errors in waveguides due to process variations have been investigated in earlier studies. Adar *et al.* introduced the concept of coherence length to characterize phase errors in silica core waveguides using Mach-Zehnder interferometers (MZIs) [46]. Yang *et al.* characterized the phase errors in silicon waveguides fabricated on a 220 nm thick SOI wafer using 248 nm lithography and measured coherence lengths of 4.17 mm and 1.61 mm for strip and rib waveguides, respectively [98]. These values are significantly lower than those for silica waveguides (up to 27 m [46]) due to the very strong refractive index contrast of silicon waveguides. SiN waveguides should have longer coherence lengths, but this has not been reported to date.

The performance of an AWG (e.g., insertion loss and crosstalk) is determined by how well the image from the input waveguide aperture is reproduced and captured at the aperture of the output waveguides. Variations in waveguide geometry result in effective index fluctuations and phase errors in the waveguide array. The accumulation of phase errors in arrayed waveguides deteriorates the image quality and performance of an AWG [47; 99; 100]. Umar *et al.* studied the impact of changes in waveguide width and length in Si AWGs and examined the effects of phase errors in the arrayed waveguides. Their work confirms that longer waveguides lead to higher insertion loss and crosstalk in AWGs and that wider waveguides can help suppress them [47]. Okamoto introduced an empirical expression relating the phase error as a function of the length of the middle waveguide in the array to the crosstalk of the AWG [94]. Based on these earlier reports, we see that the performance of AWGs is very sensitive to waveguide phase errors caused by process variations. Direct characterization of process variations at the wafer level, e.g., by using techniques such as scanning electron microscopy and ellipsometry, is a tedious process. It is more practical to extract behavioral parameters, such as waveguide propagation loss and coherence length, through optical measurements of compact structures, such as waveguide spirals and interferometers. However, there still exists a gap between these parameters and the quantitative analysis of more complex photonic circuits such as AWGs, which is essential for yield prediction based on important figures of merit.

In this chapter, we present a statistical analysis of the performance of SiN AWGs. As shown in Fig. 4.1, our method involves three steps: 1) wafer-level testing of MZIs to extract the parameter (coherence length in this case) characterizing the waveguide phase errors due to process variations; 2) modeling of AWGs taking into account phase errors; and 3) Monte Carlo simulations and yield prediction. The remainder of this chapter is organized as follows. In Section 4.4, we review the concept of coherence length and present an analytical method to measure the phase errors and coherence length of optical waveguides based on the free spectral

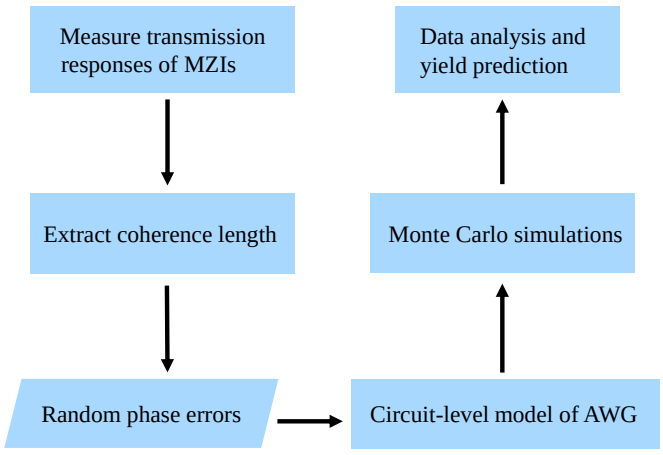


Figure 4.1: Procedure for the proposed statistical analysis.

range (FSR) and the resonant wavelength variations in MZIs. Using this method, we measure the coherence length of sub-micrometer-thick SiN waveguides fabricated using a 200 mm wafer process. In Section 4.5, we present an AWG model parameterized according to the coherence length for statistical analysis. Using this model, we perform Monte Carlo simulations in which the response of AWGs is calculated a thousand times with a semi-analytical model with random phase errors generated for each iteration. The results are used to examine the impact of process variations on the insertion loss, crosstalk, and non-uniformity of 1 × 4 AWGs with 200 GHz and 100 GHz channel spacings. Lastly, conclusions are presented in Section 4.6.

4.4 Phase error and coherence length

4.4.1 Fundamental principle

Fabrication variations result in changes in the waveguide geometrical parameters, such as thickness and width, and in variations of the refractive index of the cladding and core material [45]. The variations in the effective index of the waveguide result mainly from fluctuations in the core layer thickness due to non-uniformity in the deposition process and in the waveguide width due to etching imperfections. Wavelength sensitive components usually show large performance variations across a wafer (e.g., resonance wavelength) due to thickness variations (typically on the nanometer scale across an 8" or 12" wafer [94]). However, the correlation length L_c of wafer thickness variations is much longer than that of width or angle variations [45]. Therefore, the impact of width variations is more significant within a die. Despite the relatively complex circuit configuration, the footprint of a SiN AWG is typically on the scale of square millimeter, an area in which thickness variations are at least one order of magnitude smaller than width variations. This work focuses on a die-level analysis of figures of merit, such as insertion loss and crosstalk, that are more sensitive to phase errors within a chip. In this case, the characteristic waveguide length is much longer than the correlation length of fabrication variations, and thus the variations of phase errors can be assumed to increase linearly with the waveguide length. A mathematical model of phase errors with respect to the correlation length is derived in Appendix 4.7.

The effective index n_{eff} is modelled as a combination of an average value and a random error δn_{eff}:

$$n_{eff} = \langle n_{eff} \rangle + \delta n_{eff} \tag{4.1}$$

The value of δn_{eff} is assumed to have a Gaussian distribution with a zero mean based on the central limit theorem [101]. The accumulated phase over a length L is given by:

$$\phi(L) = k_0 \int_0^L n_{eff}(x)\, \mathrm{d}x \tag{4.2}$$

where k_0 is the wave number and equals $2\pi/\lambda$, and λ is the wavelength of light. The phase error accumulated in this waveguide is given by:

$$\delta\phi(L) = k_0 \int_0^L \delta n_{eff}(x)\, \mathrm{d}x \tag{4.3}$$

In order to characterize the fabrication variations and the accumulated phase errors, we use the coherence length L_{coh}, which is defined by:

$$\left\langle e^{\delta\phi_{L_{coh}}} \right\rangle = e^{-1} \tag{4.4}$$

Due to the small correlation length, the phase errors are assumed to have a Gaussian distribution with a zero mean with the width determined by its variance (in rad^2) or standard deviation (in rad). The phase error has a possibility of 0.68 of being within one standard deviation of the mean. Because the correlation length of δn_{eff} is small compared to L, we can approximate the autocorrelation of δn_{eff} as a delta function: $\langle \delta n_{eff}(x)\delta n_{eff}(x')\rangle = D\delta(x-x')$, where D is a constant characterizing the magnitude of the fluctuations. The variance of the phase errors accumulated over L is thus given by [46]:

$$\left\langle \delta\phi(L)^2 \right\rangle = \left(\frac{2\pi}{\lambda}\right)^2 \int_0^L \mathrm{d}x \int_0^L \mathrm{d}x' \left\langle \delta n_{eff}(x)\delta n_{eff}(x') \right\rangle = D\left(\frac{2\pi}{\lambda}\right)^2 L \equiv \frac{2L}{L_{coh}} \tag{4.5}$$

4.4.2 Characterization method

We extracted the phase error in waveguides based on measured spectra of MZIs, each consisting of two unbalanced waveguides (MZI arms) connected by a pair of 3 dB couplers. The lengths of the MZI arms are L_1 and $L_2 = L_1 + \Delta L_{MZI}$, respectively. For simplicity, we neglect the losses of the arms. The measured power intensity of the fabricated MZIs can be found by [46]:

$$I_o = I_i |r_1 e^{i\phi_1 + \delta\phi_1} + r_2 e^{i\phi_2 + \delta\phi_2}|^2 \tag{4.6}$$

where I_i is the input intensity, I_o is the output intensity, $\delta\phi = \delta\phi_1 + \delta\phi_2$ is the total phase error due to effective index variations, and $r_1 : r_2$ represents the splitting ratio of the couplers. The phase fluctuations accumulated in the arms induce shifts in resonant wavelengths of the MZI spectrum.

We extract the phase errors by measuring the wavelength fluctuations of the null power intensity at the MZI output port. The null intensity appears when the phase difference between the two arms equals $\Delta\phi = \phi_1 - \phi_2 = 2(k+1)\pi$, where k is an integer. The average phase is given by:

$$\langle \Delta\phi_{MZI} \rangle = k_0 \langle n_{eff} \rangle \Delta L_{MZI} \tag{4.7}$$

The phase error accumulated in the MZI arms results in a wavelength deviation $\delta\lambda$ between identical MZIs. The phase error deviation is given by:

$$\langle \delta\phi^2 \rangle = \frac{4\pi^2 n_g^2 \langle \delta\lambda^2 \rangle \Delta L_{MZI}^2}{\lambda^4} = \frac{4\pi^2 \langle \delta\lambda^2 \rangle}{\Delta\lambda_{FSR-MZI}^2} \tag{4.8}$$

where $\delta\lambda = |\lambda_i - \langle\lambda\rangle|$ and $\Delta\lambda_{FSR-MZI}$ and n_g are the free-spectral range of the MZI and the waveguide group index, respectively.

Since we measured the phase errors with MZIs, both arms contributed to the total phase error caused by fabrication variations. Combing Eq. 4.8 and 4.5, the coherence length is given by:

$$L_{coh} = \frac{\Delta\lambda_{FSR-MZI}^2 (L_1 + L_2)}{2\pi^2 \langle \delta\lambda^2 \rangle} \tag{4.9}$$

4.4.3 Simulation and experimental results

A schematic of the SiN waveguide considered in this study is shown in Fig. 4.2(a). The effective index of the waveguide changes with variations in its geometry, including its thickness t, width w and sidewall angle a. The effective index of the TE mode is calculated using MODE Solution from Ansys Lumerical. The reference waveguide has a nominal bottom width of 865 nm, a thickness of 450 nm and a sidewall angle of 88°. The effective index of the SiN waveguide is in red and its dependence on variations of the geometry is in blue, as shown in Fig. 4.2(b), (c) and (d). The effective index dependence on the waveguide width has a peak at 470 nm, and the sensitivity to width variations is lower in wider waveguides, as shown in Fig. 4.2(b). As shown

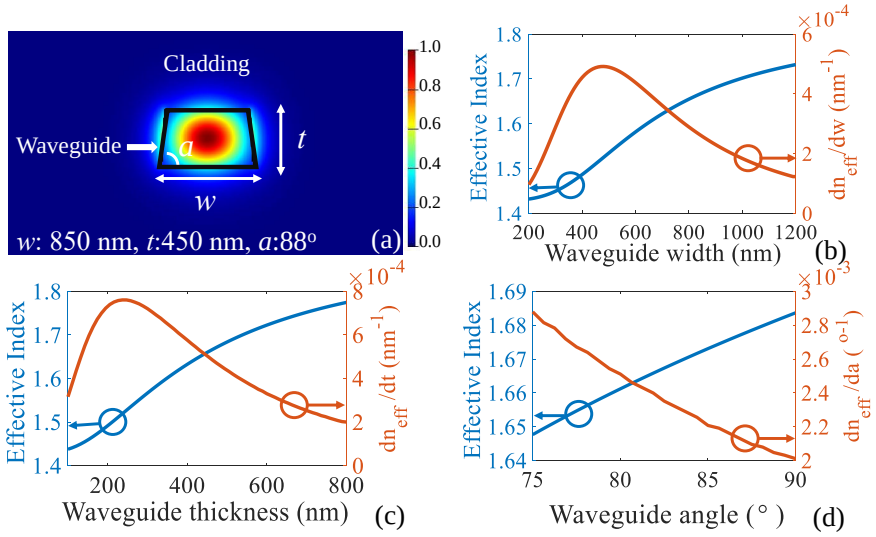


Figure 4.2: (a) Schematic of the SiN waveguide; effective index of the waveguide and its dependence at 1550 nm on (b) the waveguide width, (c) waveguide thickness and (d) waveguide sidewall angle.

in earlier studies, wider waveguides reduce the impact of fabrication variations since their effective index is less sensitive to changes in dimensions [47]. In the simulations, the effective index dependence on the waveguide thickness dn_{eff}/dt has a peak value of 7.9×10^{-4} nm^{-1}, whereas the one of the waveguide width dn_{eff}/dw is 5.1×10^{-4} nm^{-1}. For the reference waveguide used in this work, its effective index shows a dependence of 5.0×10^{-4} nm^{-1} on the thickness, of 2.6×10^{-4} nm^{-1} on the width, and of $2.1 \times 10^{-3} \circ^{-1}$ on the sidewall angle. Comparing Fig. 4.2(b), (c) and (d), the effective index shows a stronger dependence on the waveguide thickness than width, indicating that the effective index variation is more sensitive to the thickness for the TE mode. However, as discussed earlier, in this work, we focus on a die-level analysis of phase errors, where width variations dominate over thickness variations.

We performed 100 Monte Carlo simulations to assess the impact of phase errors on the resonant wavelengths of MZIs where random phase errors were applied to various characteristic coherence lengths following Eq. 4.5. The transmission spectra of MZIs with various coherence and waveguide lengths are shown in Fig. 4.3(a) and (b). The splitting ratio $r_1 : r_2$ has a mean of 48.95 : 51.05 and a standard deviation of 0.35 for the splitter-combiners in the MZIs, which is extracted from our measurement results. As indicated in Eq. 4.9, higher fabrication variations result in shorter coherence lengths. We examine the coherence lengths of 100 mm, 20 mm and 10 mm to cover recent improvement in fabrication process quality. The spectrum of MZIs with infinite coherence length and ideal waveguide fabrication are shown with black lines in both figures. We see that the resonant wavelengths of MZIs that have a total length of 2 mm shift across a wider range when their coherence length is shorter, as shown in Fig. 4.3(a). The phase errors accumulated in the waveguides and the resonant wavelengths have higher variations in longer waveguides. In Fig. 4.3(b), we examine the transmission spectra as the total arm length ($L_1 + L_2$) increases from 2 to 8 mm with a coherence length of 20 mm. We can see that the resonant wavelength spreads over a broad spectral range for longer waveguides due to larger accumulated phase errors.

We designed and fabricated 3 groups of MZIs on 7 dies. Each group has 6 nominally identical MZIs. The total waveguide lengths of the 3 groups are 2.3 mm, 2.5 mm, and 2.8 mm, respectively. The transmission spectra of all MZIs from the 7 dies across the wafer were measured to guarantee statistical significance. Figure 4.3(c) shows the intra-die spectral response of identical MZIs with a waveguide length of 2.3 mm. The wavelength resolution of the spectra is 0.1 nm. In order to extract the resonant wavelengths, we fitted a Lorentzian response to the data. Based on the method explained previously, we calculated the phase error variance, as shown in Fig. 4.3(d). The variance increases with longer waveguides. It ranges from 0.22 rad^2 to 0.27 rad^2 for the 2.3 mm-long waveguide. The phase errors are proportional to the square root of the waveguide length $\sqrt{L}$. We used a linear function to fit the distribution of the phase error variances as a function of the waveguide length. The blue line presents the average phase error variance of all dies. The average variance and standard deviation of the

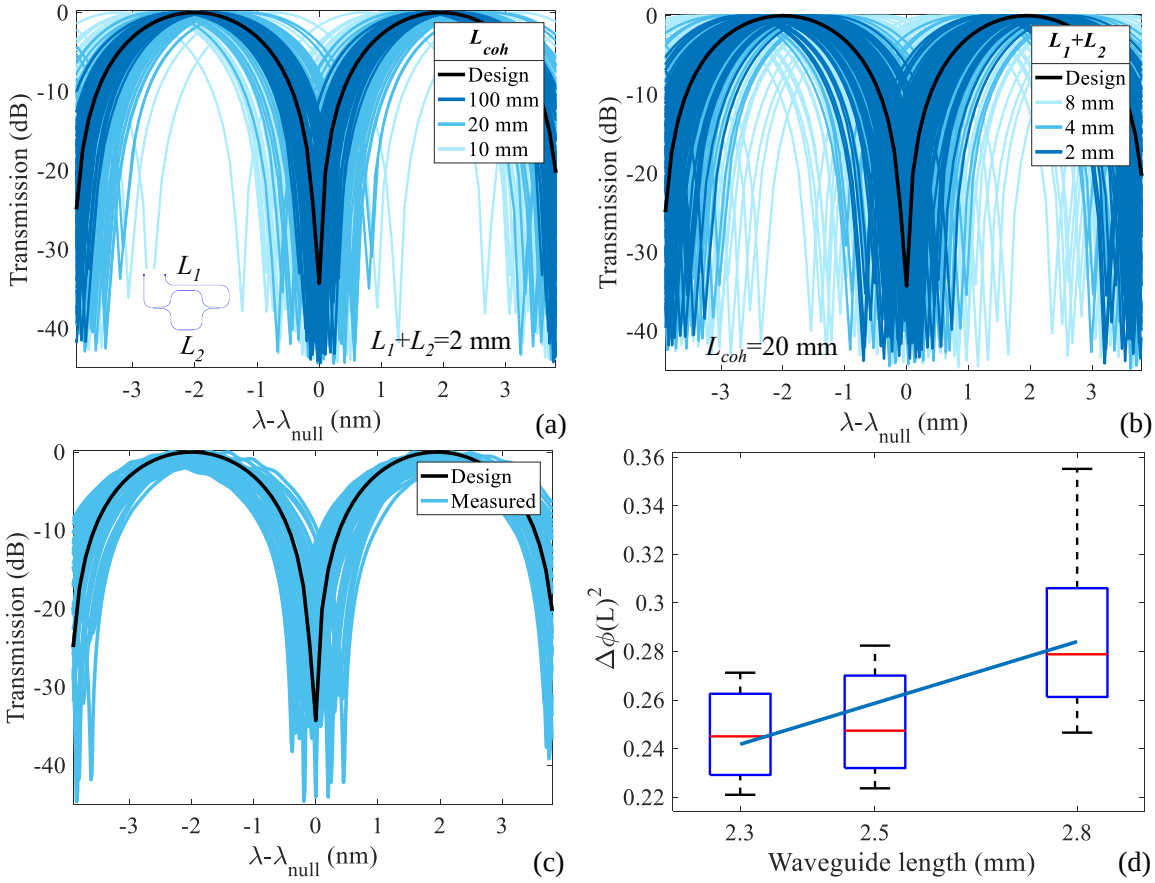


Figure 4.3: Simulated spectra of MZIs in 100 Monte Carlo samples with (a) various coherence lengths and a total waveguide length $L_1 + L_2 = 2$ mm, (b) various waveguide lengths and a coherence length $L_{coh} = 20$ mm; (c) measured spectra of the MZI with waveguide length of 2.3 nm within dies; (d) the phase error variances extracted based on MZIs.

phase error in 2.3 mm-long waveguides are 0.24 rad^2 and 0.49 rad, respectively. According to Eq. 4.9, the average coherence length of the dies is 23.7 mm in this fabrication run. Using Eq. 4.17, the phase error variance is estimated to be 0.24 rad^2, indicating a width variation of about 5 nm assuming $dn_{eff}/dw = 2.6 \times 10^{-4}$ nm^{-1} as calculated above (the variation of the sidewall angle is neglected here) and $L_c = 3.8$ μm, which agrees well with observations in typical silicon photonic fabrication processes [94; 102].

4.5 Statistical analysis of AWGs

4.5.1 AWG model

An AWG consists of an array of waveguides connecting two star couplers, as depicted in the schematic of Fig. 4.4(a). The length increment between adjacent waveguides ΔL is chosen to equal an integer multiple of the target center wavelength:

$$\Delta L = m\frac{\lambda_c}{n_c} \tag{4.10}$$

where m is the diffraction order, λ_c is the center wavelength and n_c is the value of the effective index of the waveguide mode at the center wavelength.

As light propagates along the array, it accumulates phase proportional to the waveguide length before emerging into the output star coupler. The phase at the end of the qth waveguide is given by:

$$\phi_q = k_0 n_{eff} L_q \tag{4.11}$$

where $L_q = L_0 + q\Delta L$ is the total waveguide length.

At the center wavelength ($\lambda = \lambda_c$), the relative phase difference at the end of each waveguide is a multiple of 2π. The coherent superposition of the light from every waveguide in the output star coupler thus reproduces the conditions of the input star coupler, effectively reconstructing the input mode field along the output interface. As the wavelength of the light departs from λ_c, a linear phase gradient is introduced between the waveguides which shifts the mode field image along the output interface. This behavior is critical to the operation of the AWG.

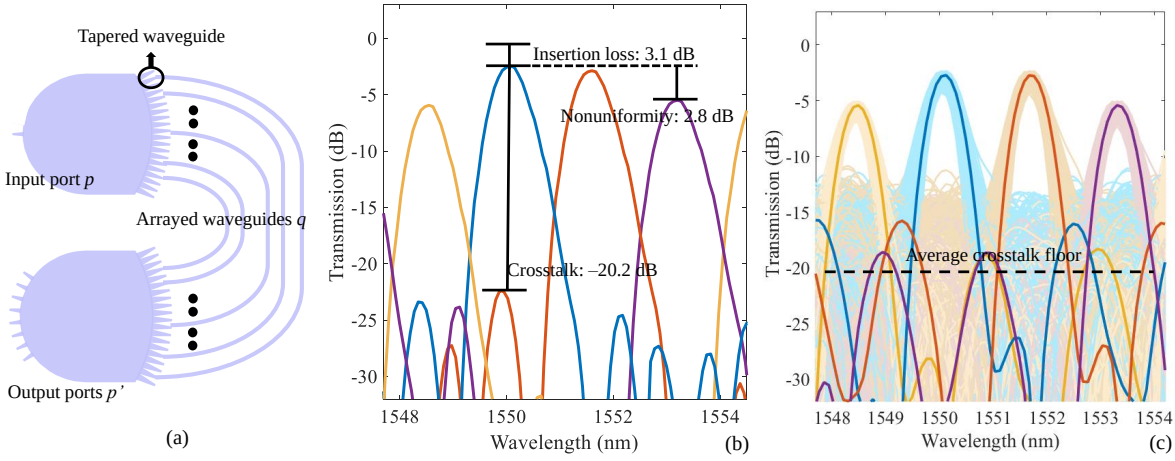


Figure 4.4: (a) Schematic of an AWG; (b) measured spectrum for the 1 × 4 200 GHz AWG and (c) 1000 Monte Carlo calculated responses in light color, one specific response in dark color, and average crosstalk floor shown with a black dash line.

Due to unavoidable fabrication imperfections, slight deviations in the waveguide geometry affecting the mode effective index are introduced along the waveguide resulting in an overall random offset in the output phases. These phase errors lead to a distortion of the interference pattern in the output star coupler that negatively impacts the device performance, as in the MZIs above.

The normalized transmission spectrum of a 1 × 4 200 GHz AWG with 17 arrayed waveguides is shown in Fig. 4.4(b). The spectrum was measured on a vertical coupling setup where light is coupled to the chip through surface grating couplers (SGCs). At the center wavelength, the insertion loss is approximately 3.1 dB. The crosstalk in the center channel is around −20.2 dB. The non-uniformity between the center and outer channels is around 2.8 dB. The degradation in performance is mainly attributed to the phase errors caused by fabrication variations.

To account for the phase error contributions, Eq. 4.11 is modified by making use of Eqs. 4.1 and 4.2:

$$\phi_q = k_0 \langle n_{eff} \rangle L_q + \delta\phi_q \tag{4.12}$$

where $\delta\phi_q = k_0 \int_0^{L_q} \delta n_{eff} dx$ is the local phase error obtained from Eq. 4.3. As shown above, the result of taking the integral in Eq. 4.12 leads to a random variable that is normally distributed around zero with a variance given by Eq. 4.5. In this way, knowing the coherence length of the target fabrication process allows to accurately predict the random phase error effect on the overall device performance. The complex transmission from an input waveguide p to an output waveguide p' is then described by (see section 1.33):

$$t_{p,p'} = \sum_{q=1}^{N} C_{p,q} C_{q,p'} \exp\left(-jk_0 \langle n_{eff} \rangle L_q\right) \exp\left(-j\delta\phi_q\right) \tag{4.13}$$

where N is the number of arrayed waveguides, and $C_{p,q}$ and $C_{q,p'}$ are the input and output coupling coefficients, respectively.

Statistical predictions of the impact of phase errors on crosstalk and other parameters can be extracted from Monte Carlo simulations (shown in Fig. 4.4(c)). This is done by repeatedly simulating the AWG transmission for randomly generated phase error $\delta\phi_q$ in each waveguide. The random values are carefully chosen from a gaussian distribution with the variance set according to the target coherence length. A coherence length of 20 mm is close to the one measured in Fig. 4.3, while 100 mm and 10 mm are chosen for comparison. The result of these simulations for various coherence lengths are presented in the following sub-sections.

4.5.2 Insertion loss

In the following numerical studies, we consider AWGs with one input and four output channels with spacings of 100 GHz and 200 GHz. We assume $\Delta\lambda_{FSR} = N_o \Delta\lambda$ in the design, where N_o is number of the output channels and $\Delta\lambda$ is the channel spacing. The length difference between adjacent waveguides in the array is 346.5 μm for the 100 GHz design and 173.3 μm for the 200 GHz one, respectively. The FSR is 3.2 nm for the 100 GHz design and 6.4 nm for the one with a 200 GHz channel spacing.

The insertion loss of an AWG includes the waveguide scattering loss, the bending loss, and the transition loss between the star couplers and the arrayed waveguides [22]. The bending loss can be minimized by using a large bending radius of more than 100 μm for the SiN waveguides presented in Fig. 4.2(a). However, long bent waveguides increase the overall length and footprint of AWGs. Tapered waveguides are used at the junctions between the arrayed waveguides and the star couplers in order to minimize transition losses, as shown in Fig. 4.4(a). Propagation losses in arrayed waveguides are primarily due to the roughness of the waveguide sidewalls created during the lithography and etching processes. The values reported in the literature range from sub- to a few dB/cm [22]. In our SiN AWG model, the shortest path is set to 400 μm, including the bent and straight sections.

Figure 4.5 shows the impact of the number of waveguides in the array on the insertion loss for various levels of propagation loss. The solid lines show how the insertion loss changes with negligible propagation loss in the arrayed waveguides for the 1 $\times$ 4 AWG. The insertion loss is around 3.3 dB with 10 waveguides in the array. It decreases to 2.2 dB when the number of waveguides increases to 21. The reduction in insertion loss results from the improvement in spatial resolution and also from the higher overall coupling efficiency between the star coupler and the arrayed waveguides since the waveguides cover a wider area of the star coupler. Beyond 21 waveguides, the insertion loss remains at a low and constant levels for both cases.

The difference in insertion loss of the 200 GHz and 100 GHz AWGs is obvious when the waveguides are lossy. The insertion losses with a propagation loss of 2 dB/cm as a function of the number of arrayed waveguides are shown with dashed lines in Fig. 4.5. A similar trend appears for both AWGs when there are only a few waveguides in the array. In this case, the

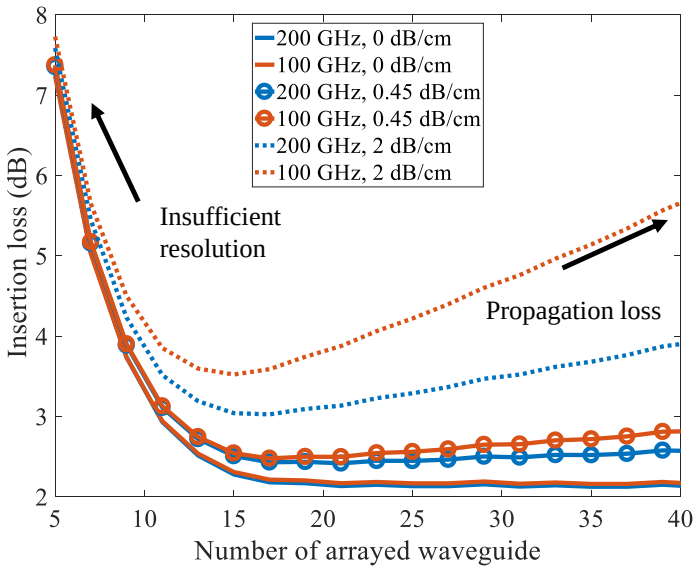


Figure 4.5: Insertion loss for 200 GHz and 100 GHz AWGs with different propagation loss.

resolution is insufficient and the coupling losses at the interface with the coupler are high. On the other hand, for an AWG with a large number of arrayed waveguides, the insertion loss increases with the addition of waveguides since the average length of the array also increases. Therefore, propagation losses become the dominant source of losses. The insertion loss is 3.0 dB for the 200 GHz AWG with 17 waveguides, and 3.5 dB for the 100 GHz AWG with 15 waveguides. When the shortest waveguide length in the array is set to 400 μm for both channel spacings, the 100 GHz AWG has longer waveguides and an higher insertion loss than the 200 GHz AWGs with the same number of arrayed waveguides. The insertion loss follows the same trends for both AWGs when the propagation loss is reduced to 0.45 dB/cm. For AWGs with 40 arrayed waveguides, the difference in loss is approximately 0.24 dB between the two AWGs configurations, when the propagation loss is 0.45 dB/cm, and it increases to 1.7 dB when the propagation loss is 2 dB/cm.

Figure 4.6 shows the impact of the number of arrayed waveguides on the insertion loss as a function of the coherence length. The phase errors accumulated in the arrayed waveguides, which increases as the coherence length decreases, shift the focal points in the output star coupler. As a result, part of the power may be coupled into other channels, which increases the power floor of the AWG spectral response. Therefore, the phase errors also introduce excess losses in the AWGs. Figure 4.6(a) and (b) show how the insertion losses change as a function of the coherence length when the propagation loss is 0.45 dB/cm in the arrayed waveguides. The insertion losses were calculated for 1000 Monte Carlo simulations, and the results are shown with the light color lines. The average insertion losses as a function of the number

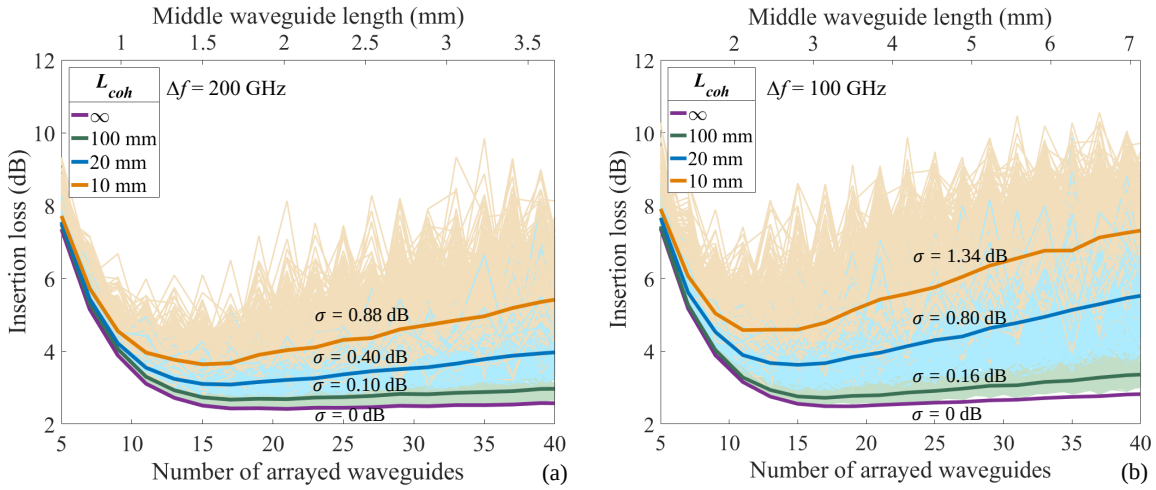


Figure 4.6: (a) and (b) show the simulated insertion loss as a function of the number of waveguides in the array (the equivalent length of the middle waveguide is also shown) at various coherence lengths for the AWG with a channel spacing of (a) 200 GHz and (b) 100 GHz. Each scatter plot contains 1000 Monte Carlo simulations. The standard deviations at 25 arrayed waveguides are labeled.

of waveguides in the array are plotted with dark color lines in both figures. The coherence lengths considered are infinite, 100 mm, 20 mm and 10 mm. The curves for the AWGs with an infinite coherence length are the same as those for the 0.45 dB/cm propagation loss in Fig. 4.5. In AWGs with a few waveguides where the middle waveguide is short, the propagation loss and random phase errors have a lower impact on the insertion loss. As the number of arrayed waveguides increases, so does the insertion loss in all cases. The fluctuations of the insertion loss are stronger for shorter coherence lengths. The standard deviation of the insertion loss is up to 0.88 dB for a coherence length of 10 mm, whereas it is 0.40 dB and 0.10 dB for coherence lengths of 20 mm and 100 mm, respectively, in 200 GHz AWGs with 25 arrayed waveguides. For 100 GHz AWGs, the standard deviation of the fluctuations reaches 1.34 dB for a coherence length of 10 mm in a device with 25 arrayed waveguides. These fluctuations result from random variations in the phase errors. The variations in phase errors increase as light propagates in longer imperfect waveguides. The length of the middle waveguide (equal to the average length of the arrayed waveguides), which increases with the number of waveguides, is indicated on the top axis of Fig. 4.6. The standard deviation of the fluctuations broaden as the length of the middle waveguide increases for all the coherence lengths considered.

The average insertion losses are higher for shorter coherence lengths for both AWGs with the same number of arrayed waveguides. The insertion loss is as low as 2.6 dB for a 200 GHz AWG with 15 waveguides and a coherence length of 100 mm, whereas they are 3.1 and 3.6 dB for a device with same number of waveguides and a coherence length of 20 mm and 10 mm, respectively. In comparison with 200 GHz AWGs, the insertion losses for 100 GHz AWGs increase more rapidly due to the longer middle arrayed waveguide. It results in higher insertion loss for 100 GHz AWGs than for 200 GHz AWGs with the same number of arms, as shown in Fig. 4.6.

4.5.3 Crosstalk

In addition to the insertion loss, another important performance metric is the crosstalk between channels. One contribution to crosstalk comes from the coupling between output channels through the exponential tails of the field distributions [59]. This source of crosstalk can be suppressed to a low level by increasing the separation between the output channels [59; 63]. Crosstalk can also be introduced by phase distortions due to parasitic coupling in the arrayed waveguides. Superlattice AWGs, where waveguides of different widths are used in the array to suppress parasitic coupling, and thus crosstalk, were demonstrated on a SiN platform in our previous work [28]. In this work, the dominant contribution to crosstalk lies in the phase errors accumulated in the arrayed waveguides. As discussed in the previous section, the accumulated phase errors increase with the length of the arrayed waveguides. As a result, crosstalk tends to be stronger in AWGs with longer arrayed waveguides [47].

We calculated the crosstalk from 1000 Monte Carlo simulations using the AWG model pre-

sented in Section 4.4.3. The average crosstalk obtained in 4 output channels of AWGs with 17 arrayed waveguides with a coherence length of 20 mm is shown in Fig. 4.7(a) and (b). The average crosstalk ranges from −8 dB to −32 dB with a mean of −17.1 dB for the 200 GHz AWGs. For the 100 GHz AWGs, the average crosstalk ranges from −3 dB to −28 dB with a mean of −14.2 dB. According to the crosstalk extracted from the Monte Carlo simulations, AWGs with a large FSR should provide better crosstalk performance due to their short arrayed waveguides. Figure 4.7(c) and (d) show the probability that an AWG provides a given level of crosstalk. We choose the normalized probabilities of 0.5, 0.68 and 0.97 to assess the crosstalk performances. There is a probability of 0.68 to obtain a crosstalk lower than −15.4 dB in 200 GHz AWGs and −12.6 dB in 100 GHz AWGs.

The average crosstalk as a function of the number of waveguides in the array for various coherence lengths is shown in Fig. 4.8. Small arrays provide low resolution and increase the power in the side bands of the field distribution, which results in a large crosstalk. As shown with the purple lines, the crosstalk decreases as the number of waveguides in the array increases. The crosstalk can be as low as −52 dB with 25 waveguides when there is no phase

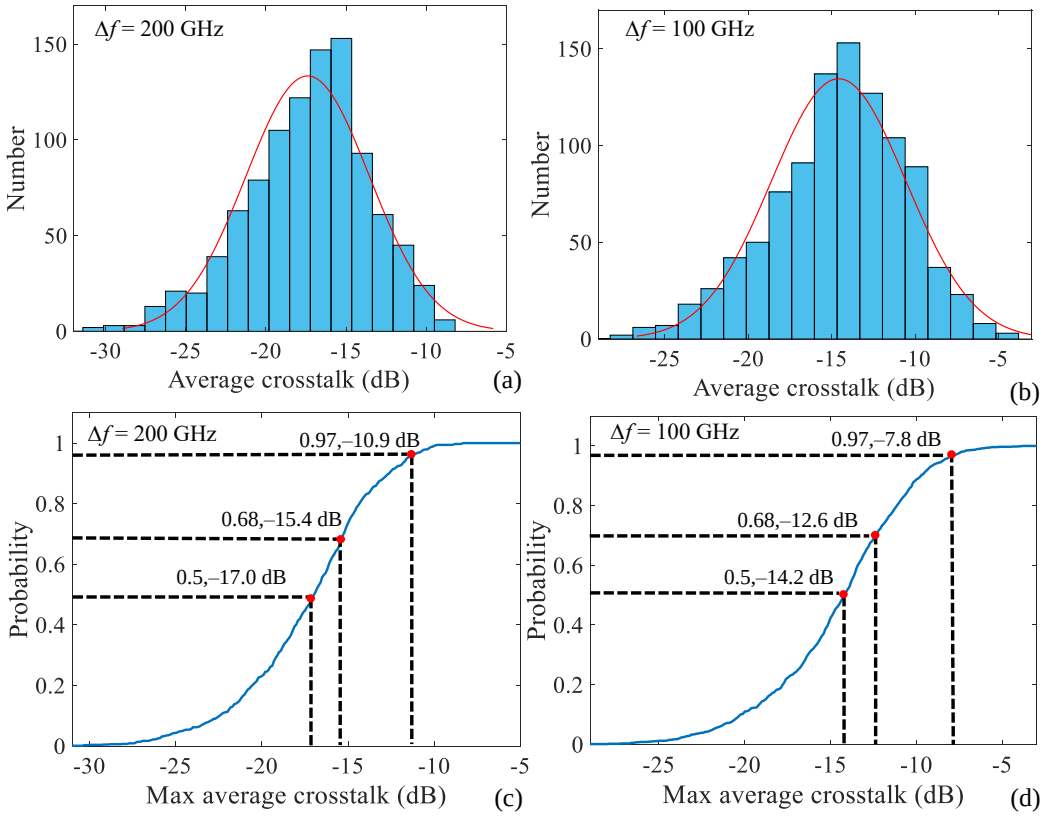


Figure 4.7: Crosstalk distributions of AWGs with 17 waveguides based on a coherence length of 20 mm in 1000 Monte Carlo samples for (a) the 200 GHz AWG and (b) the 100 GHz AWG; probability of 0.5, 0.68 and 0.97 to obtain the maximum crosstalk for (c) the 200 GHz AWG and (d) the 100 GHz AWG.

errors. As expected, the phase errors due to fabrication variations strongly affect the crosstalk in AWGs. The sampled average crosstalk in 1000 Monte Carlo simulations for 200 GHz and 100 GHz AWGs are shown with light color lines in Fig. 4.8(a) and (b). When both AWGs with channel spacing of 100 GHz and 200 GHz have middle waveguides with the same length, the average crosstalk is higher for shorter coherence lengths. The mean average crosstalk is −23.8 dB, −16.8 dB and −13.2 dB for the 200 GHz AWGs with 17 arrayed waveguides when the coherence length is 100 mm, 20 mm and 10 mm, respectively.

The average crosstalk increases with the number of waveguides in the array since the average

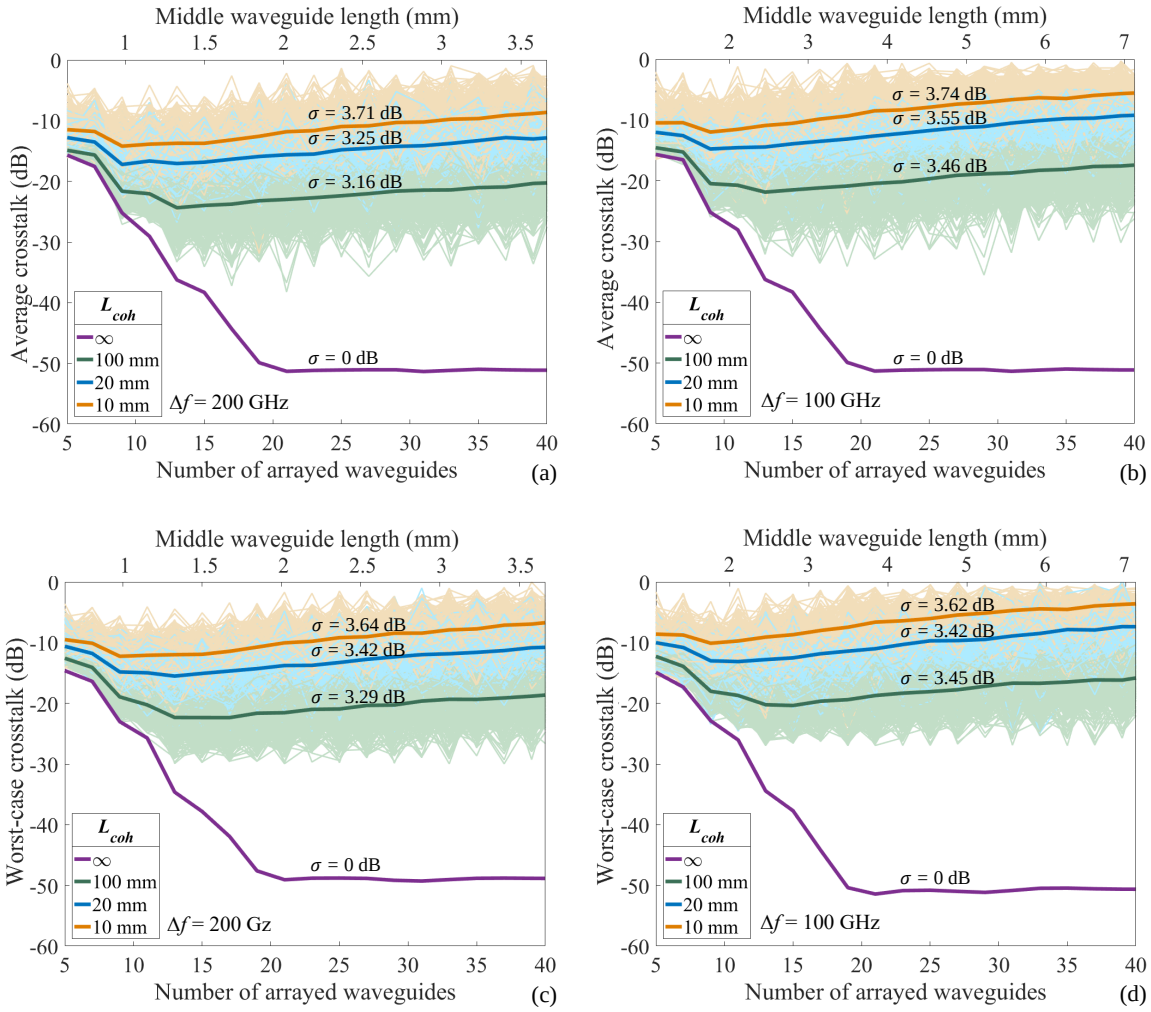


Figure 4.8: (a) and (b) show the simulated average crosstalk as a function of the number of waveguides in the array (the equivalent length of the middle waveguide is also shown) at various coherence lengths for the AWG with a channel spacing of (a) 200 GHz and (b) 100 GHz. (c) and (d) show the worst-case crosstalk for (c) the 200 GHz AWG and (d) the 100 GHz AWG. Each scatter plot contains 1000 Monte Carlo simulations. The standard deviations at 25 arrayed waveguides are labeled.

length of the waveguides also increases. The contribution of the phase errors dominates the crosstalk in large arrays. The mean average crosstalk is −16.7 dB for the 200 GHz AWG with 17 arrayed waveguides and a coherence length of 20 mm, and −12.7 dB for an AWG with 40 arrayed waveguides. Since the middle waveguide is twice as long in the 100 GHz AWG compared to the 200 GHz AWG for the same number of waveguides, the AWG with the narrower channel spacing suffers from a higher level of crosstalk, as shown in Fig. 4.8. For instance, when both have a coherence length of 20 mm, the crosstalk in the 200 GHz AWG with 40 arrayed waveguides is −12.7 dB whereas it is −9.2 dB for the 100 GHz AWG. The worst-case crosstalk in 4 channels AWG is presented in Fig. 4.8(c) and (d). For both AWGs with channel spacing of 100 GHz and 200 GHz, the worst-case crosstalk also increases with shorter coherence lengths and longer middle waveguides.

4.5.4 Non-uniformity

Non-uniformity in an AWG is defined as the difference in power between the channel with the highest output and the one with the lowest output (see Fig. 4.4). Typically, tapered waveguides are used to connect the arrayed waveguides and star couplers to minimize coupling losses. Furthermore, the far field distribution appearing at the end of the arrayed waveguides has a near Gaussian shape. Therefore, a similar field distribution appears at the image plane in the output star coupler. This near Gaussian shaped field distribution results in the outer channel having a higher insertion loss than the center channel [64; 65].

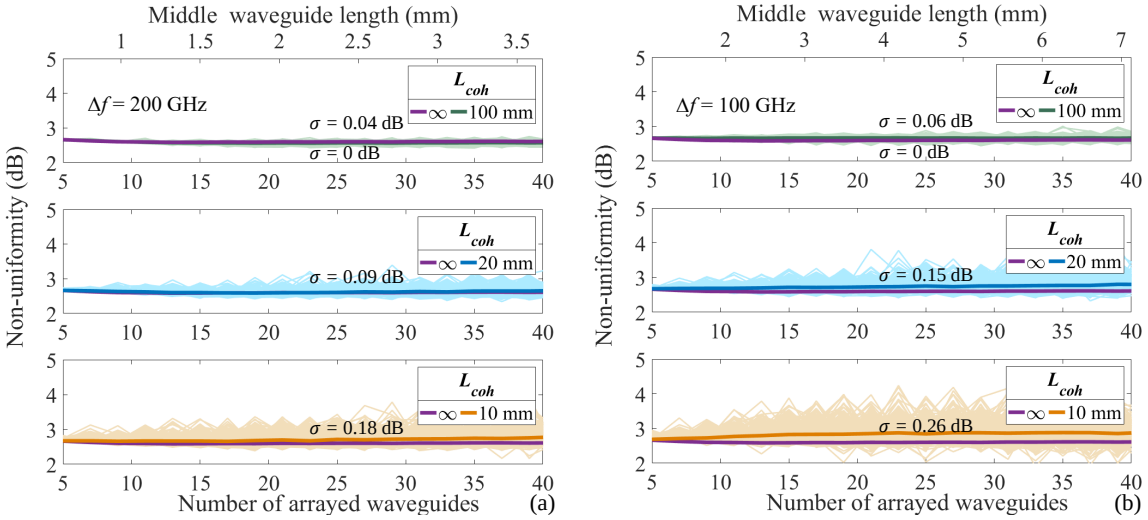


Figure 4.9: (a) and (b) show the simulated non-uniformity as a function of the number of waveguides in the array (the equivalent length of the middle waveguide is also shown) at various coherence lengths for AWG with a channel spacing of (a) 200 GHz and (b) 100 GHz. Each scatter plot contains 1000 Monte Carlo simulations. The standard deviations at 25 arrayed waveguides are labeled.

The non-uniformities for various coherence lengths based on a statistical analysis for 200 GHz

and 100 GHz AWGs are shown in Fig. 4.9(a) and (b). The analysis of 1000 Monte Carlo simulations with random phase errors were performed and the results are plotted with light color lines and the average non-uniformity values are shown with dark color lines. In this study, the $\Delta\lambda_{FSR} = N_o\Delta\lambda$, and the calculated non-uniformity without phase errors is about 2.8 dB in both the 200 GHz and 100 GHz AWGs. The average non-uniformity shows an increasing trend with higher phase errors. The average non-uniformity in 200 GHz AWGs with 25 arrayed waveguides and a coherence length of 20 mm is 2.8 dB, whereas the one for 100 GHz AWGs is 2.9 dB. The non-uniformity fluctuations also increase with higher phase errors. The standard deviations of non-uniformity for 100 GHz AWGs with 25 arrayed waveguides and coherence lengths of 100 mm, 20 mm and 10 mm are 0.06 dB, 0.16 dB and 0.26 dB, respectively.

4.6 Conclusion

In summary, we presented a method to study the impact of phase errors on the performances of AWGs used as DWDM multiplexers, including their insertion loss, crosstalk, and non-uniformity, through a statistical analysis. In our method, the phase errors of the waveguides are characterized by their coherence length, which is extracted by measuring MZIs with varying arm lengths. The coherence length of the SiN strip waveguides fabricated using a wafer scale process is measured to be 23.7 mm. The experimentally measured coherence length is then used to take into consideration the phase error accumulated in waveguides with an efficient circuit-level AWG model used to performance Monte Carlo simulations and statistical analysis. Higher fabrication variations increase the range over which the insertion loss and crosstalk fluctuate and their statistical average values. AWGs with large FSRs usually have arrayed waveguides with a short average length and thus accumulated less phase errors, leading to improved performance. The trade-off between spatial resolution and tolerance to fabrication variations is a crucial consideration in the optimization of an AWG, e.g., in choosing the number of arrayed waveguides. The proposed method allows one to predict yield and to optimize complex PICs (such as AWGs) using compact, ready-for-test optical metrology structures such as MZIs. It could also be used with other more complex PICs and may accelerate their development, mass-production, and real-world applications.

4.7 Appendix

The effective index deviation $\delta n_{eff}(x)$ at one position x along the longitudinal direction of a waveguide is correlated to the deviation $\delta n_{eff}(x')$ when x' is near-by x. The phase error variance caused by effective index fluctuations is given by [45]:

$$\left\langle \delta\phi^2(L) \right\rangle = k_0^2 \int_0^L \mathrm{d}x \int_0^L \left\langle \delta n_{eff}(x) \delta n_{eff}(x') \right\rangle \, \mathrm{d}x' \tag{4.14}$$

In [103; 104], an exponential model describing the autocorrelation function of the waveguide roughness is given based on the correlation length Lc, which is defined as the distance over which the autocorrelation function falls to e^{-1}. Accordingly, the autocorrelation function of the effective index deviation is given by:

$$\left\langle \delta n_{eff}(x)\delta n_{eff}(x')\right\rangle = \sigma_n^2 \exp\left(-\frac{|x-x'|}{L_c}\right) \tag{4.15}$$

where σ_n^2 is the effective index variance due to geometrical variations in the waveguide. Combining Eq. 4.14 and 4.15, the phase error variance is given by:

$$\left\langle \delta\phi^2(L)\right\rangle = 2(k_0\sigma_n L_c)^2\left(\frac{L}{L_c} + \exp\left(-\frac{L}{L_c}\right) - 1\right) \tag{4.16}$$

Within a small area, the dominant source of phase errors is normally variations in width, for which the correlation length is typically in tens of nanometer, which is much shorter than the length of SiN waveguides. [105]. In this case, the approximate phase error variance is given by:

$$\left\langle \delta\phi^2(L)\right\rangle \approx 2\left(k_0\sigma_n\right)^2 LL_c, \quad L >> L_c \tag{4.17}$$

Equation 4.17 indicates that the phase error standard deviation is proportional to $\sqrt{L}$ (i.e., the variance is linearly proportional to L), which confirms the assumption behind the coherence length and agrees well with the experimental observation in Section 4.4.3.

Conclusion and outlook

Conclusion

In this thesis, we demonstrated several novel concepts and compact AWGs on a SiN-thin-film platform for next-generation low-cost, high-capacity DWDM systems.

In Chapter 1, we explained why we choose to design and fabricated AWGs on a SiN platform. Comparing silicon and silicon nitride, SiN waveguides show better performances in the wavelength range of interest, including lower propagation losses and sensitivity to fabrication variations. We simulated SiN waveguide devices using FDTD Solutions and MODE Solutions from Ansys Lumerical. The operating principle of the star coupler and AWGs were presented. The impacts of the parameters of SiN AWGs on their performances were also studied based on a semi-analytical model. Based on the study, researchers can design AWGs with high performance based on PIC technology.

Most AWGs are only designed to work with the fundamental TE mode, which limits the improvement of transmission capacity in WDM systems. In Chapter 2, we achieved a novel polarization insensitive AWG built with SiN waveguides for the first time. Both channel spacing and center wavelength polarization insensitivity were achieved within a single AWG. The channel spacing polarization insensitivity was obtained by optimizing the geometry of the arrayed waveguides whereas the central wavelength polarization insensitivity was obtained by splitting the two polarization states and adjusting their angle of incidence at the input coupler. A 1×8 AWG with the channel spacing of 100 GHz was designed and demonstrated experimentally.

In Chapter 3, we proposed and demonstrated an ultra-compact SiN AWG enabled by a novel concept of waveguide superlattice. In the device presented in Chapter 2, gaps wider than 10 μm between adjacent identical SiN waveguides were used to suppress parasitic coupling, which limited how compact the AWG could be. To reduce the gap between waveguides while minimizing parasitic coupling, a periodic waveguide superlattice consisting of waveguides of alternating widths was introduced to form the arrayed waveguides. Thanks to the dense waveguide superlattice, the 1×8 100 GHz SL-AWG had a compact footprint of 4.3 mm $\times$ 0.6 mm, which is more than two times smaller than conventional AWGs. Furthermore, the

superlattice AWG had a low insertion loss of 3.4 dB and a crosstalk level of −18 dB. A large FSR decreases the length difference and further shrinks the gaps between adjacent waveguides. The superlattice AWGs can be implemented to cover a wide FSR while maintaining high performance in DWDM systems. Therefore, the transmission capacity could be increased at a low cost. The microspectrometers with a high resolution and a large number of channels can also be implemented with superlattice AWGs to shrink their footprint.

In Chapter 4, we developed for the first time a statistical analysis of the performance of SiN AWGs in the presence of phase errors in arrayed waveguides. The phase errors were characterized using the coherence length. Beside the coupling between waveguides discussed in Chapter 3, crosstalk can also be caused by phase errors resulting from fabrication variations. In order to quantify the impacts of fabrication variations on the performances of AWGs, performance figures, including insertion loss, crosstalk and non-uniformity were parameterized. A die-level coherence length of 23.7 mm was measured for SiN waveguides. We calculated the impacts of phase errors on the performance of 1×4 AWGs as a function of the number of arrayed waveguides and middle waveguide lengths, through Monte Carlo simulations. This system of statistical analysis can not only predict the performance of AWGs based on a SiN platform but also on other material systems, and allows to extend the predictions to other optical devices at other wavelength ranges.

Outlook and future work

Our research paved the way for the development of low-cost, high-performance AWGs on an integrated CMOS-compatible, nano-photonics platform. Looking to the future, the following studies and research are suggested:

- The AWG model can be improved for more accurate prediction of performance. Our model described in Chapter 1 showed good agreement with the experimental results. However, it does not include coupling between the waveguides. AWGs with a large FSR usually have narrow gaps between the waveguides. In this case, the coupling may have a non-negligible impact on the crosstalk performance, especially for the SiN platform based on the waveguides with a relatively low refractive index contrast compared to SOI.

- The optical properties of materials are often sensitive to temperature. Therefore, thermal control is usually needed for integrated photonic circuits [106]. To compensate for the impacts of phase errors caused by fabrication variations, one can use heaters or other tuning mechanisms onto the arrayed waveguides for improved crosstalk.

- As shown in Chapter 4, the crosstalk of an AWG is very sensitive to fabrication errors and wafer non-uniformity. Therefore, improving the fabrication process (e.g., better control over lithography and side-wall roughness) can substantially enhance the performance of

an AWG. A thinner and wider waveguide can also be used to decrease the impacts of sidewall roughness and fabrication variations, as discussed in Chapter 1 and 4.

- The AWGs integrated with interleavers can be designed to cover a broad band in ultra-dense WDM systems. An interleaver can help AWGs to obtain a narrower channel spacing meanwhile maintain a good performance. Both AWGs and interleavers with a flat-top response can be optimized to minimize the impacts of phase errors.

- A flattop response can be obtained by designing an MZI-AWG or an MMI-AWG [107; 108; 109]. The impacts of phase errors should be studied. A comparison between these designs will be helpful to allow designers to choose the most suitable structures for specific applications.

- Integration with other components such as lasers, modulators, and photo-detectors, as well as other passive optical components and electric control system is our motivation of developing a high-performance AWG on the photonic integrated circuits platform. It will be of great interest to demonstrate such a fully integrated chip-scale photonic system in the future work.

Bibliography

[1] L. Vivien and L. Pavesi, *Handbook of silicon photonics*. Taylor & Francis, 2016.

[2] D. Ozcelik, J. W. Parks, T. A. Wall, M. A. Stott, H. Cai, J. W. Parks, A. R. Hawkins, and H. Schmidt, "Optofluidic wavelength division multiplexing for single-virus detection," *Proceedings of the National Academy of Sciences*, vol. 112, no. 42, pp. 12 933–12 937, 2015.

[3] D. de Felipe, C. Zawadzki, Z. Zhang, A. Maese, M. Wenzel, H. Li, G. Przyrembel, A. Sigmund, M. Möhrle, N. Keil, N. Grote, and M. Schell, "Hybrid InP/polymer optical line terminals for 40-channel 100-GHz spectrum-sliced WDM-PON," in *39th European Conference and Exhibition on Optical Communication (ECOC 2013)*, 2013, pp. 1–3.

[4] S. Tsuda and V. L. da Silva, "Transmission of 80 × 10 Gbit/s WDM channels with 50 GHz spacing over 500 km of LEAF® fiber," in *Optical Fiber Communication Conference*. Optica Publishing Group, 2000, p. TuJ6.

[5] Y.-S. Hurh, G.-S. Hwang, J.-Y. Jeon, K.-G. Lee, K.-W. Shin, S. S. Lee, K. Y. Yi, and J.-S. Lee, "1-Tb/s (100/spl times/12.4 gb/s) transmission of 12.5-GHz-spaced ultradense WDM channels over a standard single-mode fiber of 1200 km," *IEEE Photonics Technology Letters*, vol. 17, no. 3, pp. 696–698, 2005.

[6] A. Srivastava, S. Radic, C. Wolf, J. Centanni, J. Sulhoff, K. Kantor, and Y. Sun, "Ultradense WDM transmission in L-band," *IEEE Photonics Technology Letters*, vol. 12, no. 11, pp. 1570–1572, 2000.

[7] R. Ramaswami, "Optical fiber communication: from transmission to networking," *IEEE Communications Magazine*, vol. 40, no. 5, pp. 138–147, 2002.

[8] K. Yamada, T. Shoji, T. Tsuchizawa, T. Watanabe, J.-i. Takahashi, and S.-i. Itabashi, "Silicon-wire-based ultrasmall lattice filters with wide free spectral ranges," *Optics letters*, vol. 28, no. 18, pp. 1663–1664, 2003.

[9] F. Horst, W. M. Green, S. Assefa, S. M. Shank, Y. A. Vlasov, and B. J. Offrein, "Cascaded Mach-Zehnder wavelength filters in silicon photonics for low loss and flat pass-band WDM (de-) multiplexing," *Optics express*, vol. 21, no. 10, pp. 11 652–11 658, 2013.

[10] F. Horst, W. M. Green, B. J. Offrein, and Y. A. Vlasov, "Silicon-on-insulator echelle grating WDM demultiplexers with two stigmatic points," *IEEE Photonics Technology Letters*, vol. 21, no. 23, pp. 1743–1745, 2009.

[11] J. Brouckaert, W. Bogaerts, S. Selvaraja, P. Dumon, R. Baets, and D. Van Thourhout, "Planar concave grating demultiplexer with high reflective Bragg reflector facets," *IEEE Photonics technology letters*, vol. 20, no. 4, pp. 309–311, 2008.

[12] W. Bogaerts, S. Pathak, A. Ruocco, and S. Dwivedi, "Silicon photonics non-resonant wavelength filters: comparison between AWGs, echelle gratings, and cascaded Mach-Zehnder filters," in *Integrated Optics: Devices, Materials, and Technologies XIX*, J.-E. Broquin and G. N. Conti, Eds., vol. 9365, International Society for Optics and Photonics. SPIE, 2015, pp. 42 – 53.

[13] S. I. Bozhevolnyi, V. S. Volkov, E. Devaux, J.-Y. Laluet, and T. W. Ebbesen, "Channel plasmon subwavelength waveguide components including interferometers and ring resonators," *Nature*, vol. 440, no. 7083, pp. 508–511, 2006.

[14] W. Bogaerts, P. De Heyn, T. Van Vaerenbergh, K. De Vos, S. Kumar Selvaraja, T. Claes, P. Dumon, P. Bienstman, D. Van Thourhout, and R. Baets, "Silicon microring resonators," *Laser & Photonics Reviews*, vol. 6, no. 1, pp. 47–73, 2012.

[15] K. Yamada, T. Shoji, T. Tsuchizawa, T. Watanabe, J.-i. Takahashi, and S.-i. Itabashi, "Silicon-wire-based ultrasmall lattice filters with wide free spectral ranges," *Optics letters*, vol. 28, no. 18, pp. 1663–1664, 2003.

[16] F. Horst, W. M. Green, S. Assefa, S. M. Shank, Y. A. Vlasov, and B. J. Offrein, "Cascaded Mach-Zehnder wavelength filters in silicon photonics for low loss and flat pass-band WDM (de-) multiplexing," *Optics express*, vol. 21, no. 10, pp. 11 652–11 658, 2013.

[17] D. Liu, L. Zhang, Y. Tan, and D. Dai, "High-order adiabatic elliptical-microring filter with an ultra-large free-spectral-range," *Journal of Lightwave Technology*, vol. 39, no. 18, pp. 5910–5916, 2021.

[18] S. Pathak, P. Dumon, D. Van Thourhout, and W. Bogaerts, "Comparison of AWGs and echelle gratings for wavelength division multiplexing on silicon-on-insulator," *IEEE Photonics Journal*, vol. 6, no. 5, pp. 1–9, 2014.

[19] C. Dragone, "Efficient N× N star coupler based on Fourier optics," *Electronics Letters*, vol. 24, no. 15, pp. 942–944, 1988.

[20] T. WADA, "New focusing and dispersive planar component based on an optical phased array," *Electronics letters*, vol. 24, no. 7, p. 385, 1988.

[21] C. Dragone, C. Edwards, and R. Kistler, "Integrated optics N* N multiplexer on silicon," *IEEE Photonics Technology Letters*, vol. 3, no. 10, pp. 896–899, 1991.

[22] D. Dai, Z. Wang, J. F. Bauters, M.-C. Tien, M. J. Heck, D. J. Blumenthal, and J. E. Bowers, "Low-loss Si_3N_4 arrayed-waveguide grating (de) multiplexer using nano-core optical waveguides," *Optics express*, vol. 19, no. 15, pp. 14 130–14 136, 2011.

[23] S. Kamei, Y. Inoue, T. Shibata, and A. Kaneko, "Low-loss and compact silica-based athermal arrayed waveguide grating using resin-filled groove," *J. Lightwave Technol.*, vol. 27, no. 17, pp. 3790–3799, Sep 2009.

[24] H. Li, Y. Wang, Y. Sun, S. Zhang, Z. An, S. Zhang, C. Zhang, Z. Zhang, Q. Mao, and J. D. P. García, "Design and fabrication of SU-8 polymer arrayed waveguide gratings based on flexible PDMS substrates," *Applied Optics*, vol. 61, no. 9, pp. 2213–2218, 2022.

[25] Y. Barbarin, X. Leijtens, E. Bente, C. Louzao, J. Kooiman, and M. Smit, "Extremely small AWG demultiplexer fabricated on InP by using a double-etch process," *IEEE Photonics Technology Letters*, vol. 16, no. 11, pp. 2478–2480, 2004.

[26] Y. Wu, T.-n. T. Lang, J. Song, and J.-J. He, "A 8 × 8 arrayed waveguide grating router based on SOI platform," in *2017 16th International Conference on Optical Communications and Networks (ICOCN)*, 2017, pp. 1–3.

[27] S. Wang, T. Lang, Y. Wu, and J.-J. He, "16 × 16 awg router based on silicon oxynitride waveguide platform," in *2018 Conference on Lasers and Electro-Optics Pacific Rim (CLEO-PR)*, 2018, pp. 1–2.

[28] Q. Han, M. Ménard, and W. Shi, "Superlattice Arrayed Waveguide Grating in Silicon Nitride," *IEEE Photonics Technology Letters*, vol. 32, no. 22, pp. 1411–1414, 2020.

[29] I. Kiyat, A. Aydinli, and N. Dagli, "A compact silicon-on-insulator polarization splitter," *IEEE Photonics Technology Letters*, vol. 17, no. 1, pp. 100–102, 2005.

[30] A. M. Prabhu, A. Tsay, Z. Han, and V. Van, "Ultracompact SOI microring add–drop filter with wide bandwidth and wide FSR," *IEEE Photonics Technology Letters*, vol. 21, no. 10, pp. 651–653, 2009.

[31] S. T. Fard, V. Donzella, S. A. Schmidt, J. Flueckiger, S. M. Grist, P. T. Fard, Y. Wu, R. J. Bojko, E. Kwok, N. A. Jaeger *et al.*, "Performance of ultra-thin SOI-based resonators for sensing applications," *Optics express*, vol. 22, no. 12, pp. 14 166–14 179, 2014.

[32] T. Iwaki, J. Covington, F. Udrea, S. Ali, P. Guha, and J. Gardner, "Design and simulation of resistive SOI CMOS micro-heaters for high temperature gas sensors," in *Journal of Physics: Conference Series*, vol. 15, no. 1. IOP Publishing, 2005, p. 005.

[33] S. Keyvaninia, G. Roelkens, D. Van Thourhout, C. Jany, M. Lamponi, A. Le Liepvre, F. Lelarge, D. Make, G.-H. Duan, D. Bordel *et al.*, "Demonstration of a heterogeneously integrated III-V/SOI single wavelength tunable laser," *Optics express*, vol. 21, no. 3, pp. 3784–3792, 2013.

[34] A. Subramanian, P. Neutens, A. Dhakal, R. Jansen, T. Claes, X. Rottenberg, F. Peyskens, S. Selvaraja, P. Helin, B. Du Bois *et al.*, "Low-loss singlemode PECVD silicon nitride photonic wire waveguides for 532–900 nm wavelength window fabricated within a CMOS pilot line," *IEEE Photonics Journal*, vol. 5, no. 6, pp. 2 202 809–2 202 809, 2013.

[35] Y. Huang, J. Song, X. Luo, T.-Y. Liow, and G.-Q. Lo, "CMOS compatible monolithic multi-layer Si_3N_4-on-SOI platform for low-loss high performance silicon photonics dense integration," *Optics Express*, vol. 22, no. 18, pp. 21 859–21 865, 2014.

[36] S. Romero-García, F. Merget, F. Zhong, H. Finkelstein, and J. Witzens, "Silicon nitride CMOS-compatible platform for integrated photonics applications at visible wavelengths," *Optics express*, vol. 21, no. 12, pp. 14 036–14 046, 2013.

[37] R. Baets, A. Z. Subramanian, S. Clemmen, B. Kuyken, P. Bienstman, N. Le Thomas, G. Roelkens, D. Van Thourhout, P. Helin, and S. Severi, "Silicon Photonics: silicon nitride versus silicon-on-insulator," in *Optical Fiber Communication Conference*. Optical Society of America, 2016, pp. Th3J–1.

[38] R. M. De Ridder, K. Warhoff, A. Driessen, P. V. Lambeck, and H. Albers, "Silicon oxynitride planar waveguiding structures for application in optical communication," *IEEE Journal of selected topics in quantum electronics*, vol. 4, no. 6, pp. 930–937, 1998.

[39] L. Wang, W. Xie, D. Van Thourhout, Y. Zhang, H. Yu, and S. Wang, "Nonlinear silicon nitride waveguides based on a PECVD deposition platform," *Optics express*, vol. 26, no. 8, pp. 9645–9654, 2018.

[40] C.-E. Morosanu, "The preparation, characterization and applications of silicon nitride thin films," *Thin Solid Films*, vol. 65, no. 2, pp. 171–208, 1980. [Online]. Available: https://www.sciencedirect.com/science/article/pii/0040609080902540

[41] R. Heideman, A. Leinse, W. Hoving, R. Dekker, D. Geuzebroek, E. Klein, R. Stoffer, C. Roeloffzen, L. Zhuang, and A. Meijerink, "Large-scale integrated optics using TriPleX waveguide technology: from UV to IR," in *Photonics Packaging, Integration, and Interconnects IX*, vol. 7221. International Society for Optics and Photonics, 2009, p. 72210R.

[42] A. Z. Subramanian, E. Ryckeboer, A. Dhakal, F. Peyskens, A. Malik, B. Kuyken, H. Zhao, S. Pathak, A. Ruocco, A. De Groote *et al.*, "Silicon and silicon nitride photonic circuits for spectroscopic sensing on-a-chip," *Photonics Research*, vol. 3, no. 5, pp. B47–B59, 2015.

[43] A. Dhakal, A. Z. Subramanian, P. Wuytens, F. Peyskens, N. Le Thomas, and R. Baets, "Evanescent excitation and collection of spontaneous Raman spectra using silicon nitride nanophotonic waveguides," *Optics letters*, vol. 39, no. 13, pp. 4025–4028, 2014.

[44] D. Martens, A. Z. Subramanian, S. Pathak, M. Vanslembrouck, P. Bienstman, W. Bogaerts, and R. G. Baets, "Compact silicon nitride arrayed waveguide gratings for very near-infrared wavelengths," *IEEE Photonics Technology Letters*, vol. 27, no. 2, pp. 137–140, 2014.

[45] T. Goh, S. Suzuki, and A. Sugita, "Estimation of waveguide phase error in silica-based waveguides," *Journal of Lightwave Technology*, vol. 15, no. 11, pp. 2107–2113, 1997.

[46] R. Adar, C. H. Henry, M. A. Milbrodt, and R. C. Kistler, "Phase coherence of optical waveguides," *Journal of Lightwave Technology*, vol. 12, no. 4, pp. 603–606, 1994.

[47] U. Khan, M. Fiers, Y. Xing, and W. Bogaerts, "Experimental phase-error extraction and modelling in silicon photonic arrayed waveguide gratings," in *Silicon Photonics XV*, G. T. Reed and A. P. Knights, Eds., vol. 11285, International Society for Optics and Photonics. SPIE, 2020, pp. 161 – 172.

[48] W. Bogaerts, P. Dumon, D. Van Thourhout, D. Taillaert, P. Jaenen, J. Wouters, S. Beckx, V. Wiaux, and R. G. Baets, "Compact wavelength-selective functions in silicon-on-insulator photonic wires," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 12, no. 6, pp. 1394–1401, 2006.

[49] T. Tsuchizawa, K. Yamada, H. Fukuda, T. Watanabe, S. Uchiyama, and S. Itabashi, "Low-loss Si wire waveguides and their application to thermooptic switches," *Japanese journal of applied physics*, vol. 45, no. 8S, p. 6658, 2006.

[50] J. F. Bauters, M. J. Heck, D. John, D. Dai, M.-C. Tien, J. S. Barton, A. Leinse, R. G. Heideman, D. J. Blumenthal, and J. E. Bowers, "Ultra-low-loss high-aspect-ratio Si_3N_4 waveguides," *Optics express*, vol. 19, no. 4, pp. 3163–3174, 2011.

[51] L. Chrostowski and M. Hochberg, *Silicon photonics design: from devices to systems*. Cambridge University Press, 2015.

[52] Y. Fu, T. Ye, W. Tang, and T. Chu, "Efficient adiabatic silicon-on-insulator waveguide taper," *Photonics Research*, vol. 2, no. 3, pp. A41–A44, 2014.

[53] H. A. Rowland, "Preliminary notice of the results accomplished in the manufacture and theory of gratings for optical purposes," *Philosophical Magazine Series 5*, vol. 13, no. 84, pp. 469–474, 1982.

[54] L. Vivien and L. Pavesi, *Handbook of Silicon Photonics*. CRC Press, 2013.

[55] H. A. Rowland, "XXIX. On concave gratings for optical purposes," *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, vol. 16, no. 99, pp. 197–210, 1883.

[56] H. Takahashi, S. Suzuki, K. Kato, and I. Nishi, "Arrayed-waveguide grating for wavelength division multi/demultiplexer with nanometre resolution," *Electronics letters*, vol. 26, no. 2, pp. 87–88, 1990.

[57] H. Takahashi, Y. Hibino, and I. Nishi, "Polarization-insensitive arrayed-waveguide grating wavelength multiplexer on silicon," *Optics letters*, vol. 17, no. 7, pp. 499–501, 1992.

[58] P. Gatkine, S. Veilleux, Y. Hu, T. Zhu, Y. Meng, J. Bland-Hawthorn, and M. Dagenais, "Development of high-resolution arrayed waveguide grating spectrometers for astronomical applications: first results," in *Advances in Optical and Mechanical Technologies for Telescopes and Instrumentation II*, vol. 9912. SPIE, 2016, pp. 2180–2191.

[59] M. K. Smit and C. Van Dam, "PHASAR-based WDM-devices: Principles, design and applications," *IEEE Journal of selected topics in quantum electronics*, vol. 2, no. 2, pp. 236–250, 1996.

[60] E. Kleijn, M. K. Smit, and X. J. M. Leijtens, "New Analytical Arrayed Waveguide Grating Model," *Journal of Lightwave Technology*, vol. 31, no. 20, pp. 3309–3314, 2013.

[61] W. Bogaerts and S. K. Selvaraja, "Compact single-mode silicon hybrid rib/strip waveguide with adiabatic bends," *IEEE Photonics Journal*, vol. 3, no. 3, pp. 422–432, 2011.

[62] W. Bogaerts, S. K. Selvaraja, P. Dumon, J. Brouckaert, K. De Vos, D. Van Thourhout, and R. Baets, "Silicon-on-insulator spectral filters fabricated with CMOS technology," *IEEE journal of selected topics in quantum electronics*, vol. 16, no. 1, pp. 33–44, 2010.

[63] S. Pathak, D. Van Thourhout, and W. Bogaerts, "Design trade-offs for silicon-on-insulator-based AWGs for (de) multiplexer applications," *Optics letters*, vol. 38, no. 16, pp. 2961–2964, 2013.

[64] Z. Sheng, D. Dai, and S. He, "Improve channel uniformity of an Si-nanowire AWG demultiplexer by using dual-tapered auxiliary waveguides," *Journal of lightwave technology*, vol. 25, no. 10, pp. 3001–3007, 2007.

[65] Y. Sakamaki, S. Kamei, T. Hashimoto, T. Kitoh, and H. Takahashi, "Loss uniformity improvement of arrayed-waveguide grating with mode-field converters designed by wavefront matching method," *Journal of lightwave technology*, vol. 27, no. 24, pp. 5710–5715, 2009.

[66] C. Nadler, E. Wildermuth, M. Lanker, W. Hunziker, and H. Melchior, "Polarization insensitive, low-loss, low-crosstalk wavelength multiplexer modules," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 5, no. 5, pp. 1407 – 1412, Sep 1999.

[67] M. R. Amersfoort, J. B. D. Soole, H. P. LeBlanc, N. C. Andreadakis, A. Rajhel, C. Caneau, M. A. Koza, U. Bhat, and C. Youtsey, "Polarization-independent InP-arrayed waveguide filter using square cross-section waveguides ," *Optical Fiber Communications. OFC '96*, vol. 2, no. 2, p. WB5, May 1996.

[68] D. Dai, Y. Shi, and S. He, "Characteristic analysis of nanosilicon rectangular waveguides for planar light-wave circuits of high integration," *Applied optics*, vol. 45, no. 20, pp. 4941–4946, 2006.

[69] Q. Han, J. St-Yves, Y. Chen, M. Ménard, and W. Shi, "Polarization-insensitive silicon nitride arrayed waveguide grating," *Optics letters*, vol. 44, no. 16, pp. 3976–3979, 2019.

[70] Y. Zhao, C. Qiu, A. Wu, H. Huang, J. Li, Z. Sheng, W. Li, X. Wang, and F. Gan, "Broadband polarization splitter-rotator and the application in WDM receiver," *IEEE Photonics Journal*, vol. 11, no. 1, pp. 1–10, 2018.

[71] P. Trinh, S. Yegnanarayanan, F. Coppinger, and B. Jalali, "Silicon-on-insulator (SOI) phased-array wavelength multi/demultiplexer with extremely low-polarization sensitivity," *IEEE Photonics Technology Letters*, vol. 9, no. 7, pp. 940–942, 1997.

[72] K. Jia, W. Wang, Y. Tang, Y. Yang, J. Yang, X. Jiang, Y. Wu, M. Wang, and Y. Wang, "Silicon-on-insulator-based optical demultiplexer employing turning-mirror-integrated arrayed-waveguide grating," *IEEE Photonics Technology Letters*, vol. 17, no. 2, pp. 378–380, 2005.

[73] P. Dumon, W. Bogaerts, D. Van Thourhout, D. Taillaert, R. Baets, J. Wouters, S. Beckx, and P. Jaenen, "Compact wavelength router based on a silicon-on-insulator arrayed waveguide grating pigtailed to a fiber array," *Optics express*, vol. 14, no. 2, pp. 664–669, 2006.

[74] S. Cheung, T. Su, K. Okamoto, and S. Yoo, "Ultra-compact silicon photonic 512 $\times$ 512 25 GHz arrayed waveguide grating router," *IEEE Journal of Selected Topics in Quantum Electronics*, vol. 20, no. 4, pp. 310–316, 2013.

[75] P. Cheben, J. Schmid, A. Delâge, A. Densmore, S. Janz, B. Lamontagne, J. Lapointe, E. Post, P. Waldron, and D.-X. Xu, "A high-resolution silicon-on-insulator arrayed waveguide grating microspectrometer with sub-micrometer aperture waveguides," *Optics express*, vol. 15, no. 5, pp. 2299–2306, 2007.

[76] J. F. Bauters, J. R. Adleman, M. J. Heck, and J. E. Bowers, "Design and characterization of arrayed waveguide gratings using ultra-low loss Si_3N_4 waveguides," *Applied Physics A*, vol. 116, no. 2, pp. 427–432, 2014.

[77] C. Doerr, L. Chen, L. Buhl, and Y.-K. Chen, "Eight-channel SiO_2/ Si_3N_4/Si/Ge CWDM receiver," *IEEE Photonics Technology Letters*, vol. 23, no. 17, pp. 1201–1203, 2011.

[78] L. Chen, C. R. Doerr, L. Buhl, Y. Baeyens, and R. A. Aroca, "Monolithically integrated 40-wavelength demultiplexer and photodetector array on silicon," *IEEE Photonics Technology Letters*, vol. 23, no. 13, pp. 869–871, 2011.

[79] P. Gatkine, S. Veilleux, Y. Hu, J. Bland-Hawthorn, and M. Dagenais, "Arrayed waveguide grating spectrometers for astronomical applications: new results," *Optics express*, vol. 25, no. 15, pp. 17 918–17 935, 2017.

[80] A. Rahim, E. Ryckeboer, A. Z. Subramanian, S. Clemmen, B. Kuyken, A. Dhakal, A. Raza, A. Hermans, M. Muneeb, S. Dhoore *et al.*, "Expanding the silicon photonics portfolio with silicon nitride photonic integrated circuits," *Journal of lightwave technology*, vol. 35, no. 4, pp. 639–649, 2017.

[81] H. Fukuda, K. Yamada, T. Tsuchizawa, T. Watanabe, H. Shinojima, and S.-i. Itabashi, "Ultrasmall polarization splitter based on silicon wire waveguides," *Optics Express*, vol. 14, no. 25, pp. 12 401–12 408, 2006.

[82] H. Sattari, "Arrayed waveguide grating-based demultiplexer with two central wavelengths," *Optik*, vol. 123, no. 9, pp. 775–778, 2012.

[83] P. Gatkine, S. Veilleux, Y. Hu, J. Bland-Hawthorn, and M. Dagenais, "Arrayed waveguide grating spectrometers for astronomical applications: new results," *Optics express*, vol. 25, no. 15, pp. 17 918–17 935, 2017.

[84] A. Kaneko, T. Goh, H. Yamada, T. Tanaka, and L. Ogawa, "Design and applications of silica-based planar lightwave circuits," *IEEE Journal of selected topics in quantum electronics*, vol. 5, no. 5, pp. 1227–1236, 1999.

[85] S. Pathak, M. Vanslembrouck, P. Dumon, D. Van Thourhout, and W. Bogaerts, "Optimized silicon AWG with flattened spectral response using an MMI aperture," *Journal of Lightwave Technology*, vol. 31, no. 1, pp. 87–93, 2013.

[86] K. Kodate and Y. Komai, "Compact spectroscopic sensor using an arrayed waveguide grating," *Journal of Optics A: Pure and Applied Optics*, vol. 10, no. 4, p. 044011, 2008.

[87] A. Sugita, A. Kaneko, K. Okamoto, M. Itoh, A. Himeno, and Y. Ohmori, "Very low insertion loss arrayed-waveguide grating with vertically tapered waveguides," *IEEE Photonics Technology Letters*, vol. 12, no. 9, pp. 1180–1182, 2000.

[88] P. Pan, J. An, J. Zhang, Y. Wang, H. Wang, L. Wang, X. Yin, Y. Wu, J. Li, Q. Han *et al.*, "Flat-top AWG based on InP deep ridge waveguide," *Optics Communications*, vol. 355, pp. 376–381, 2015.

[89] M. Diemeer, L. Spiekman, R. Ramsamoedj, and M. Smit, "Polymeric phased array wavelength multiplexer operating around 1550 nm," *Electronics letters*, vol. 32, no. 12, pp. 1132–1133, 1996.

[90] B. Yang, Y. Zhu, Y. Jiao, L. Yang, Z. Sheng, S. He, and D. Dai, "Compact arrayed waveguide grating devices based on small SU-8 strip waveguides," *Journal of lightwave technology*, vol. 29, no. 13, pp. 2009–2014, 2011.

[91] W. Song, R. Gatdula, S. Abbaslou, M. Lu, A. Stein, W. Y. Lai, J. Provine, R. F. W. Pease, D. N. Christodoulides, and W. Jiang, "High-density waveguide superlattices with low crosstalk," *Nature communications*, vol. 6, no. 1, pp. 1–9, 2015.

[92] G. Assanto, A. Laureti-Palma, C. Sibilia, and M. Bertolotti, "All-optical switching via second harmonic generation in a nonlinearly asymmetric directional coupler," *Optics communications*, vol. 110, no. 5-6, pp. 599–603, 1994.

[93] S. Pathak, "Silicon nano-photonics based arrayed waveguide gratings," Ph.D. dissertation, Ghent University, 2014.

[94] K. Okamoto, "Progress and technical challenge for planar waveguide devices: silica and silicon waveguides," *Laser & photonics reviews*, vol. 6, no. 1, pp. 14–23, 2012.

[95] R. Adar, M. Serbin, and V. Mizrahi, "Less than 1 dB per meter propagation loss of silica waveguides measured using a ring resonator," *Journal of lightwave technology*, vol. 12, no. 8, pp. 1369–1372, 1994.

[96] B. Akca, B. Považay, A. Alex, K. Wörhoff, R. De Ridder, W. Drexler, and M. Pollnau, "Miniature spectrometer and beam splitter for an optical coherence tomography on a silicon chip," *Optics express*, vol. 21, no. 14, pp. 16 648–16 656, 2013.

[97] B. I. Akca, J. Kalkman, N. Ismail, G. Sengo, F. Sun, A. Driessen, T. G. van Leeuwen, M. Pollnau, K. Wörhoff, R. M. de Ridder *et al.*, "Toward spectral-domain optical coherence tomography on a chip," *IEEE journal of selected topics in quantum electronics*, vol. 18, no. 3, pp. 1223–1233, 2011.

[98] Y. Yang, Y. Ma, H. Guan, Y. Liu, S. Danziger, S. Ocheltree, K. Bergman, T. Baehr-Jones, and M. Hochberg, "Phase coherence length in silicon photonic platform," *Optics express*, vol. 23, no. 13, pp. 16 890–16 902, 2015.

[99] P. Gatkine, N. Jovanovic, C. Hopgood, S. Ellis, R. Broeke, K. Ławniczuk, J. Jewell, J. K. Wallace, and D. Mawet, "Potential of commercial SiN MPW platforms for developing mid/high-resolution integrated photonic spectrographs for astronomy," *Applied Optics*, vol. 60, no. 19, pp. D15–D32, 2021.

[100] A. Stoll, K. Madhav, and M. Roth, "Performance limits of astronomical arrayed waveguide gratings on a silica platform," *Optics Express*, vol. 28, no. 26, pp. 39 354–39 367, 2020.

[101] J. M. Wozencraft and J. M. Jacobs, *Principles of Communications Engineering*. New York: Wiley, 1956.

[102] S. Dwivedi, A. Ruocco, M. Vanslembrouck, T. Spuesens, P. Bienstman, P. Dumon, T. Van Vaerenbergh, and W. Bogaerts, "Experimental extraction of effective refractive index and thermo-optic coefficients of silicon-on-insulator waveguides using interferometers," *Journal of Lightwave Technology*, vol. 33, no. 21, pp. 4471–4477, 2015.

[103] F. Ladouceur, J. Love, and T. Senden, "Effect of side wall roughness in buried channel waveguides," *IEE Proceedings-Optoelectronics*, vol. 141, no. 4, pp. 242–248, 1994.

[104] J. Ogilvy and J. Foster, "Rough surfaces: Gaussian or exponential statistics?" *Journal of Physics D: Applied Physics*, vol. 22, no. 9, p. 1243, 1989.

[105] J. F. Bauters, J. R. Adleman, M. J. Heck, and J. E. Bowers, "Design and characterization of arrayed waveguide gratings using ultra-low loss Si_3N_4 waveguides," *Applied Physics A*, vol. 116, no. 2, pp. 427–432, 2014.

[106] J. Punch, "Thermal challenges in photonic integrated circuits," in *2012 13th International Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems*. IEEE, 2012, pp. 1–6.

[107] S. Pathak, M. Vanslembrouck, P. Dumon, D. Van Thourhout, and W. Bogaerts, "Optimized silicon AWG with flattened spectral response using an MMI aperture," *Journal of Lightwave Technology*, vol. 31, no. 1, pp. 87–93, 2013.

[108] K. Maru and Y. Abe, "Low-loss, flat-passband and athermal arrayedwaveguide grating multi/demultiplexer," *Optics Express*, vol. 15, no. 26, pp. 18 351–18 356, 2007.

[109] K. Maru, T. Mizumoto, and H. Uetsuka, "Modeling of multi-input arrayed waveguide grating and its application to design of flat-passband response using cascaded Mach–Zehnder interferometers," *Journal of lightwave technology*, vol. 25, no. 2, pp. 544–555, 2007.